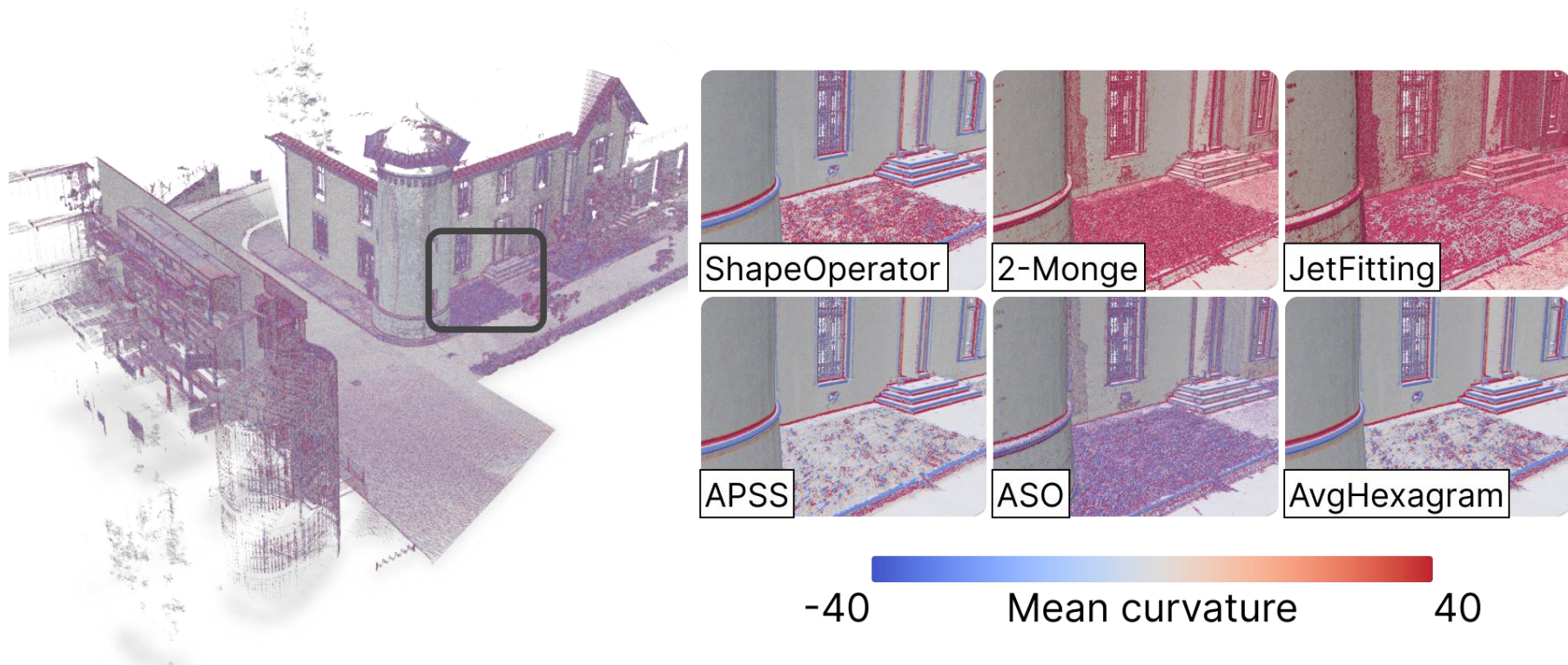


Survey on differential estimators for 3d point clouds



Survey on differential estimators for 3d point clouds

Léo Arnal--Anger¹, Thibault Lejemble², David Coeurjolly³, Loïc Barthe¹, Nicolas Mellado¹



¹ IRIT, Université de Toulouse, CNRS, Toulouse INP, UT, Toulouse, France

² Cerema, Toulouse, France

³ CNRS, Université Claude Bernard Lyon 1, INSA Lyon, LIRIS, France

Context

01

Survey on differential estimators for 3d point clouds

Survey on differential estimators for 3d point clouds

Survey on differential estimators for 3d point clouds

3D point clouds

- Capture real world geometry
- From real objects to virtual scenes



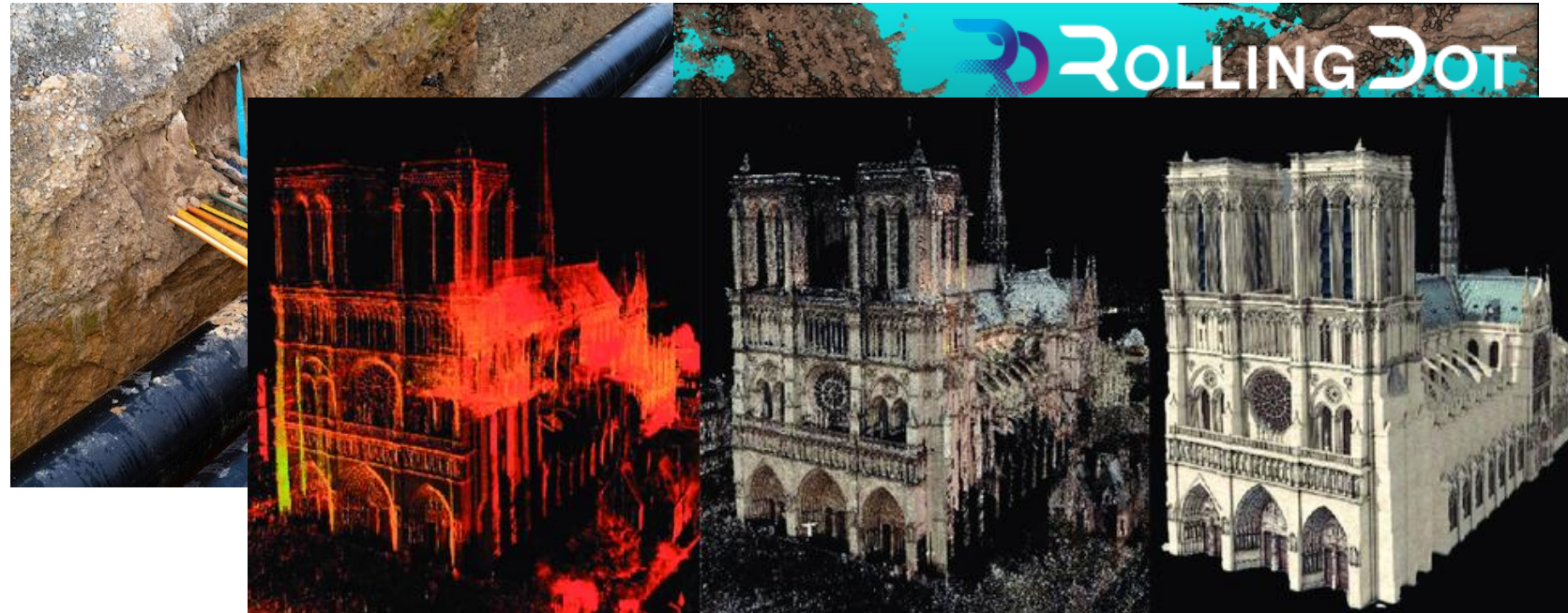
3D point clouds – Applications

- Maintenance



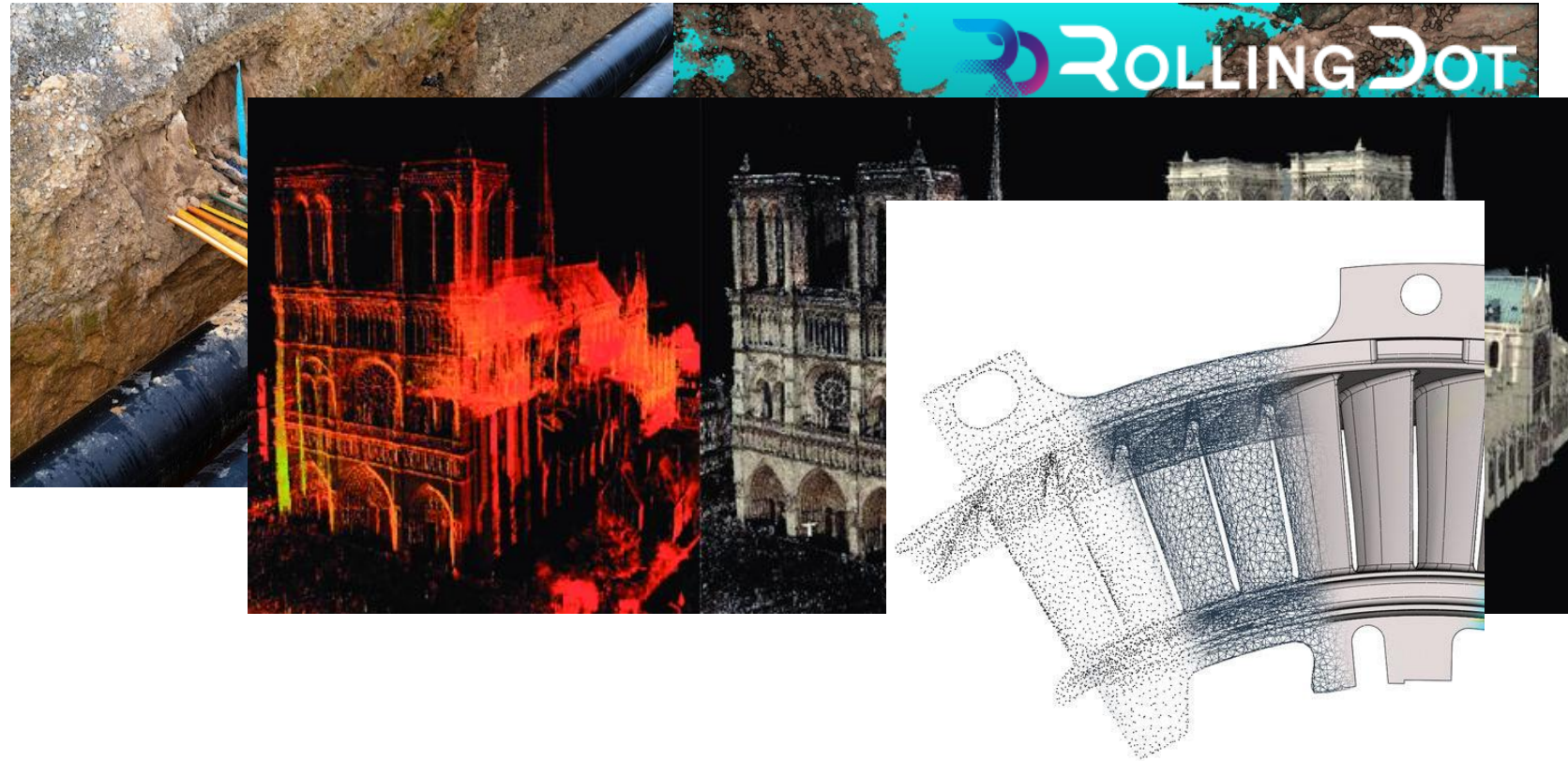
3D point clouds – Applications

- Maintenance
- Cultural Heritage



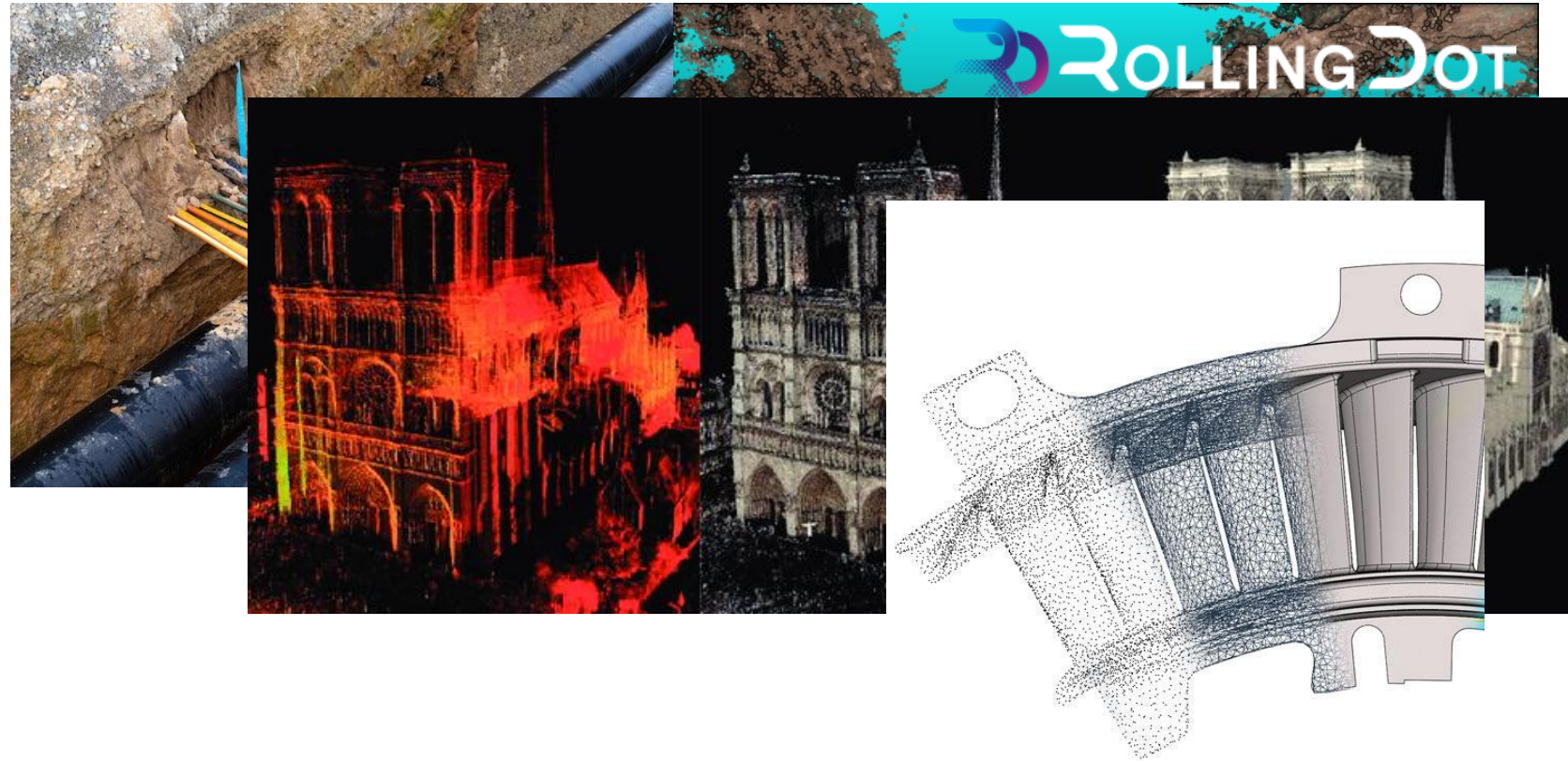
3D point clouds – Applications

- Maintenance
- Cultural Heritage
- Reverse Engineering



3D point clouds – Applications

- Maintenance
- Cultural Heritage
- Reverse Engineering
- ...
- Toward 3D digital twins



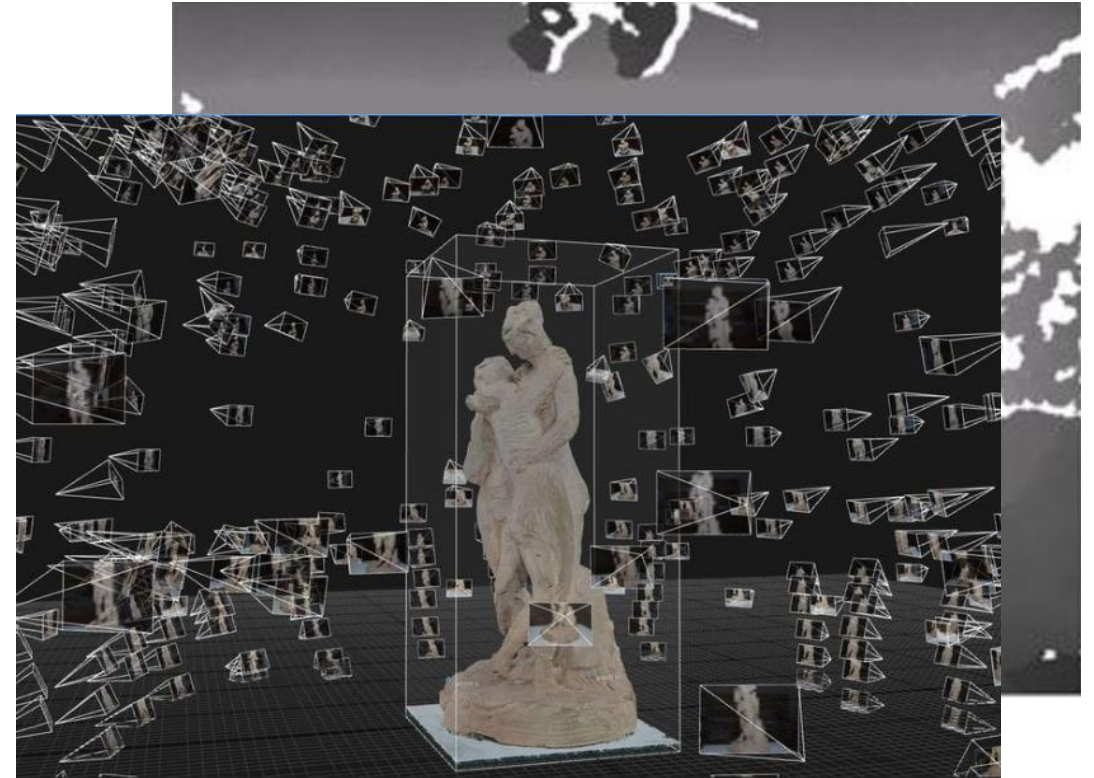
3D point clouds – Acquisition processes

- Depth map (RGBD)



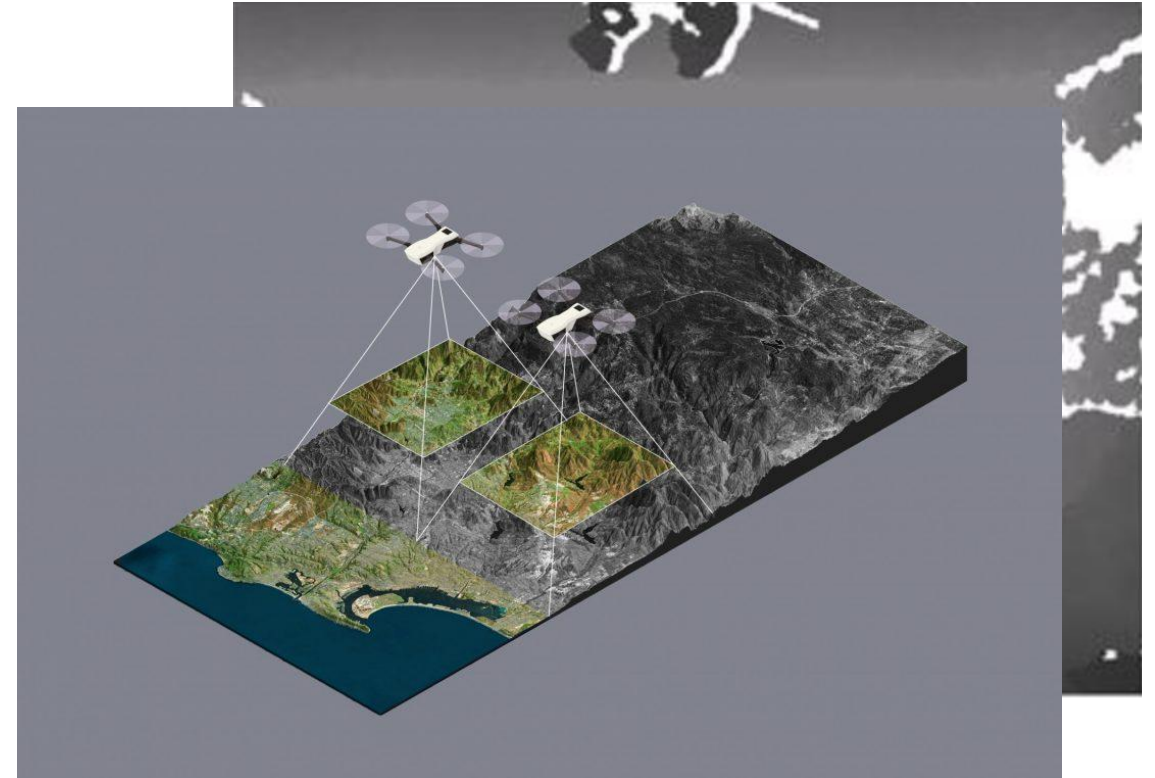
3D point clouds – Acquisition processes

- Depth map (RGBD)
- Photogrammetry



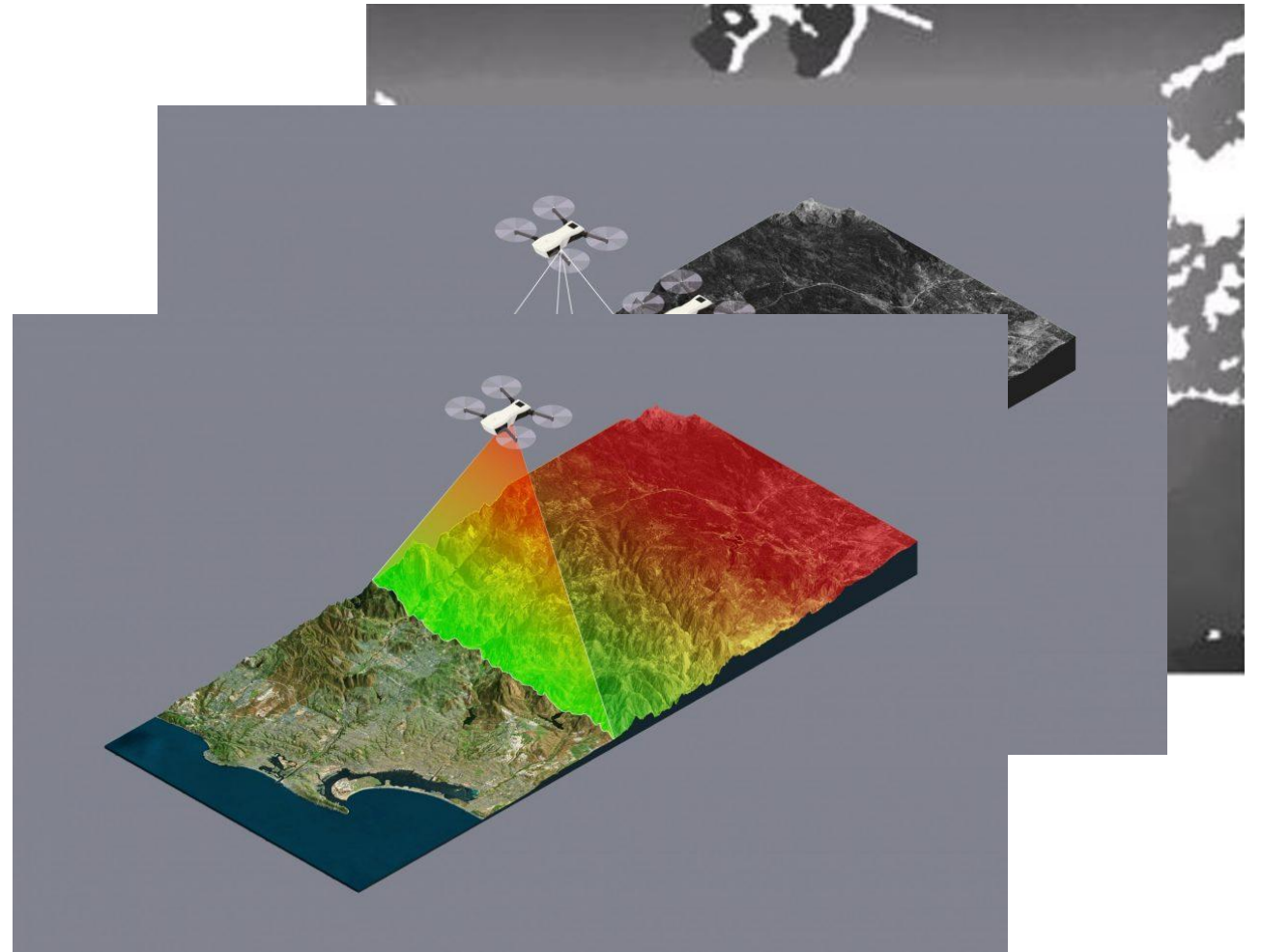
3D point clouds – Acquisition processes

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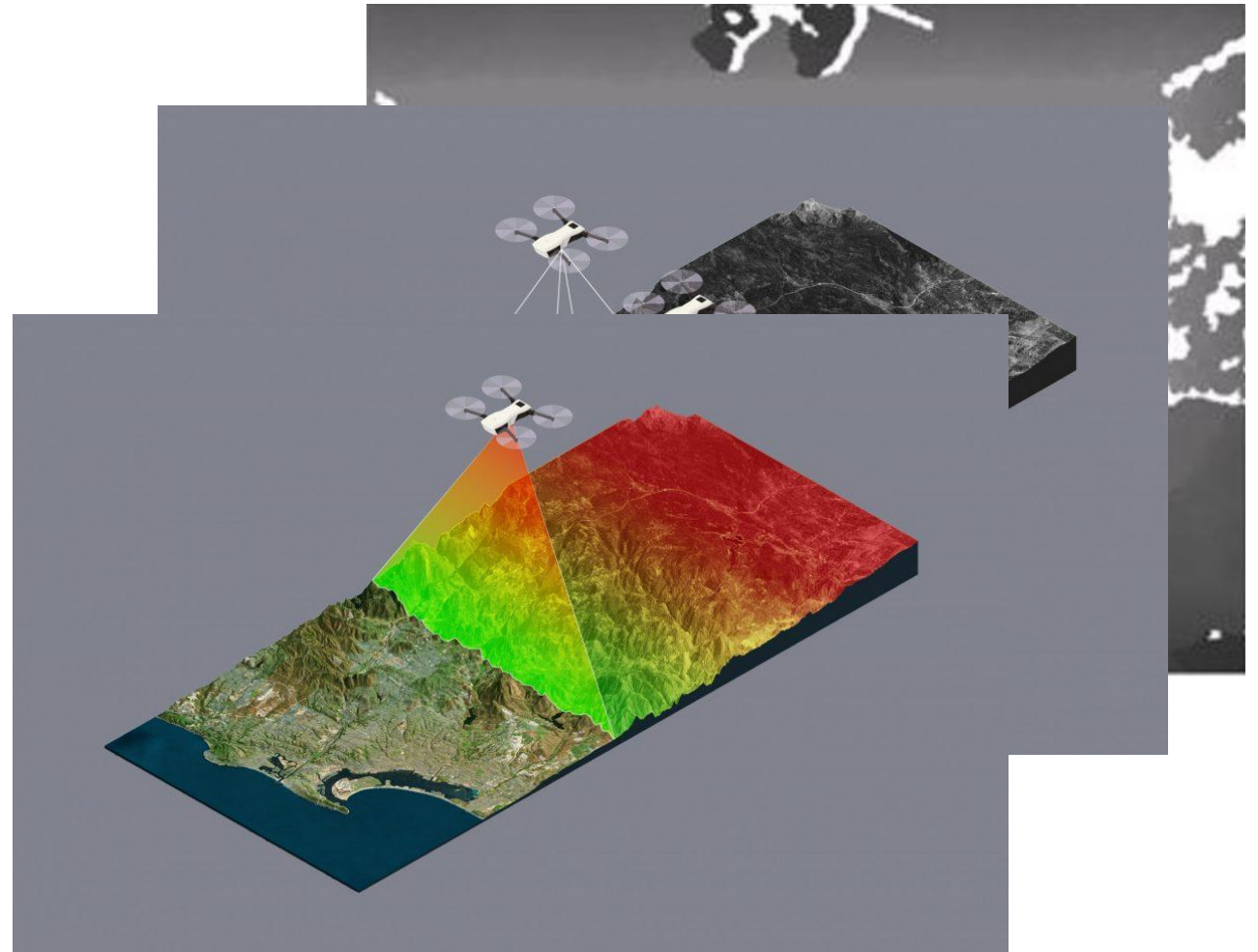
3D point clouds – Acquisition processes

- Depth map (RGBD)
- Photogrammetry
- LiDAR



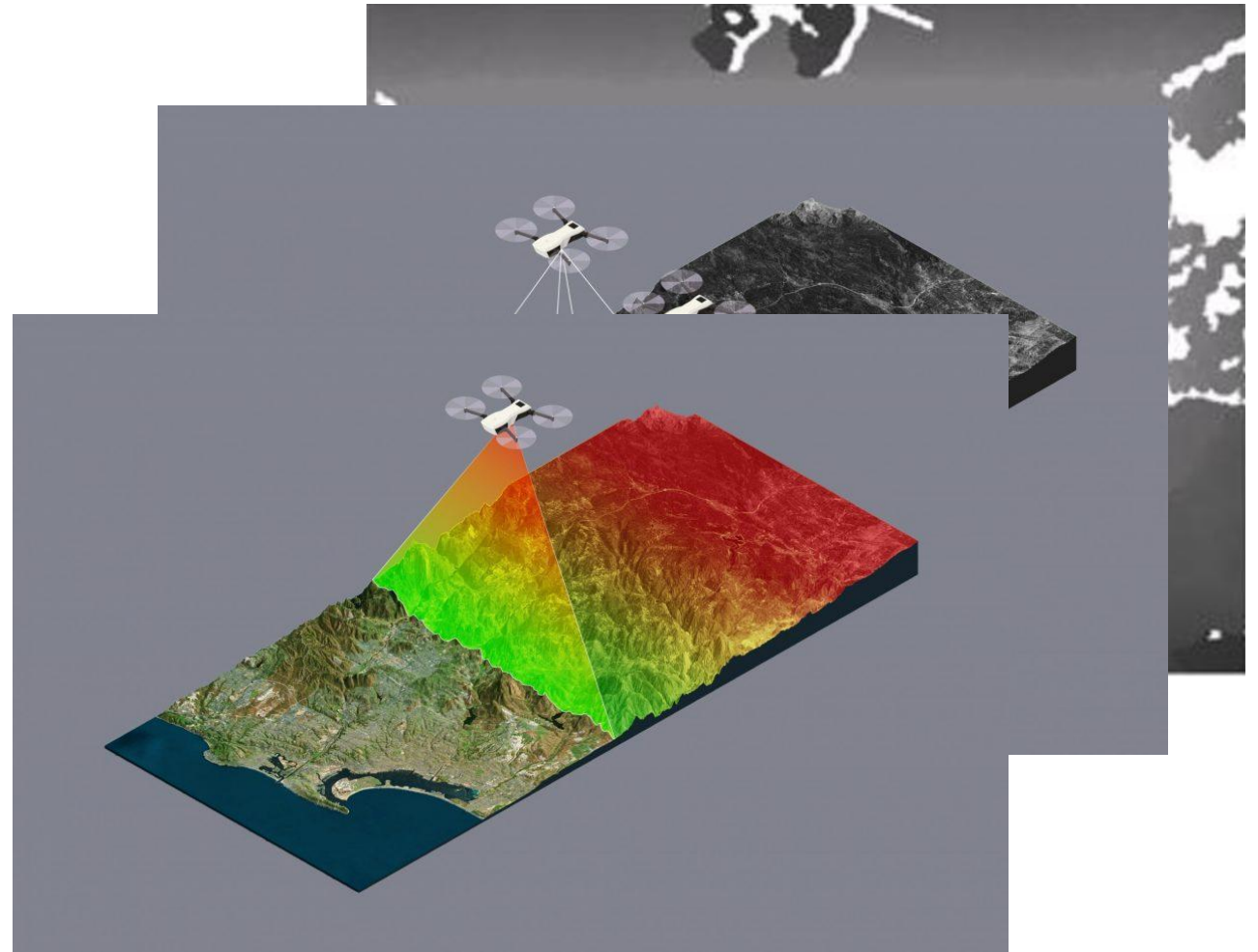
3D point clouds – Acquisition processes

- Depth map (RGBD)
- Photogrammetry
- LiDAR
- ...



3D point clouds – Acquisition processes

- Depth map (RGBD)
- Photogrammetry
- LiDAR
- ...



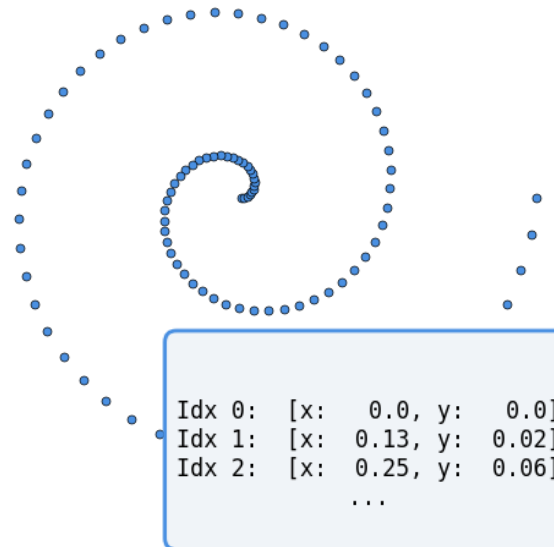
3D point clouds – Properties

- Unordered data

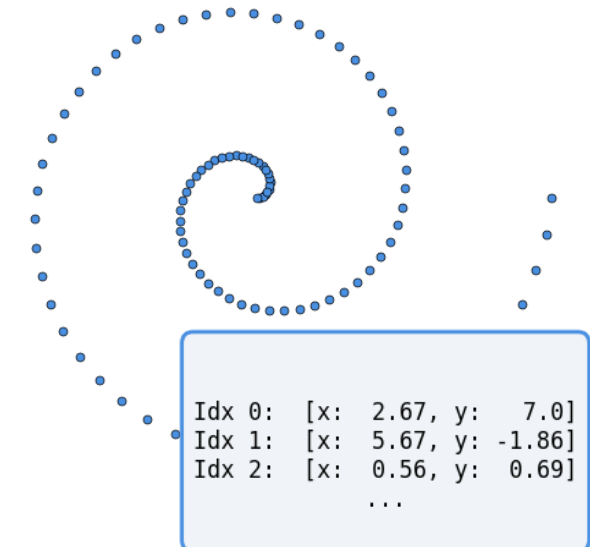


$\neq$

permuted pixels



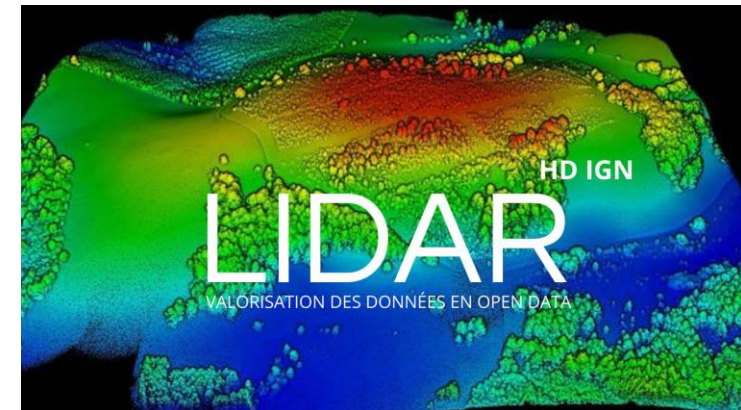
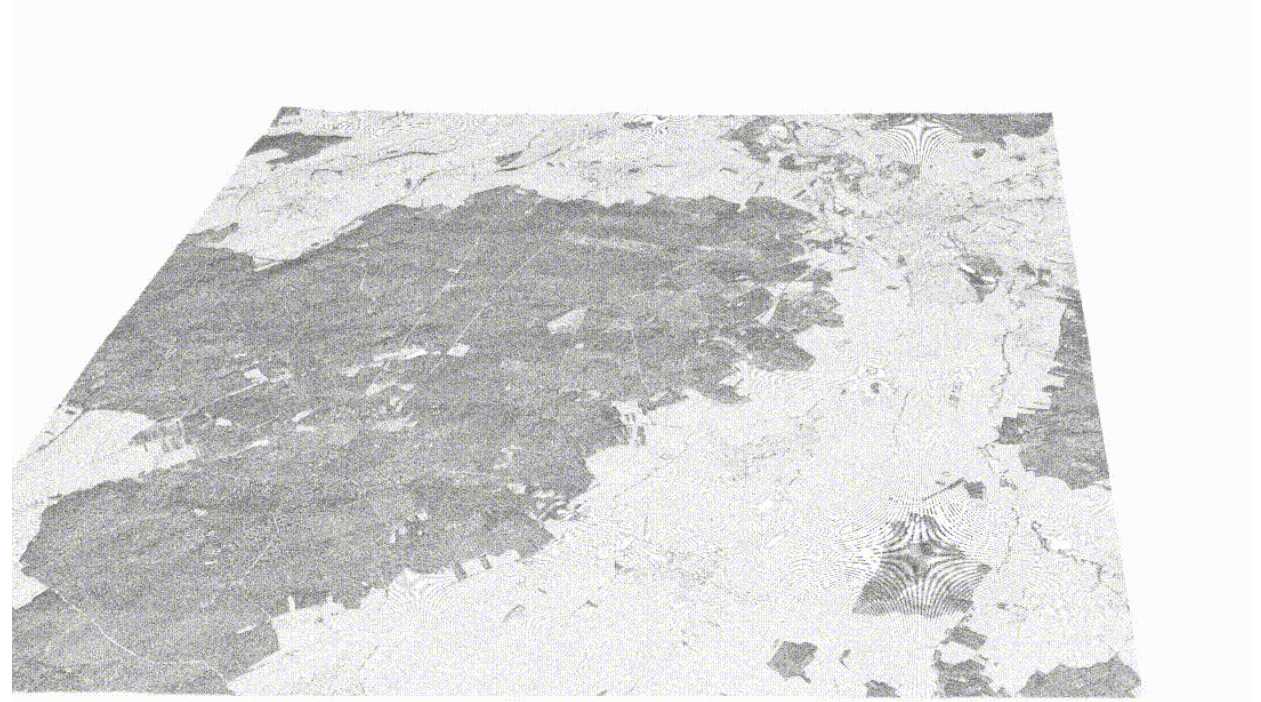
$=$



permuted points

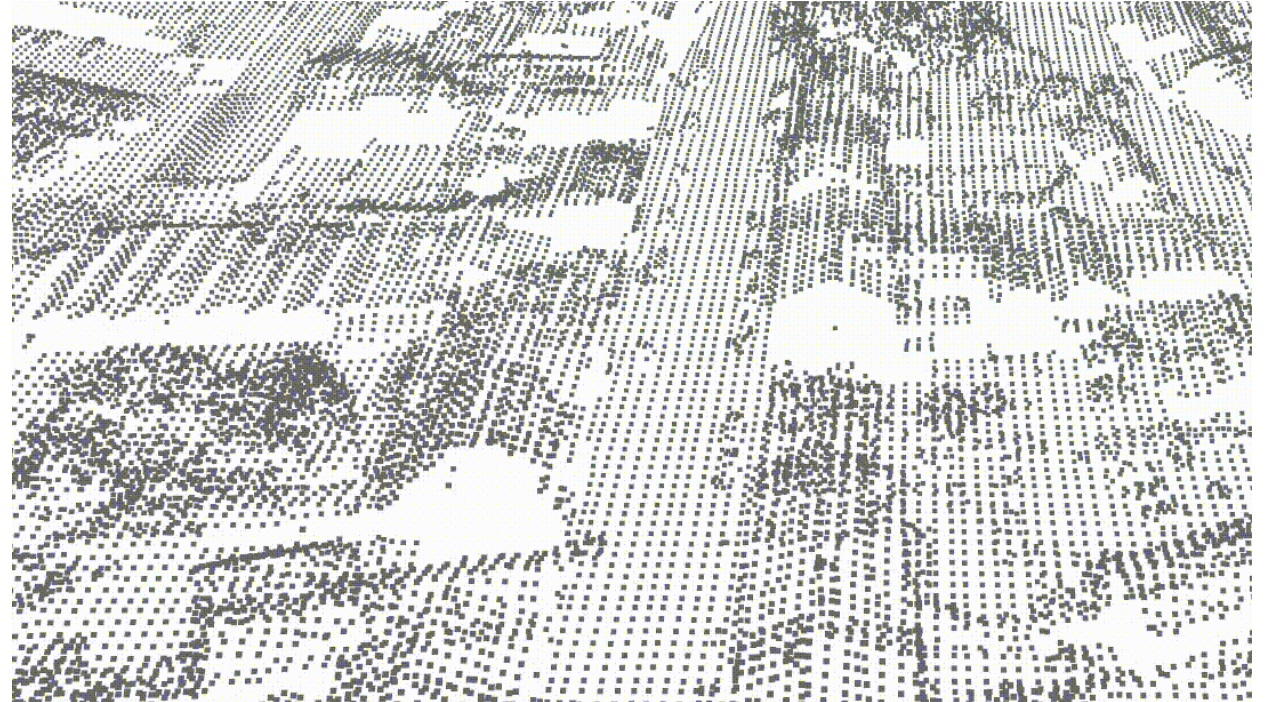
3D point clouds – Properties

- Unordered data
- Geometric complexity



3D point clouds – Properties

- Unordered data
- Geometric complexity
- Artefacts
 - Sampling variation
 - Measurement noise and outliers
 - Registration errors
 - Missing data (occlusions)



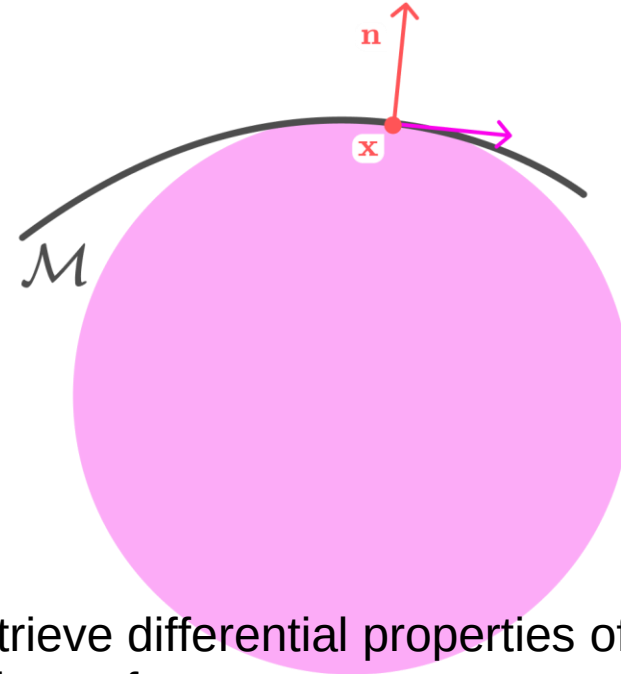
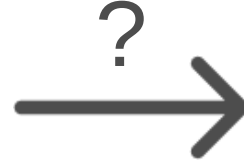
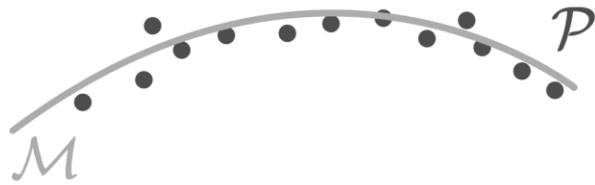
3D point clouds – Acquired data

- 3D coordinates
- Sensor positions
- Attributes (depending on the acquisition process)
 - Colors
 - Intensity / Reflectivity
 - Normal vectors (not captured but can be estimated)
 - ...

Survey on differential estimators for 3d point clouds

Survey on differential estimators for 3d point clouds

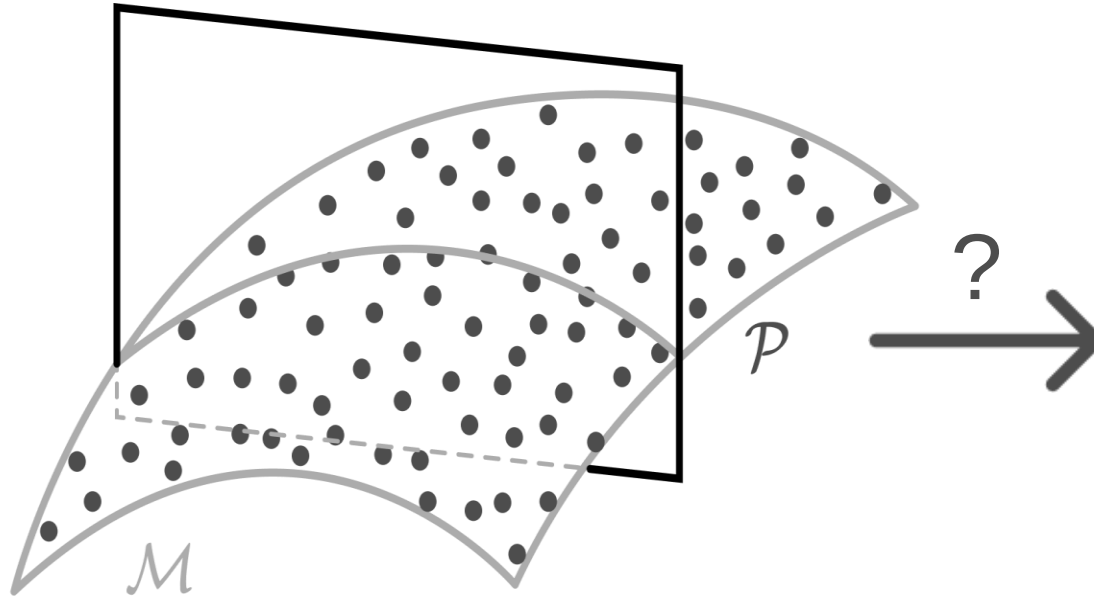
Differential



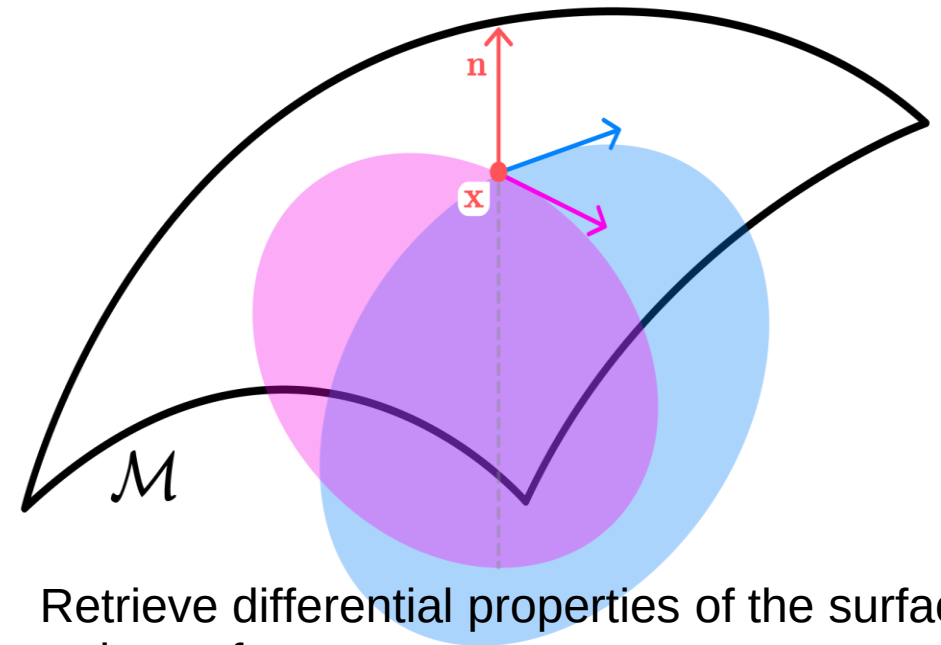
3D points sampling an unknown scanned smooth surface

Retrieve differential properties of the surface
– ie. surface curvature

Differential



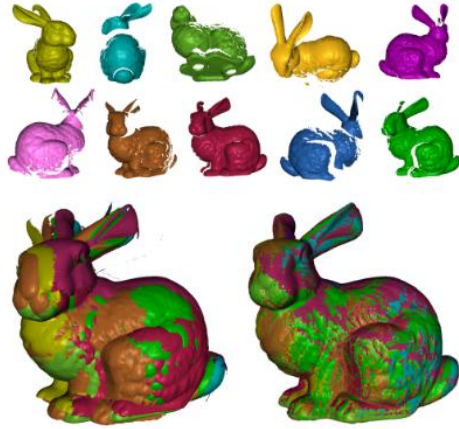
3D points sampling an unknown scanned smooth surface



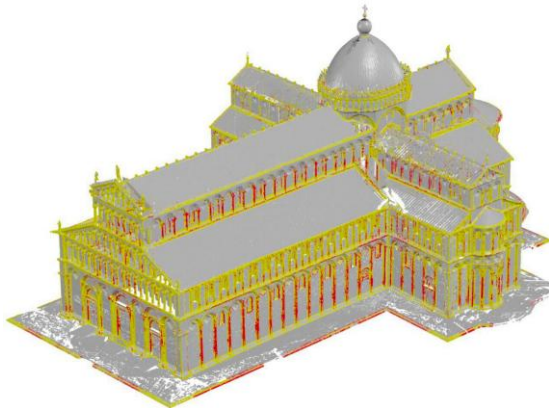
Retrieve differential properties of the surface
– ie. surface curvature

Differential – Usage Example

- Registration [GMG*05]



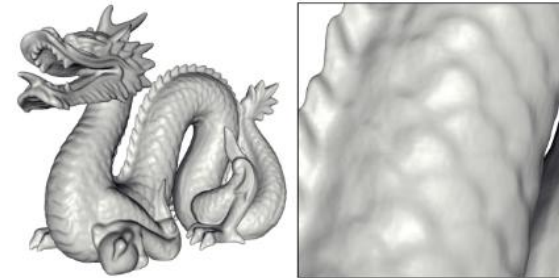
- Classification [HLP*21]



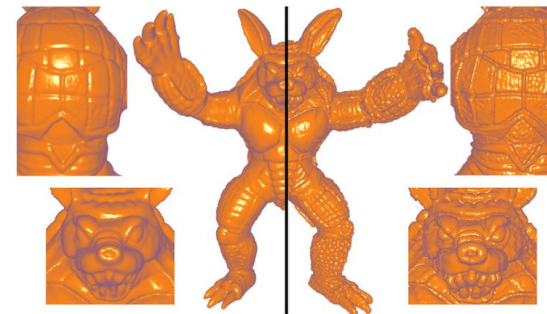
- Rendering [DFR*03]



- Meshing [KBH06]

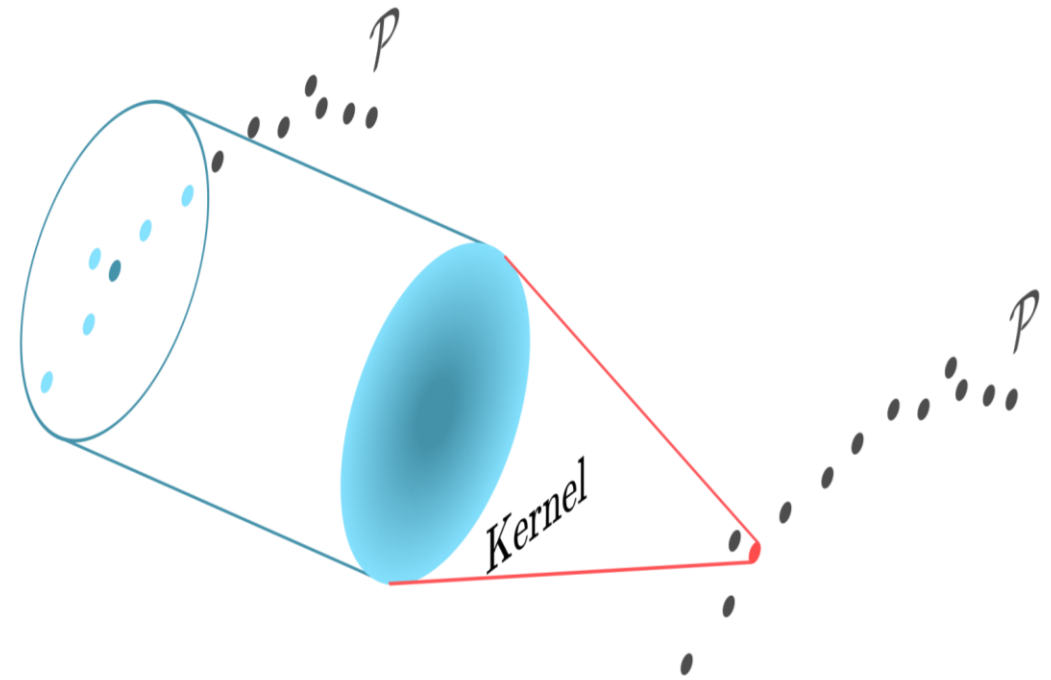
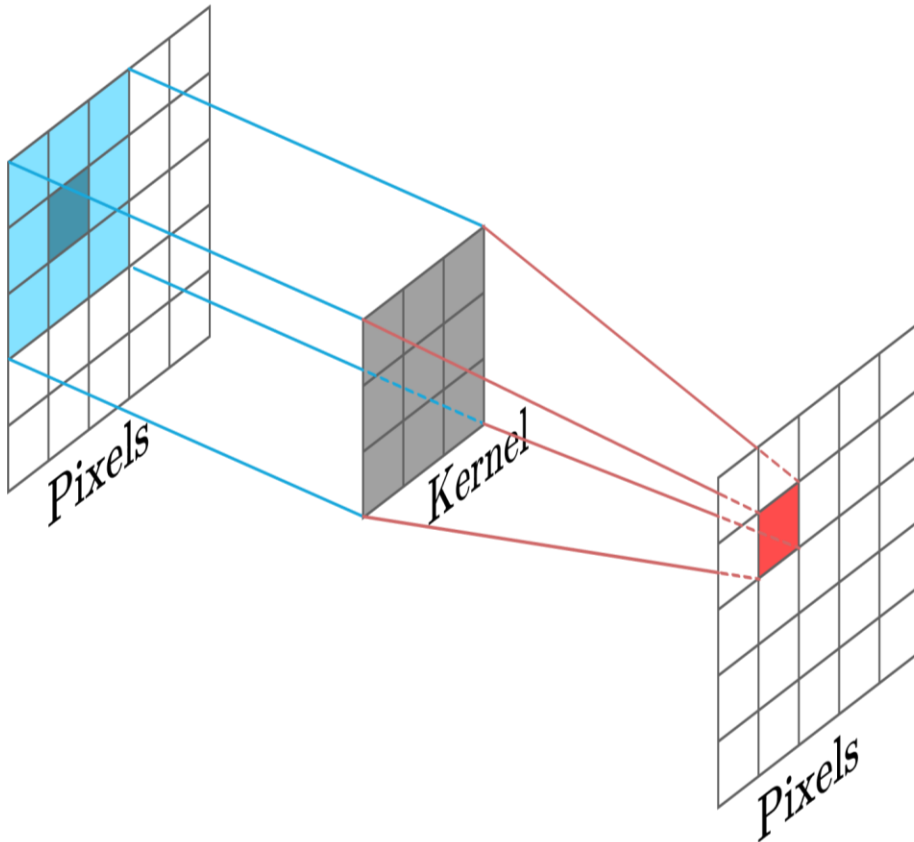


- Shape editing [BDC18]



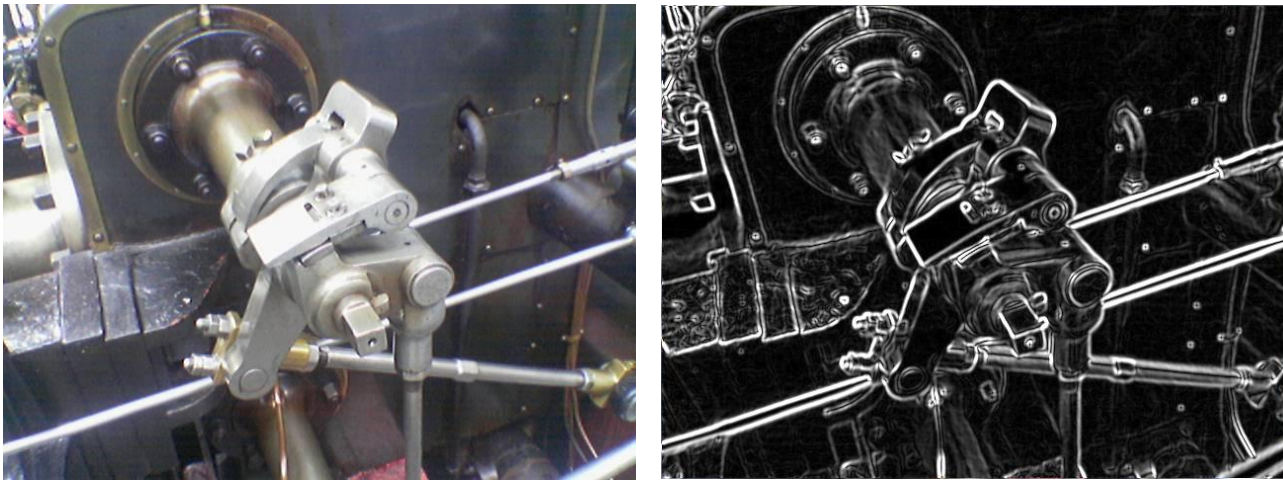
Differential – Intuitions

- Image analogy: convolution and scale



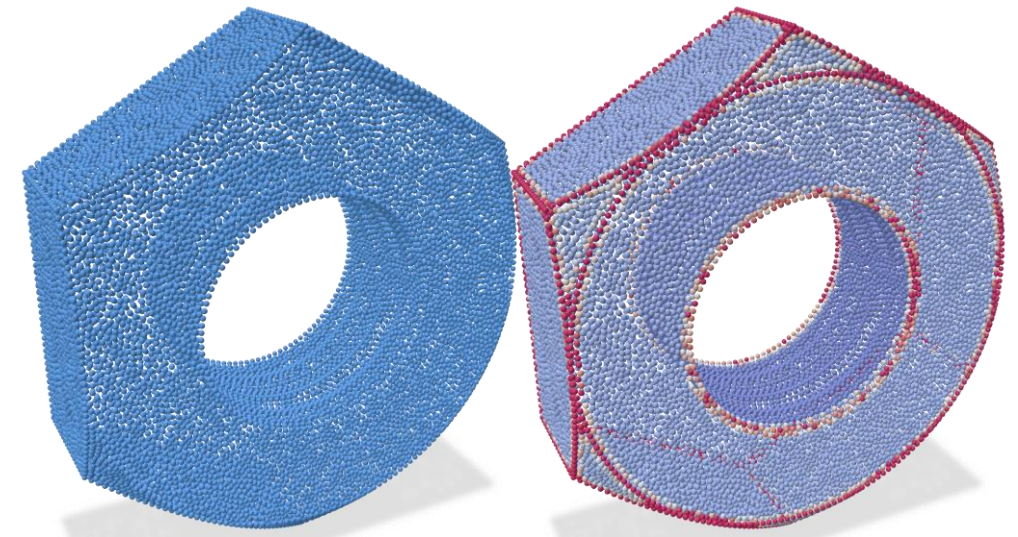
Differential – Intuitions

- Example of contour detection



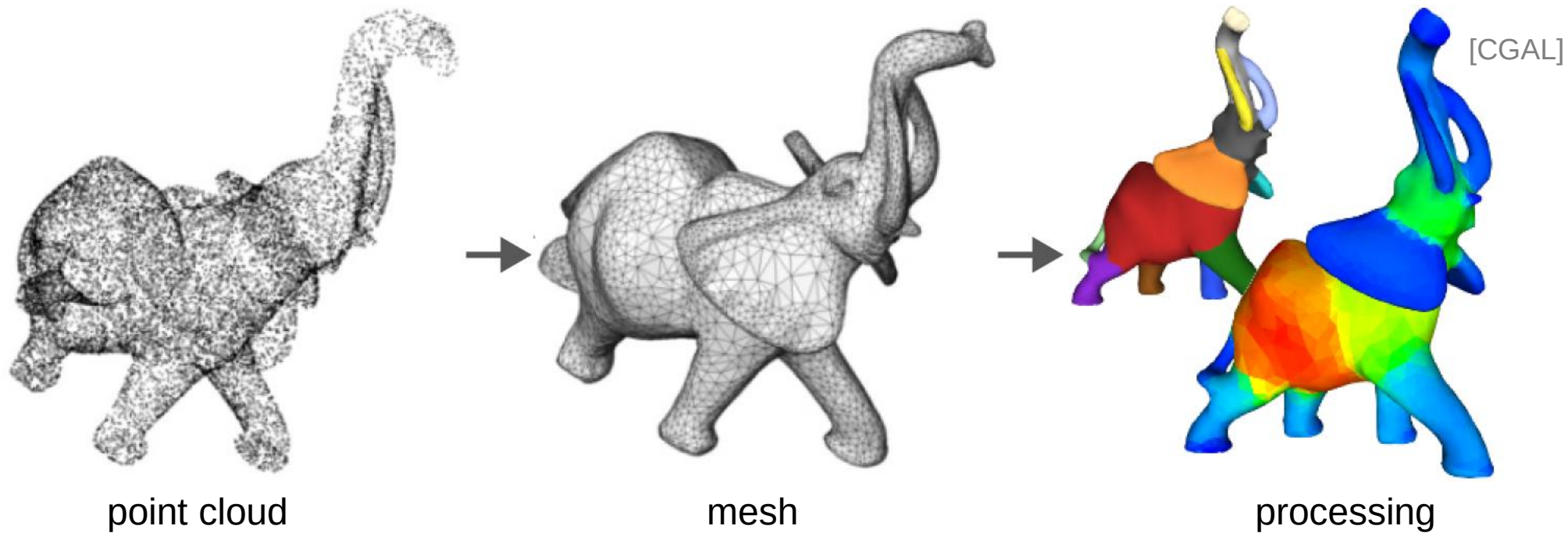
Sobel filter to compute 2D image gradients

$$\mathbf{G}_x = \begin{bmatrix} -1 & 0 & +1 \\ -2 & 0 & +2 \\ -1 & 0 & +1 \end{bmatrix} * \mathbf{A} \quad \mathbf{G}_y = \begin{bmatrix} -1 & -2 & -1 \\ 0 & 0 & 0 \\ +1 & +2 & +1 \end{bmatrix} * \mathbf{A} \quad \mathbf{G} = \sqrt{\mathbf{G}_x^2 + \mathbf{G}_y^2}$$

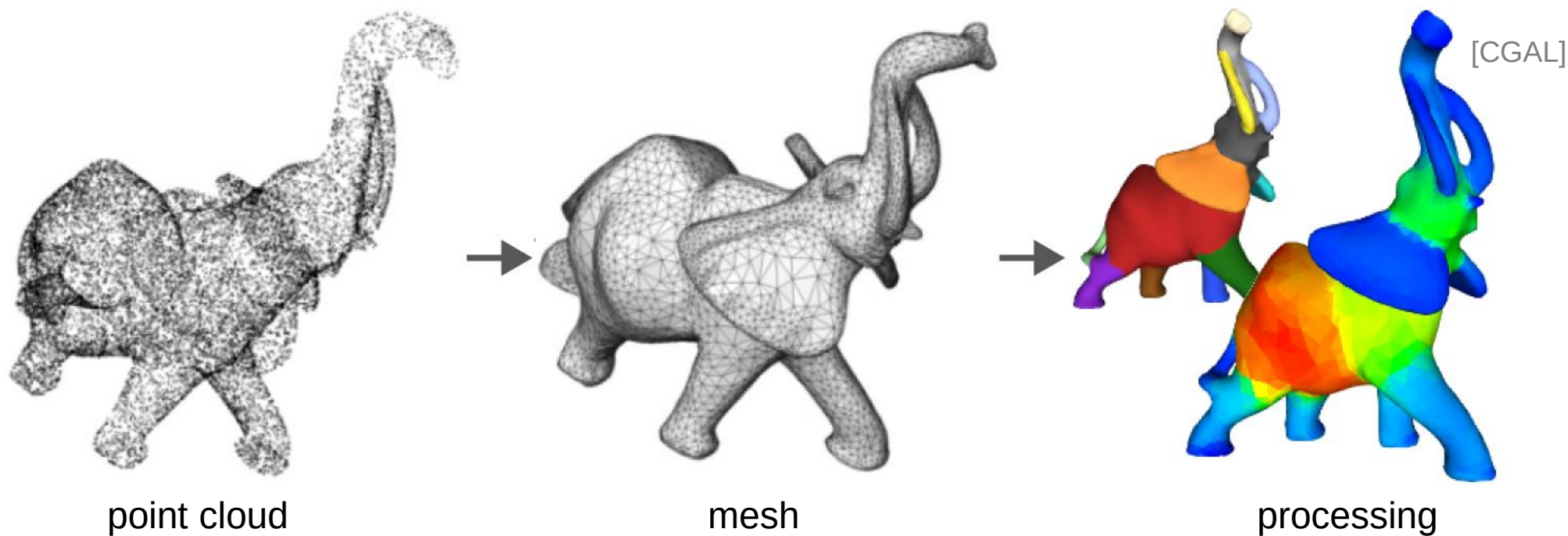


Estimate mean curvature on 3D points

Differential – Mesh-based process



Differential – Mesh-based process



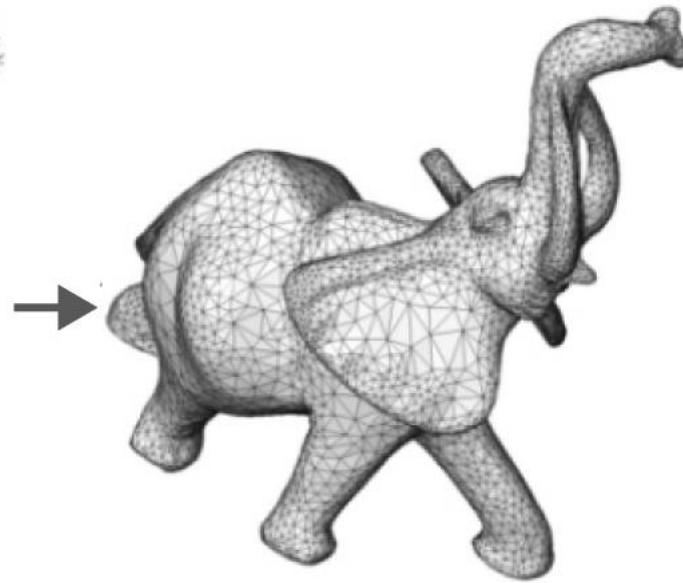
A survey of **surface reconstruction** from point clouds [BTS*17]

Mesh Statistics for Robust **Curvature Estimation** [VVP*16]

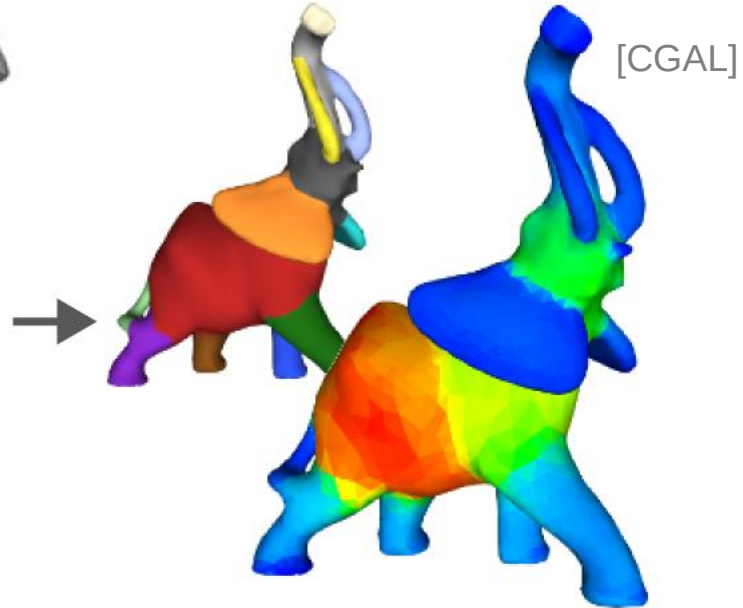
Differential – Mesh-based process



point cloud



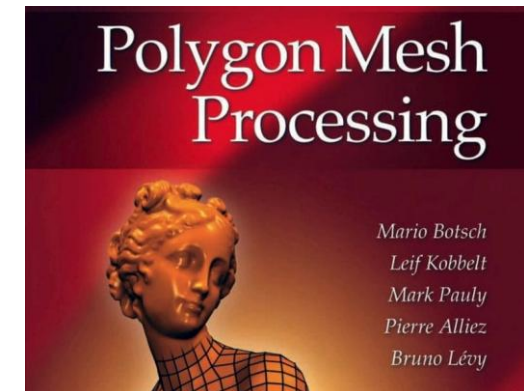
mesh



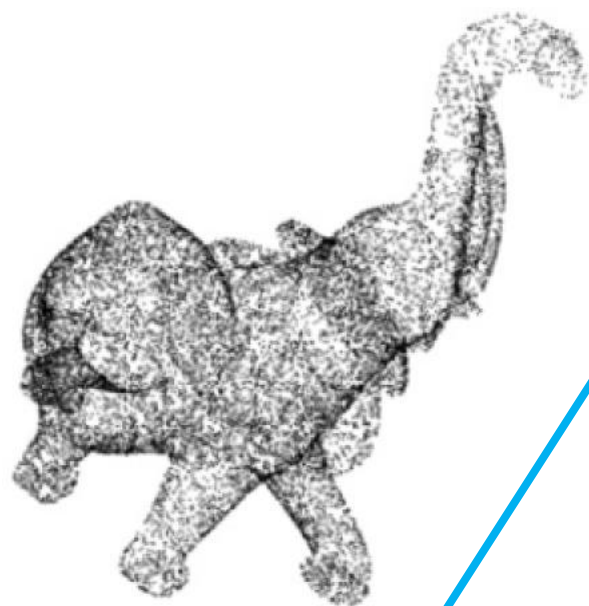
processing

A survey of **surface reconstruction** from point clouds [BTS*17]

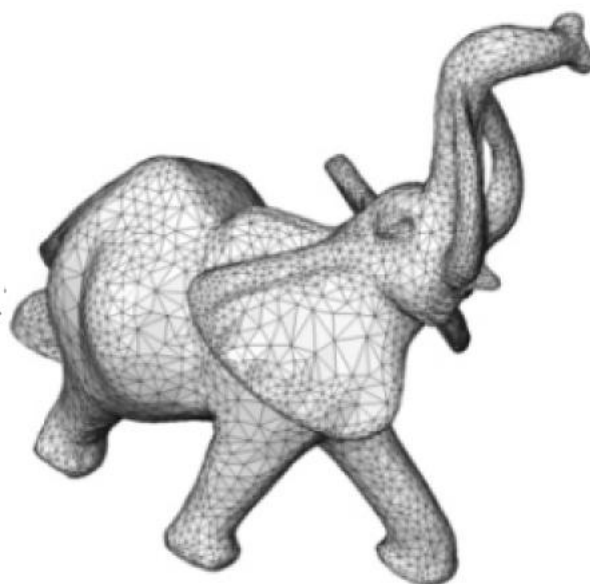
Mesh Statistics for Robust **Curvature Estimation** [VVP*16]



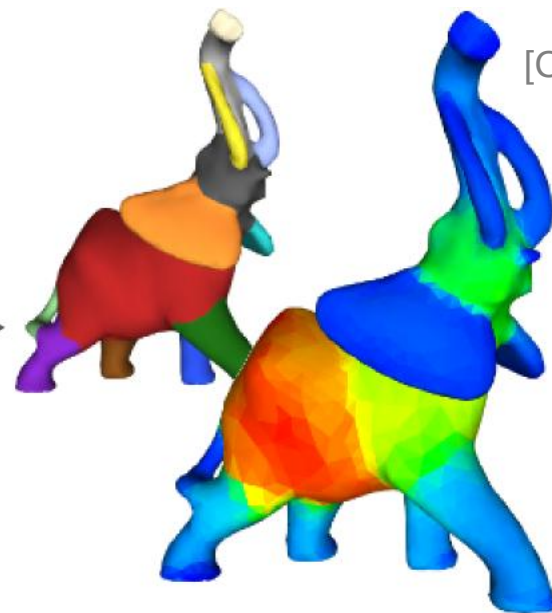
Differential – Mesh-based process



point cloud



mesh



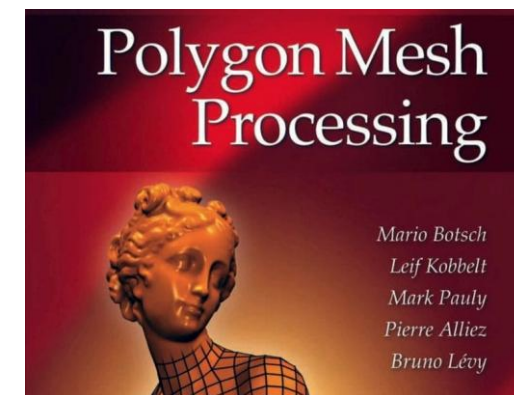
[CGAL]

processing

working on raw data

A survey of **surface reconstruction** from point clouds [BTS*17]

Mesh Statistics for Robust **Curvature Estimation** [VVP*16]



Context – Scope

- Differential properties estimation in a raw 3D point cloud
- Special focus on mathematical models
 - with theoretical guarantees
 - their use in Machine Learning models
- Unified benchmark

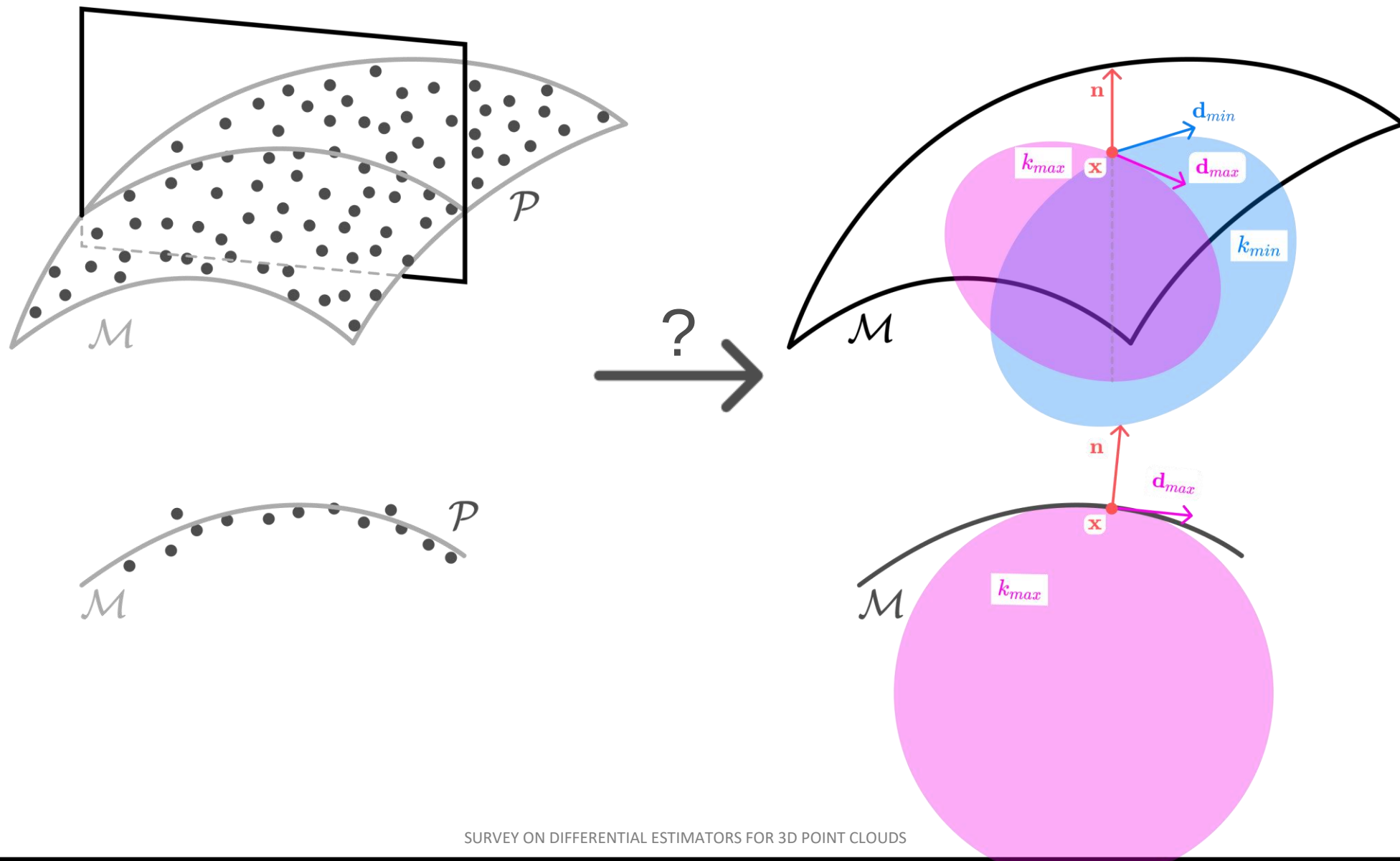
PLAN

1. Context
2. Fundamentals
3. Methods
4. Benchmark
5. Future works
6. Conclusion

Fundamentals

02

Fundamentals



Fundamentals



Normal



Mean curvature



Tangent frame

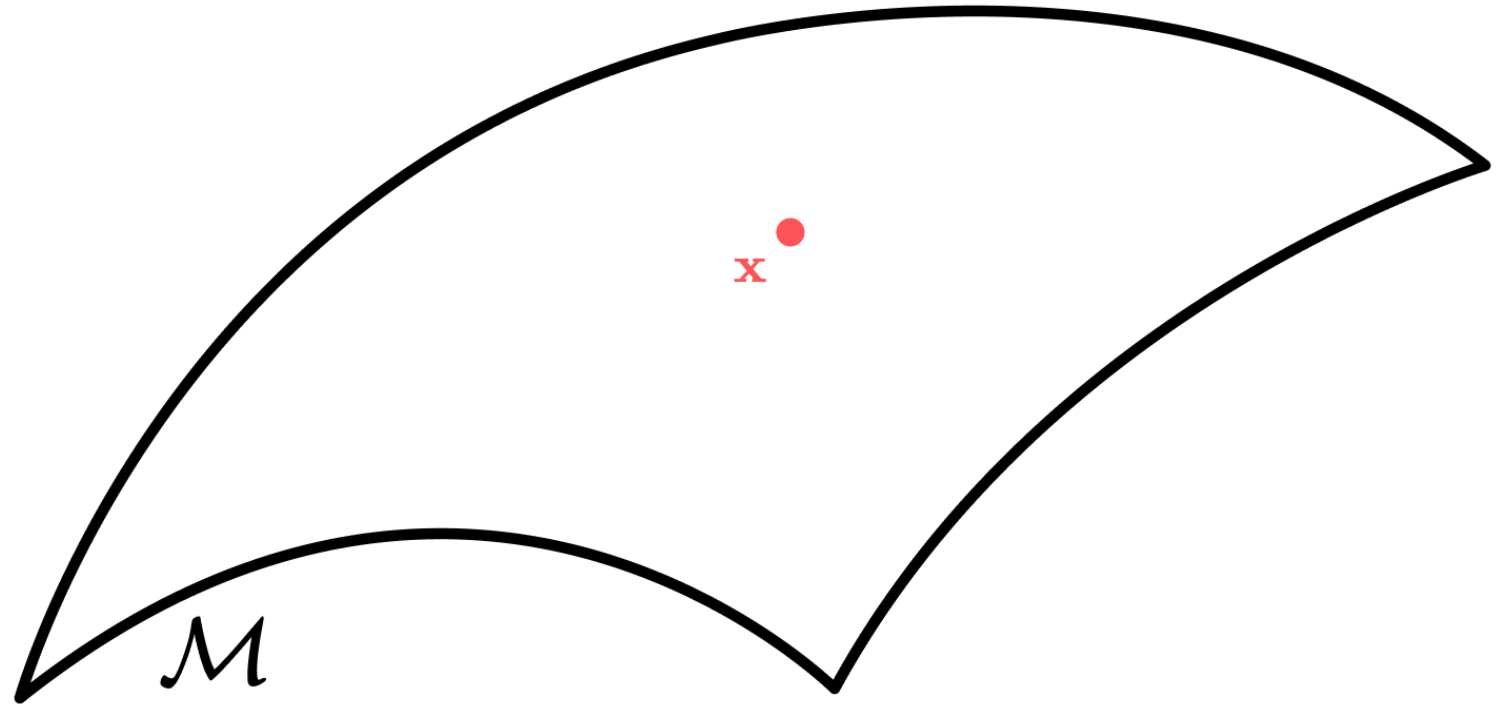


Weingarten map
(principal curvature values and directions)

Normal vector



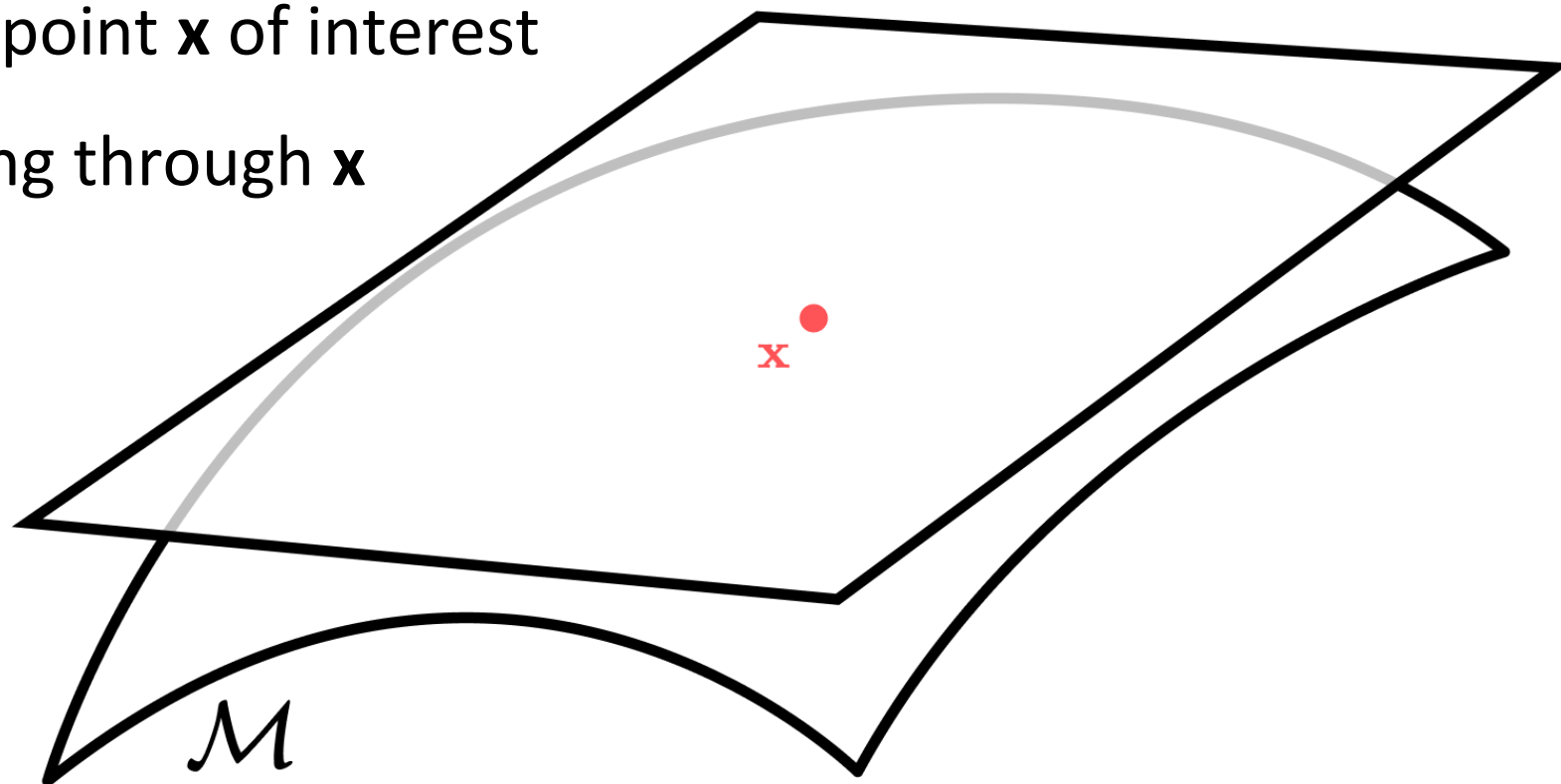
- Given a surface M and a point $\mathbf{x}$ of interest



Normal vector



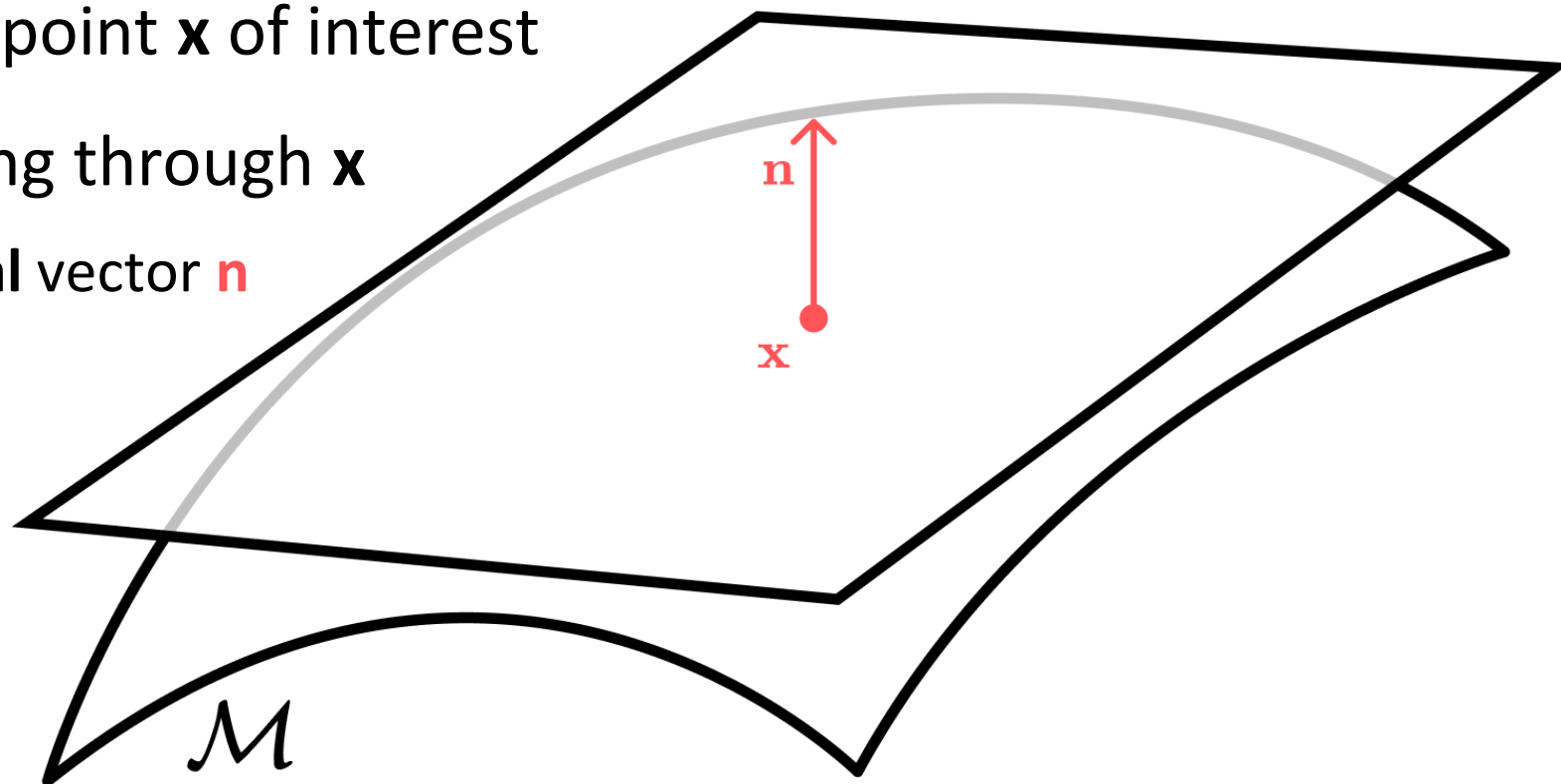
- Given a surface M and a point $\mathbf{x}$ of interest
- The **tangent plane** passing through $\mathbf{x}$



Normal vector

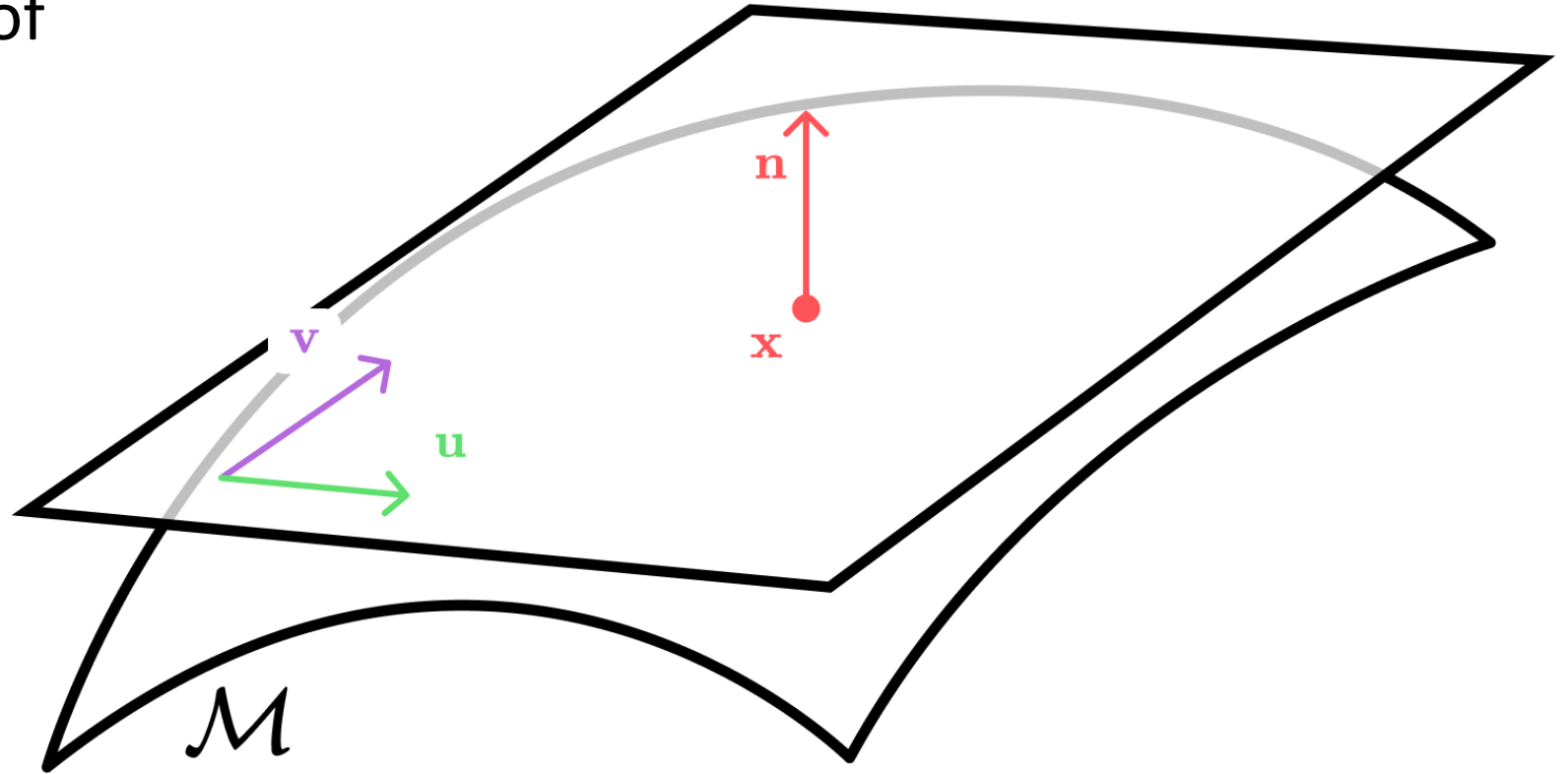


- Given a surface M and a point $\mathbf{x}$ of interest
- The **tangent plane** passing through $\mathbf{x}$
 - Orthogonal to the **normal** vector $\mathbf{n}$



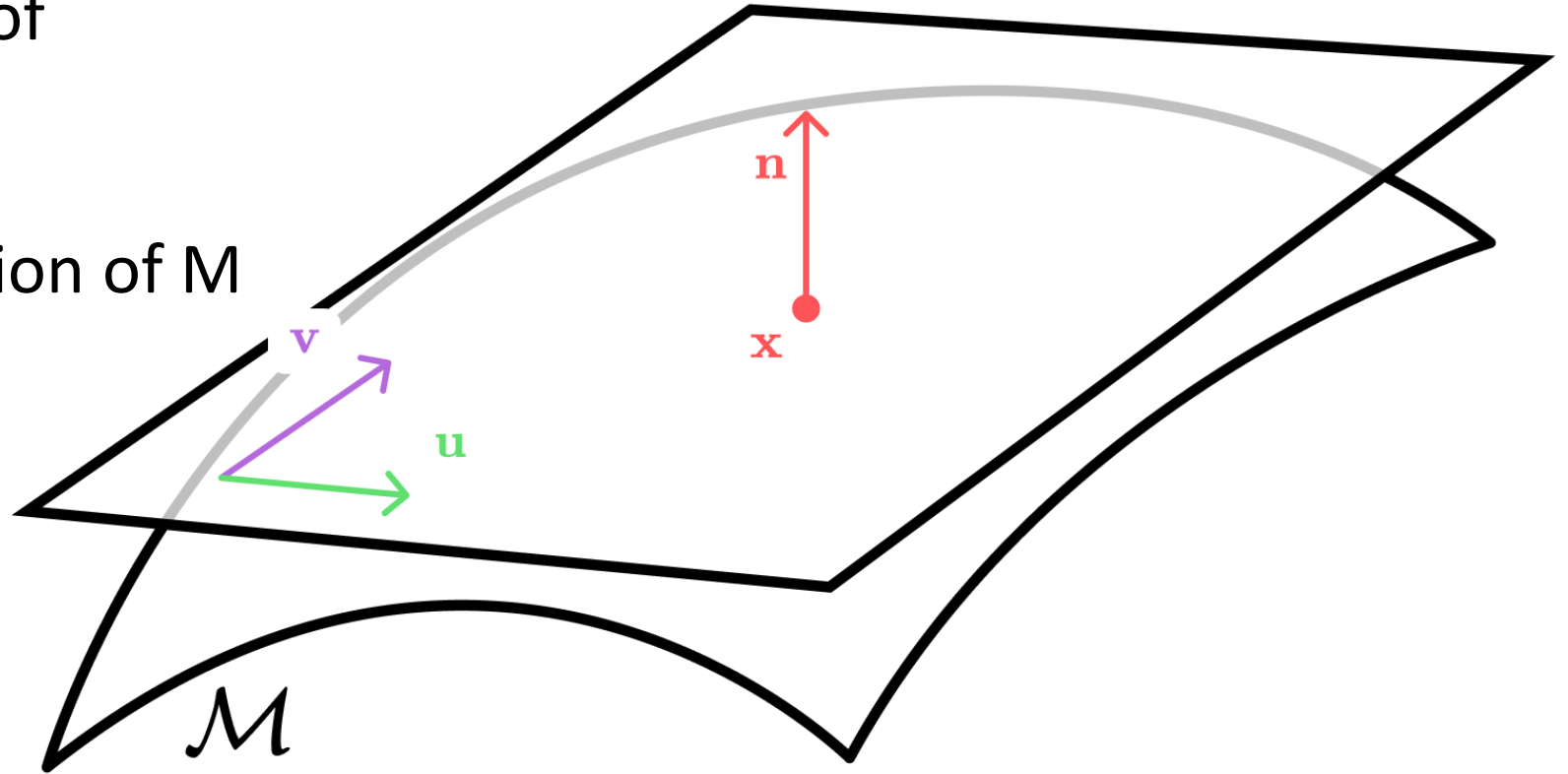
Tangent frame

- $\mathbf{u}$, $\mathbf{v}$ an orthogonal base of the **tangent frame**



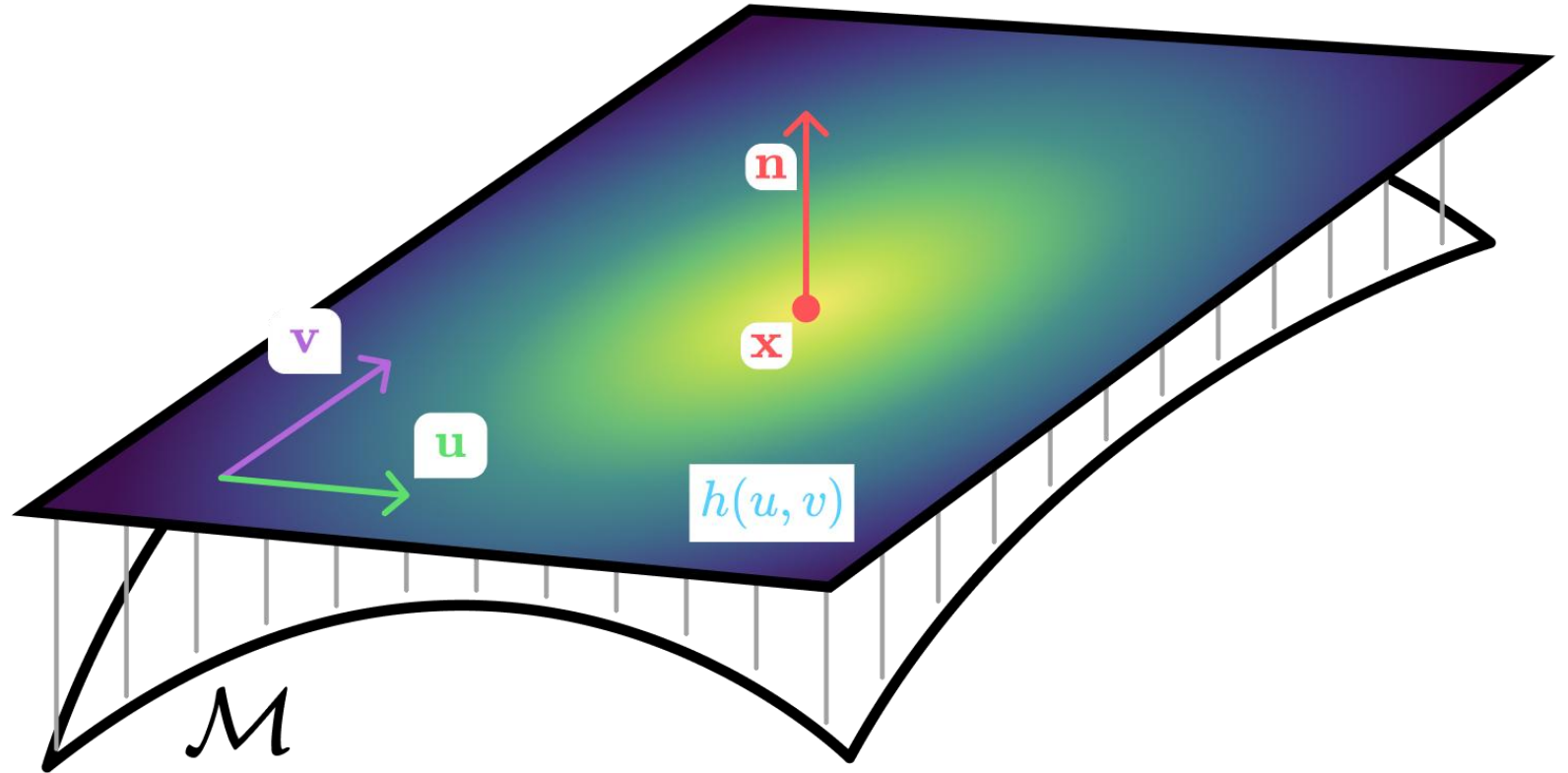
Tangent frame

- $\mathbf{u}$, $\mathbf{v}$ an orthogonal base of the **tangent frame**
- Let's find a parametrisation of M



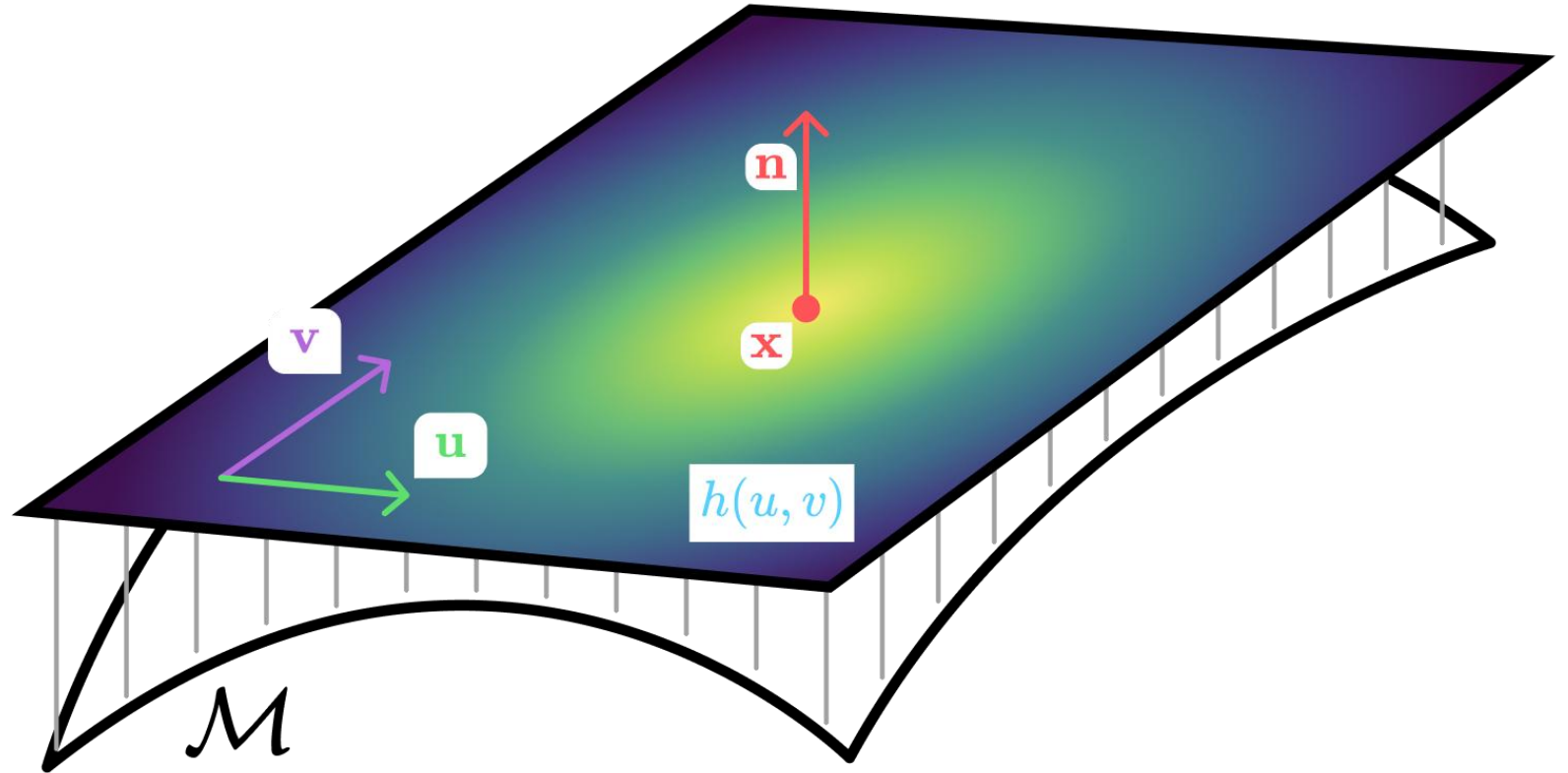
Tangent frame - parametrisation

- $h(u, v)$ the height field



Tangent frame - parametrisation

- $h(u, v)$ the height field
- Note $f(u, v, h(u, v))$
the **monge patch** of M

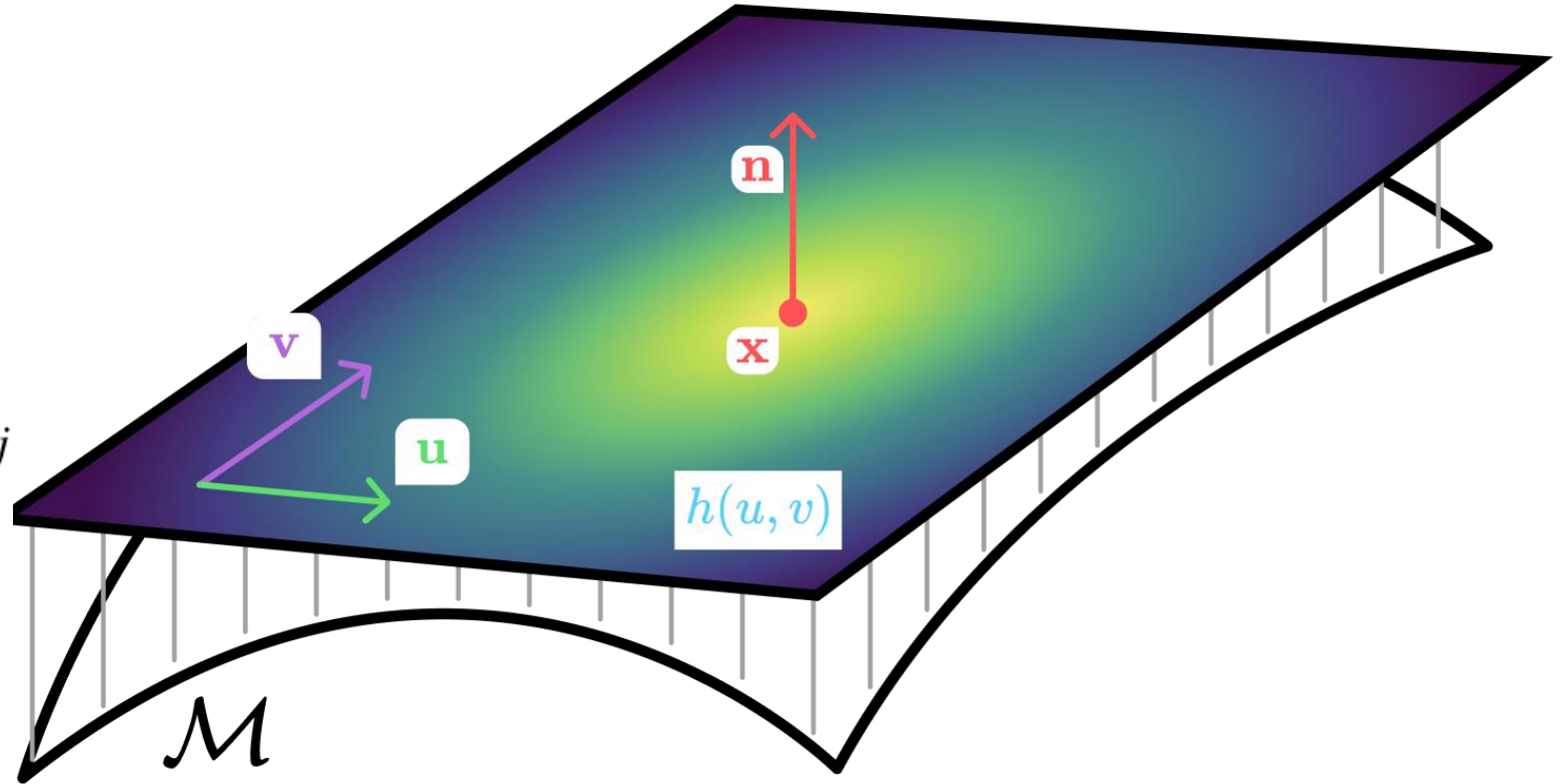


Tangent frame - parametrisation

- $h(u, v)$ the height field
- Note $f(u, v, h(u, v))$

the **monge patch** of M

$$h(u, v) = \sum_{k=0}^{\infty} \sum_{j=0}^k \frac{h_{u^k - j v^j}(0, 0)}{(k - j)! j!} u^{k-j} v^j$$



Tangent frame – Fundamental forms



- The first fundamental form
 - **Intrinsic** properties of the surface

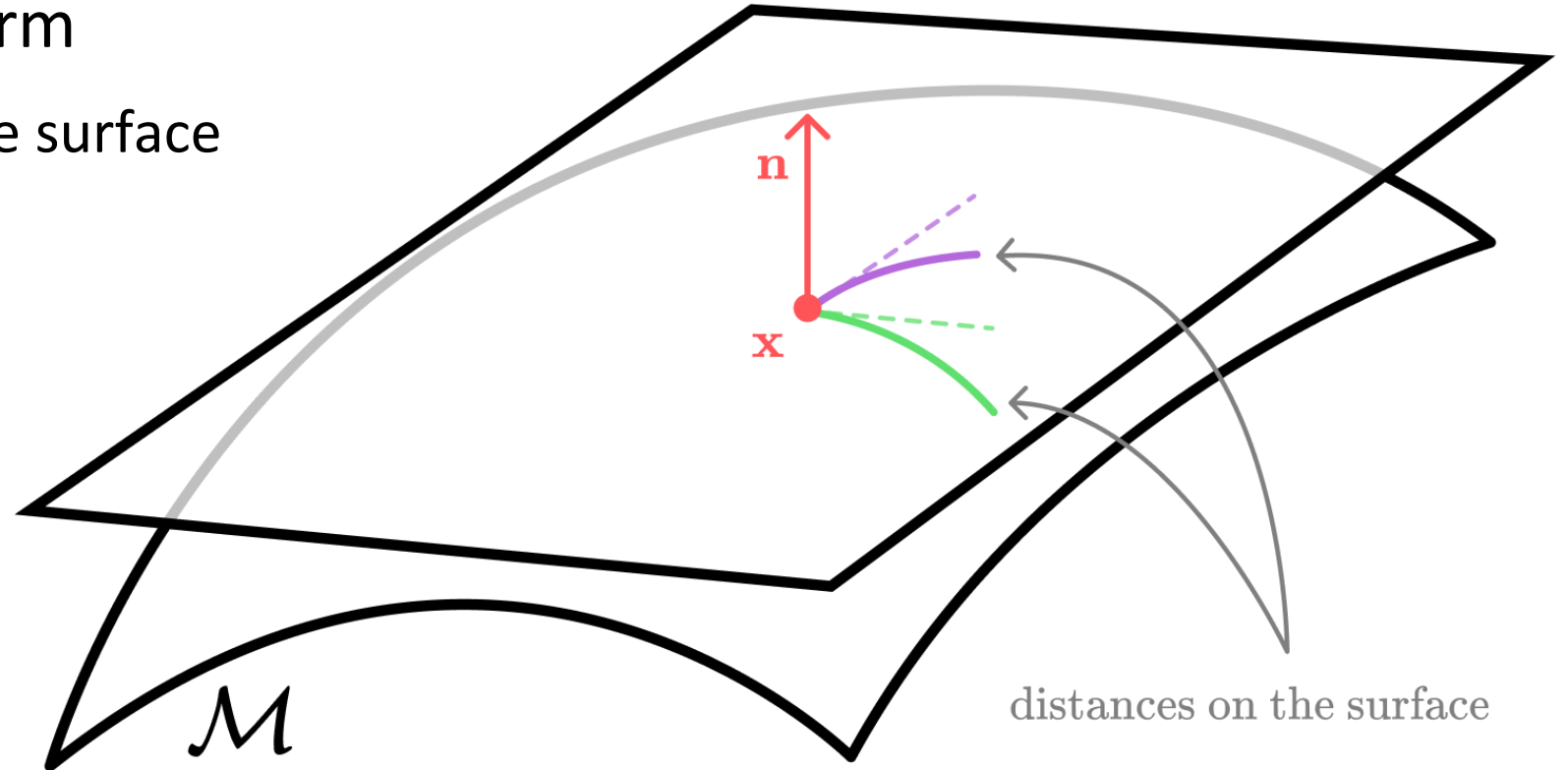
$$F_I = \begin{pmatrix} E & F \\ F & G \end{pmatrix}$$

$$E = \|f_u\|^2$$

$$F = f_u \cdot f_v$$

$$G = \|f_v\|^2$$

$$(\text{with } f_u = \frac{\partial f}{\partial u}).$$



distances on the surface

Tangent frame – Fundamental forms



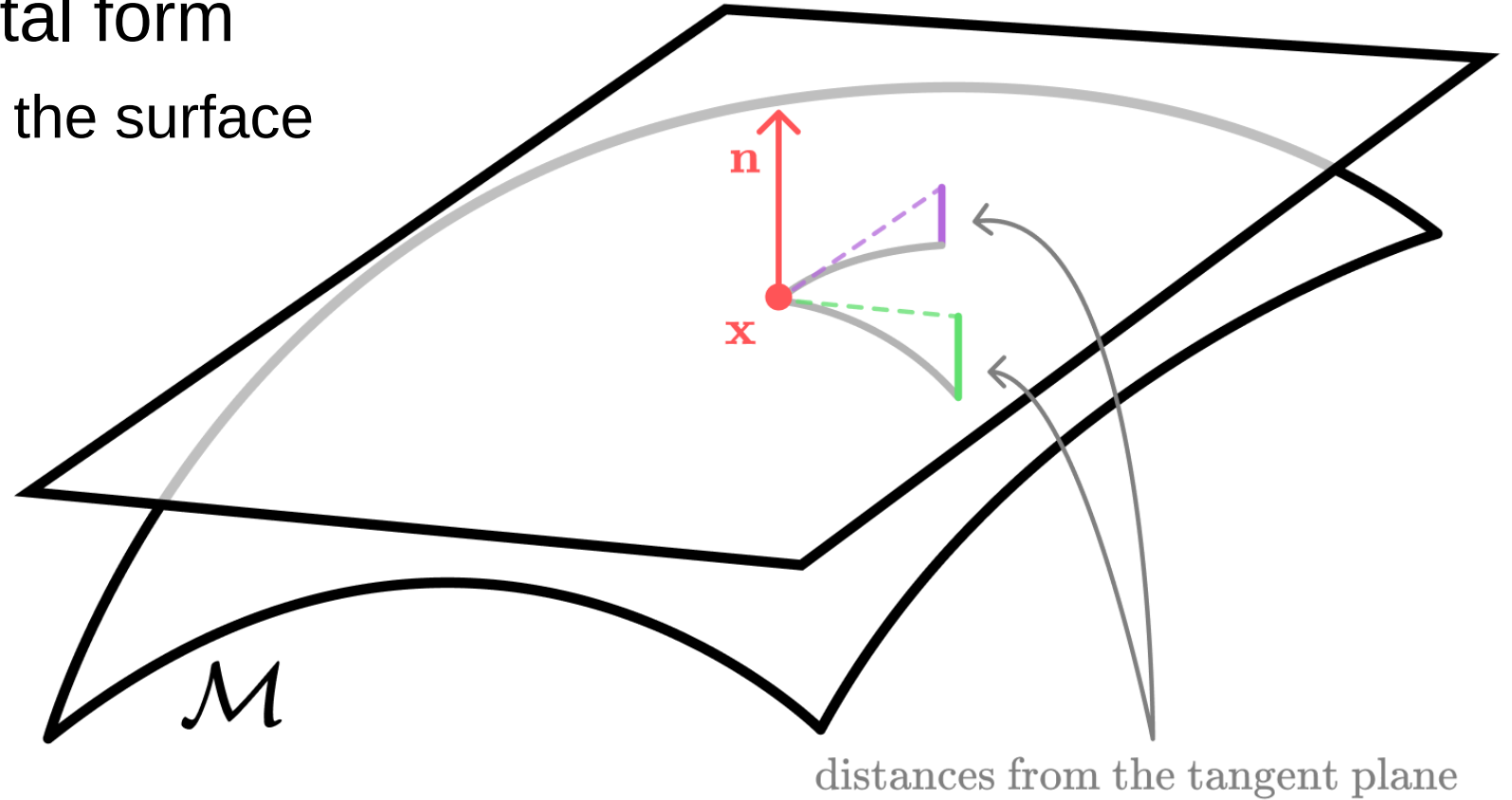
- The second fundamental form
 - **Extrinsic** properties of the surface

$$F_{\text{II}} = \begin{pmatrix} L & M \\ M & N \end{pmatrix}$$

$$L = f_{uu} \cdot \mathbf{n}_f$$

$$M = f_{uv} \cdot \mathbf{n}_f$$

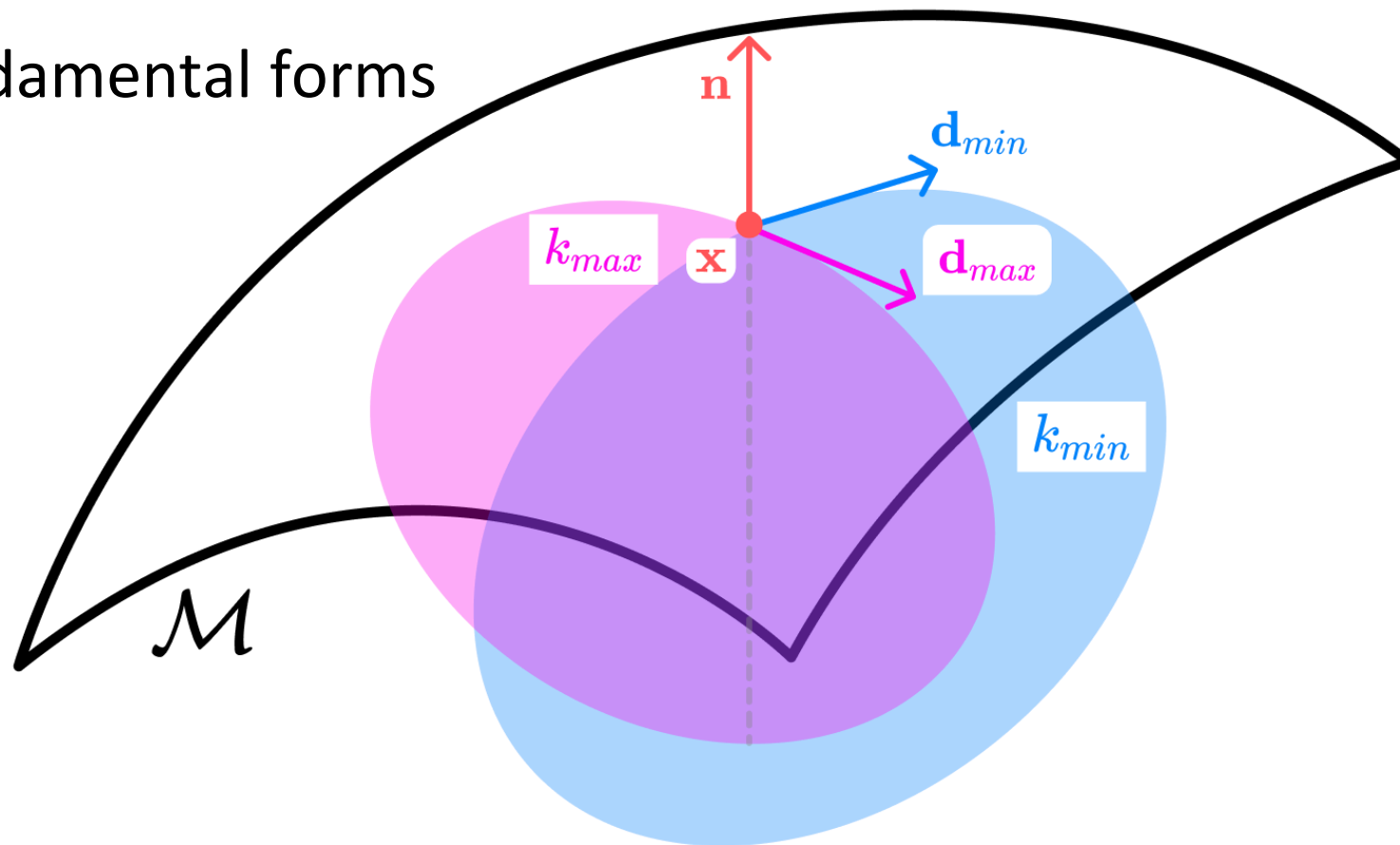
$$N = f_{vv} \cdot \mathbf{n}_f$$



Weingarten Map

- Also called **Shape Operator**
- **Multiplication** of the fundamental forms

$$W = F_I^{-1} F_{II}$$



Weingarten Map

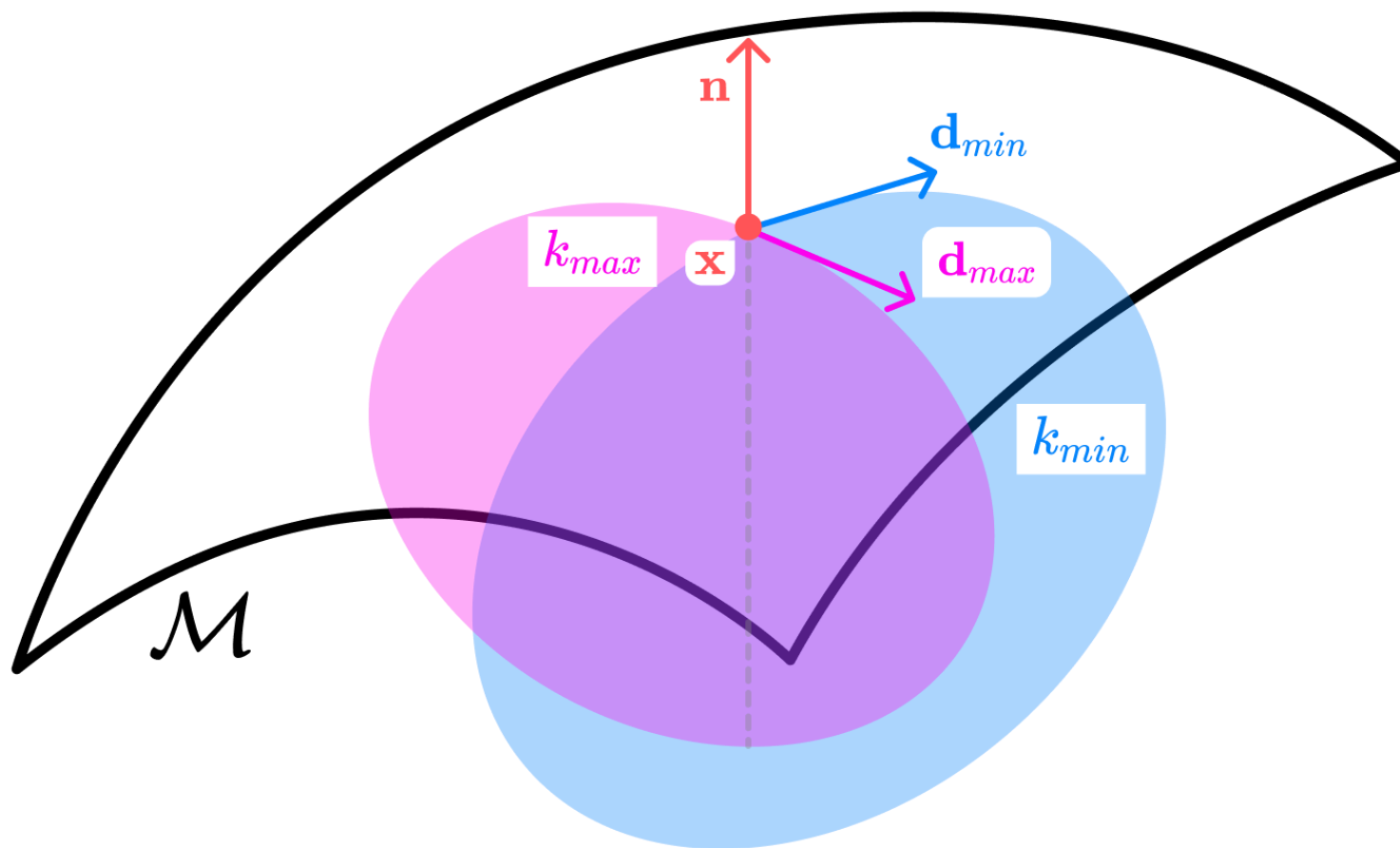


- **Decomposition** of the Weingarten map

$$W = U D U^T$$

$$U = (\mathbf{d}_{min} \quad \mathbf{d}_{max})$$

$$D = \begin{pmatrix} k_{min} & 0 \\ 0 & k_{max} \end{pmatrix}$$

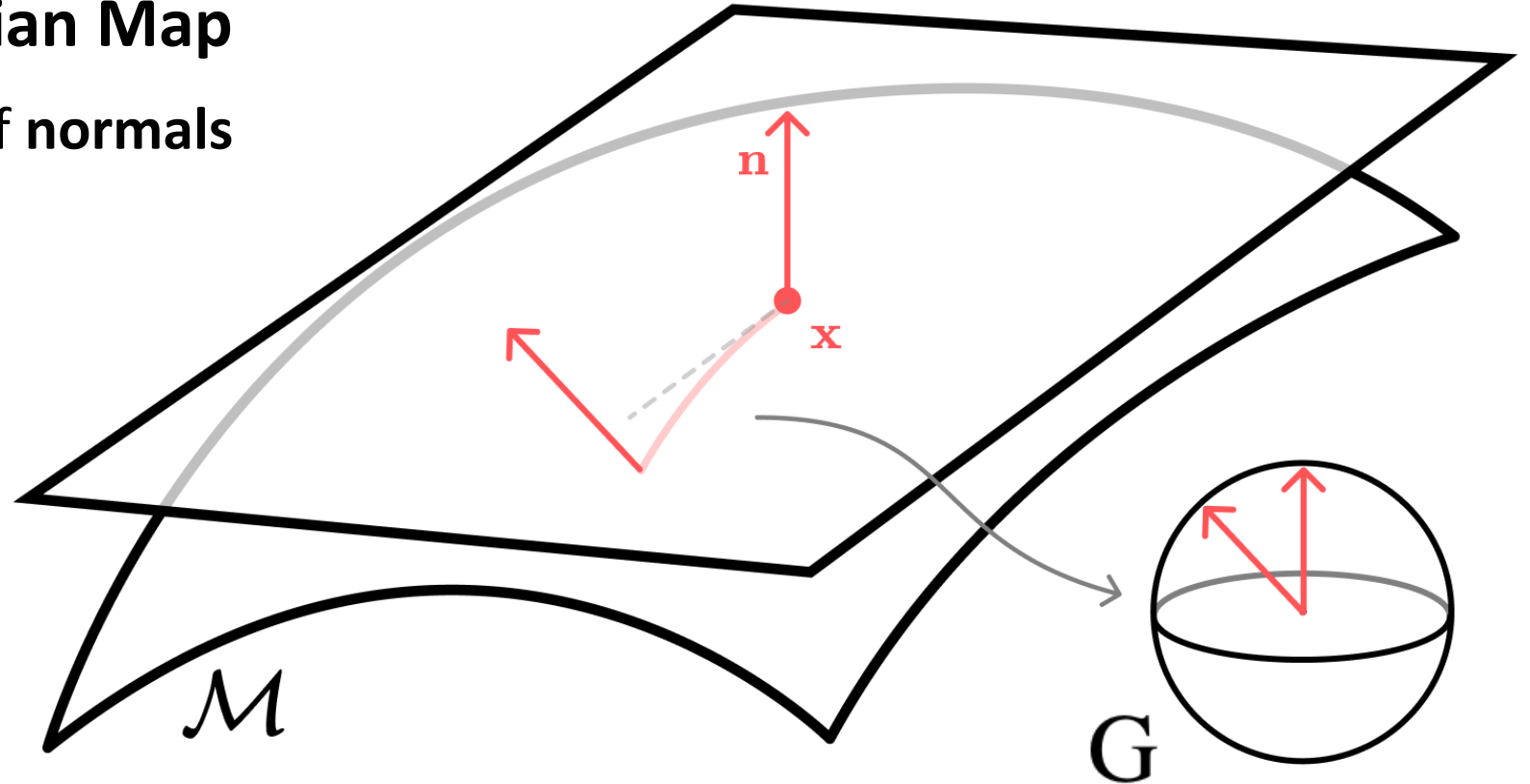


Weingarten Map – Other intuition



- Derivatives of the **Gaussian Map**
 - Observe the **variations** of normals

$$W = -D_x G$$

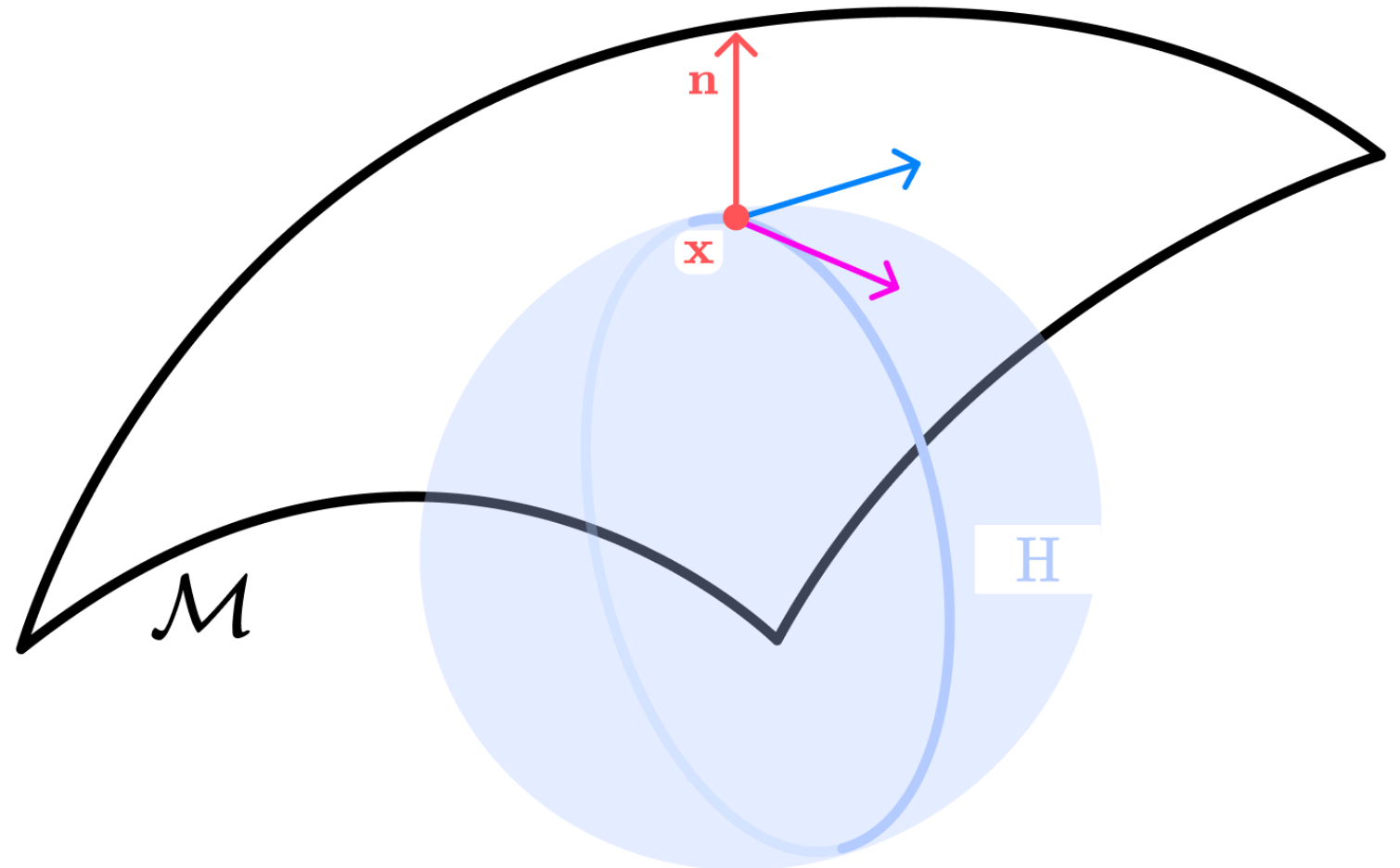


Mean Curvature



- **Averaging** the principal curvatures

$$\begin{aligned} H &= \frac{1}{2} \text{trace}(\mathbf{W}) \\ &= \frac{1}{2} \text{trace}(\mathbf{D}) \end{aligned}$$

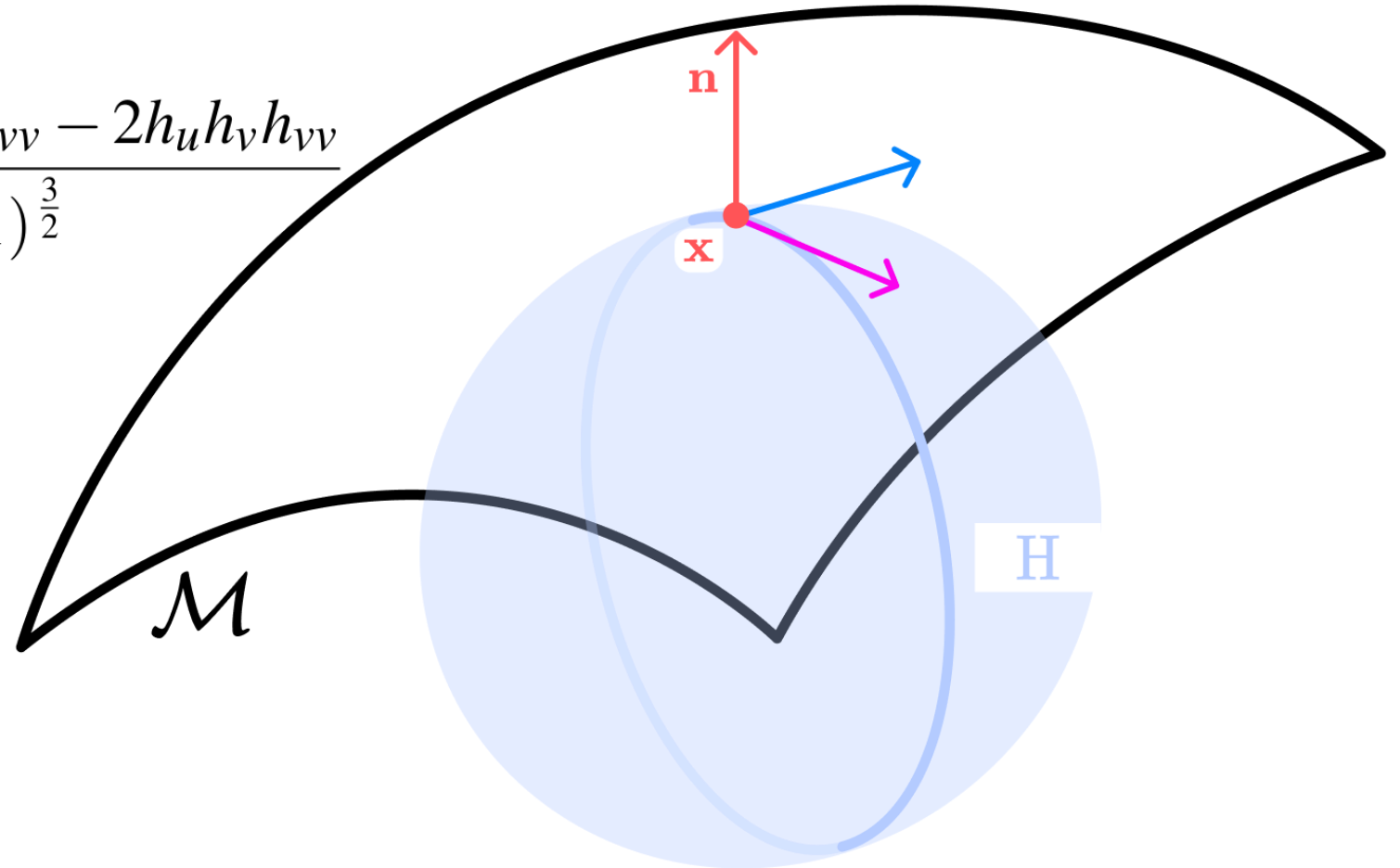


Mean Curvature

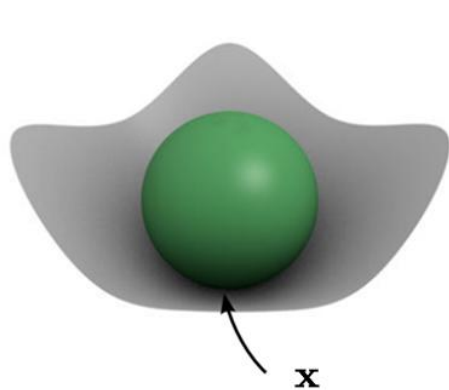


- Directly with the partial derivatives of h

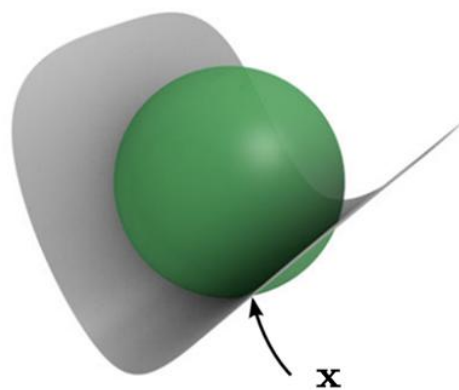
$$H = \frac{(1 + h_u^2)h_{uu} + (1 + h_v^2)h_{vv} - 2h_u h_v h_{uv}}{2(h_u^2 + h_v^2 + 1)^{\frac{3}{2}}}$$



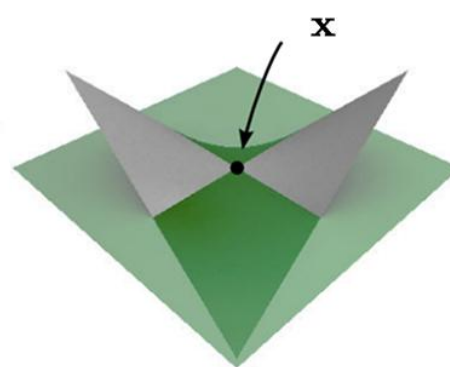
Fundamentals – Examples



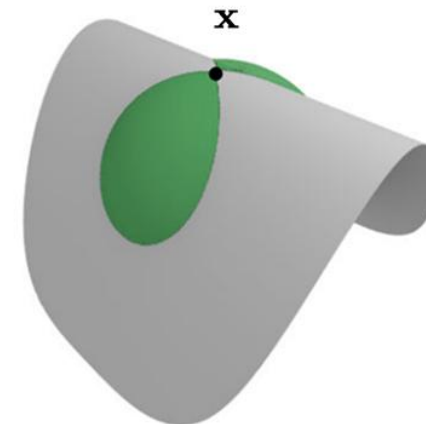
$$\begin{aligned} k_{min} &= \alpha \\ k_{max} &= \alpha \\ H &= \alpha \end{aligned}$$



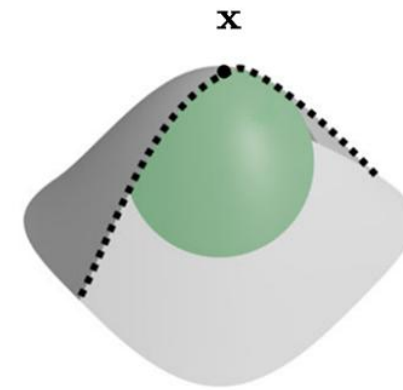
$$\begin{aligned} k_{min} &= \alpha \\ k_{max} &= 0 \\ H &= \frac{1}{2}\alpha \end{aligned}$$



$$\begin{aligned} k_{min} &= \alpha \\ k_{max} &= -\alpha \\ H &= 0 \end{aligned}$$

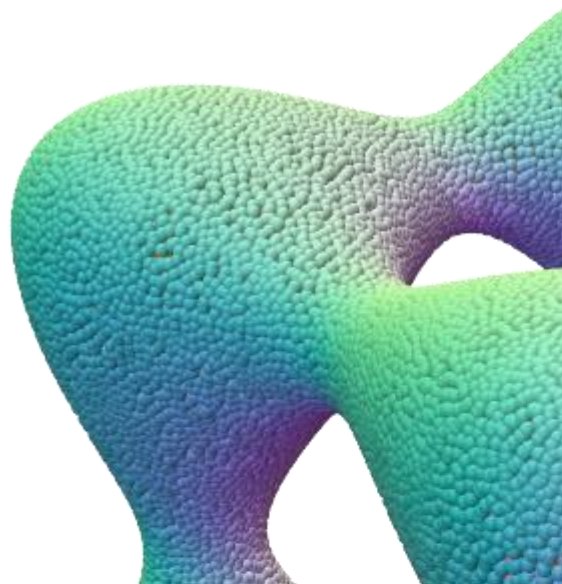


$$\begin{aligned} k_{min} &= 0 \\ k_{max} &= -\alpha \\ H &= -\frac{1}{2}\alpha \end{aligned}$$

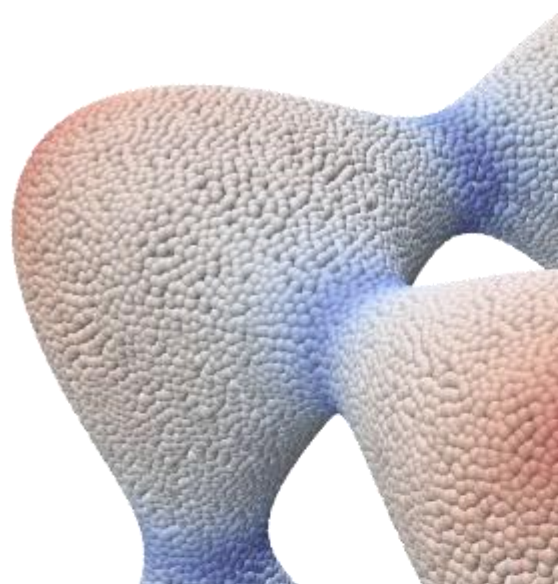


$$\begin{aligned} k_{min} &= -\alpha \\ k_{max} &= -\alpha \\ H &= -\alpha \end{aligned}$$

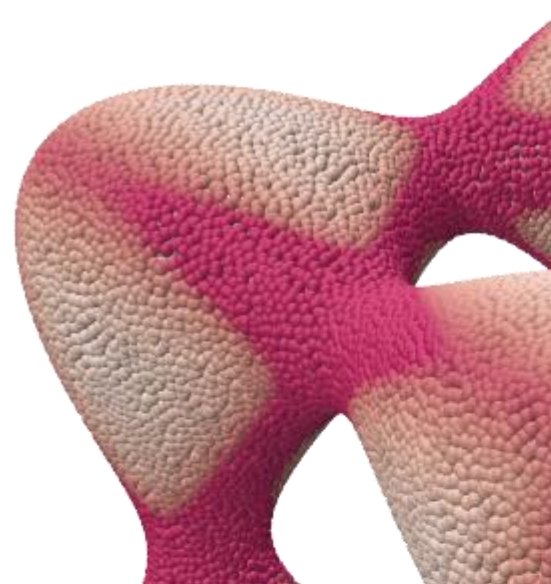
Fundamentals – Examples



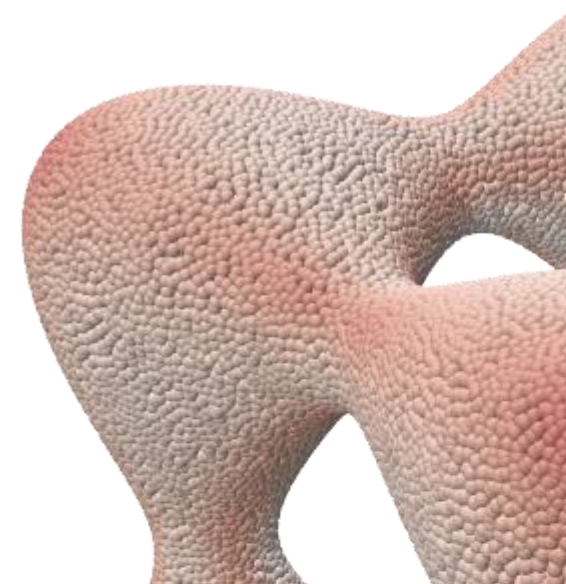
$\mathbf{n}$



k_{min}



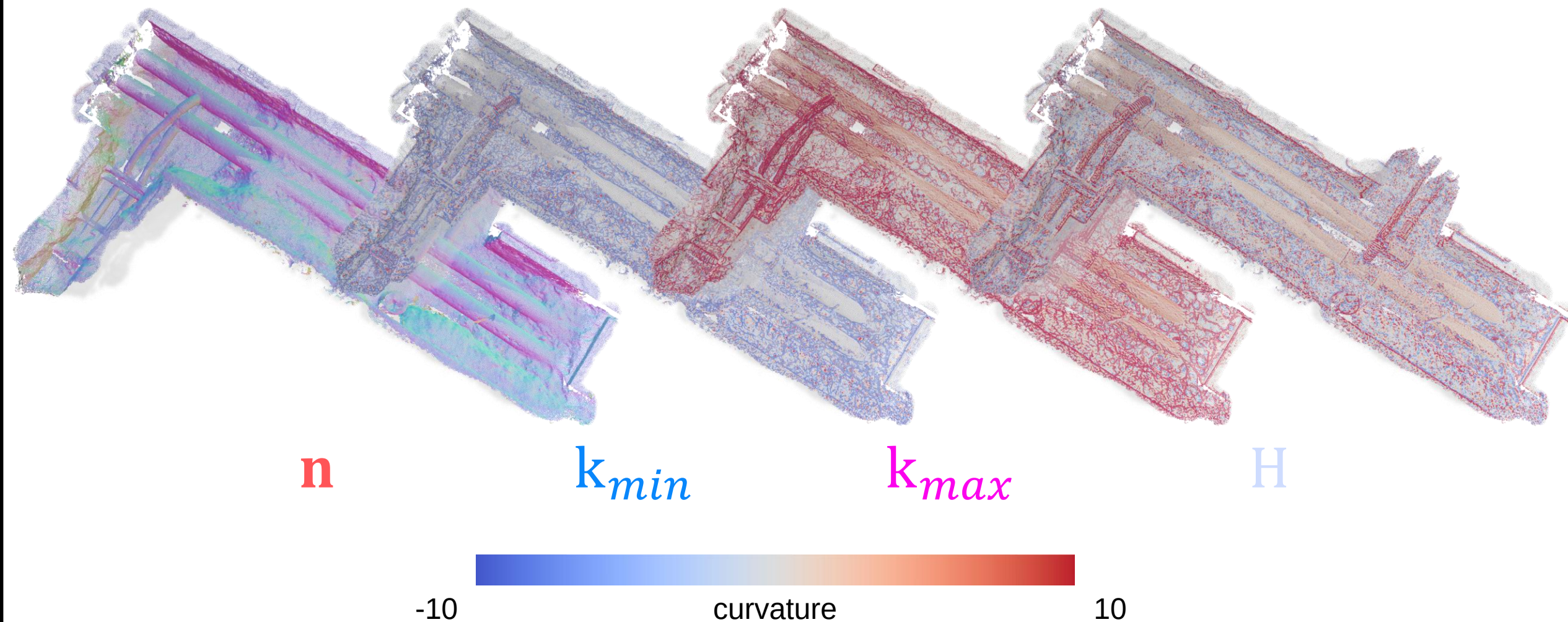
k_{max}



H



Fundamentals – Examples



QUESTION TIME

Survey on differential **estimators** for 3d point clouds

Methods

03

Methods - Notations



Normal estimation



Mean curvature estimation



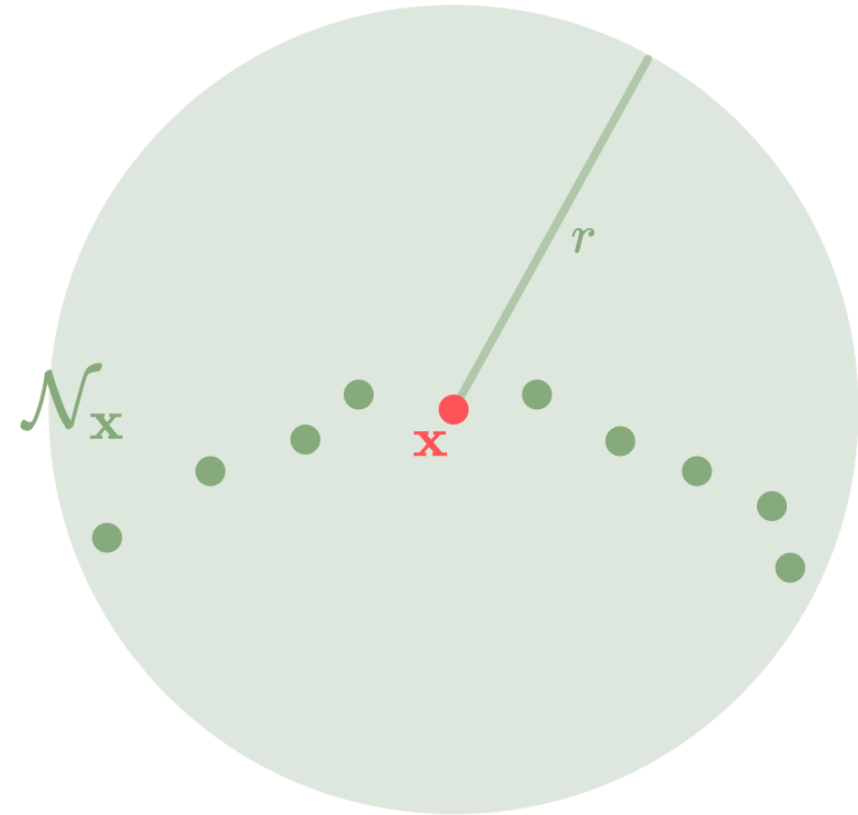
Tangent frame estimation



Weingarten map estimation
(principal curvature values and directions)

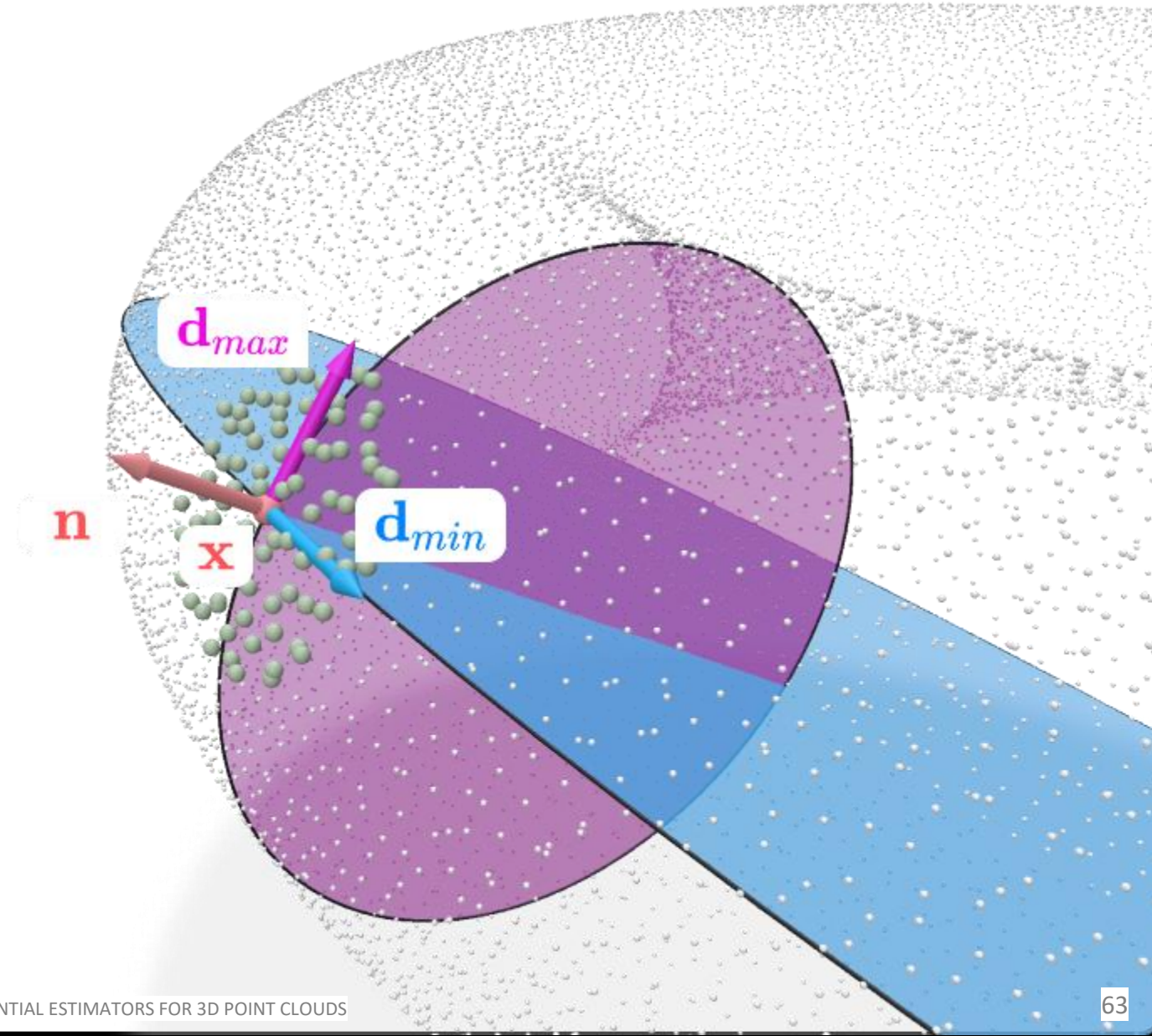
Methods – 2D example settings

- In the following sections



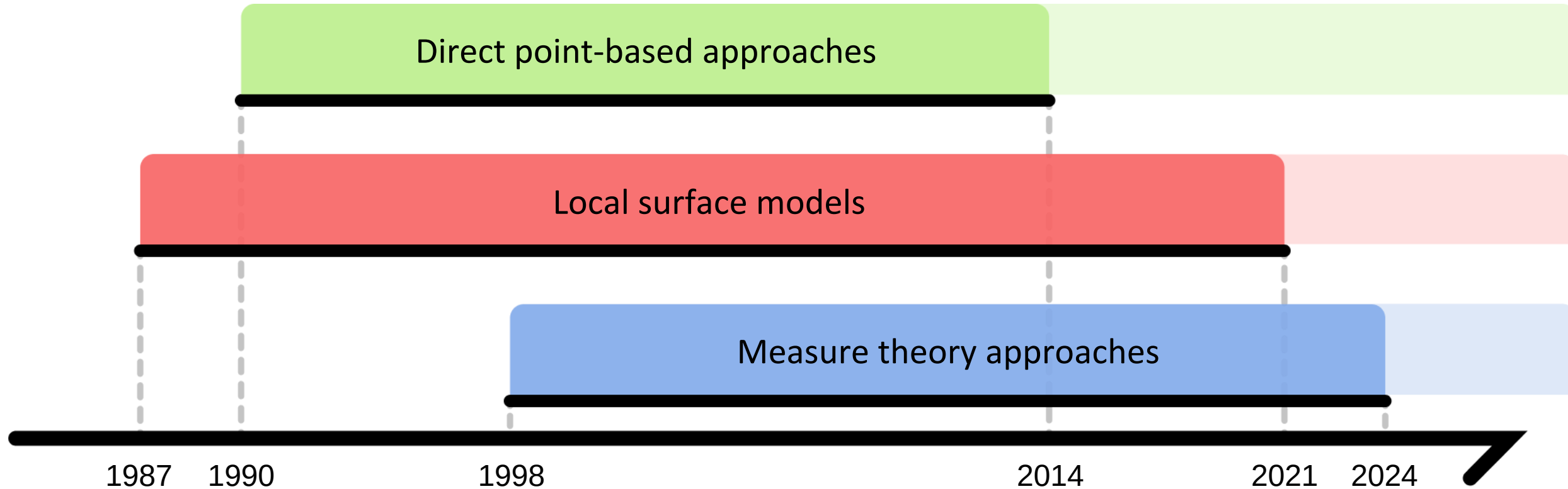
Methods – 3D example settings

- In the following sections



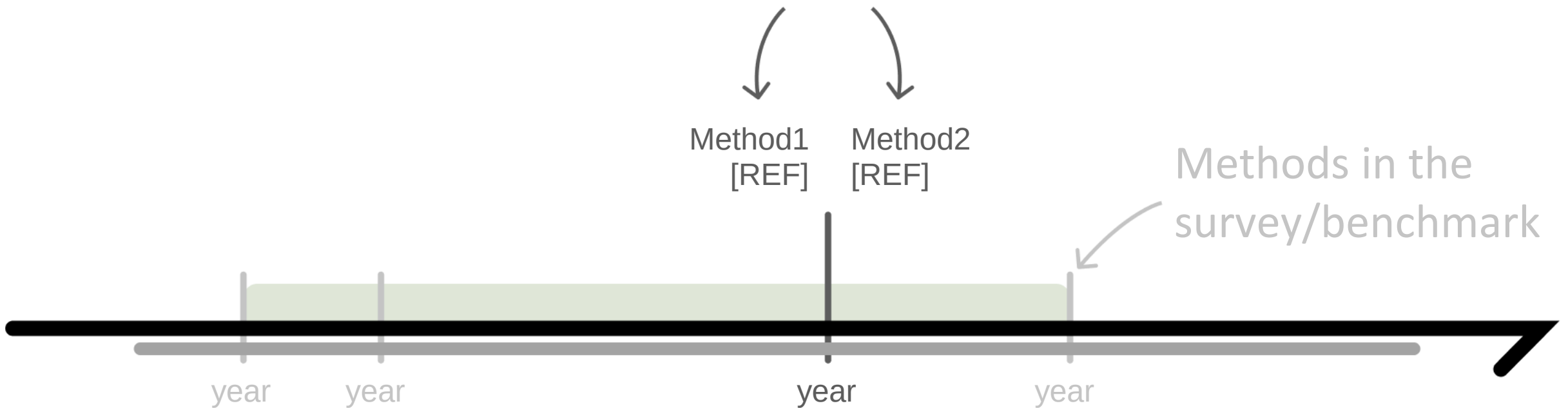
M

ethods - Overview



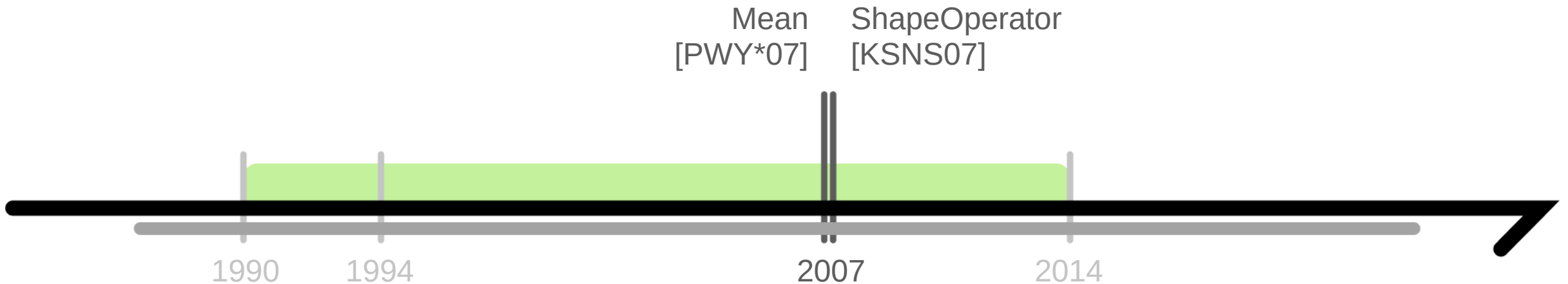
Methods – Display convention

Methods presented today



Methods – Direct Point-Based Approaches

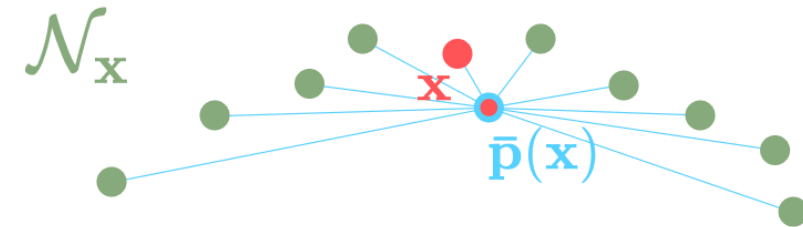
- Statistical approaches
 - Averages
 - Covariance matrices





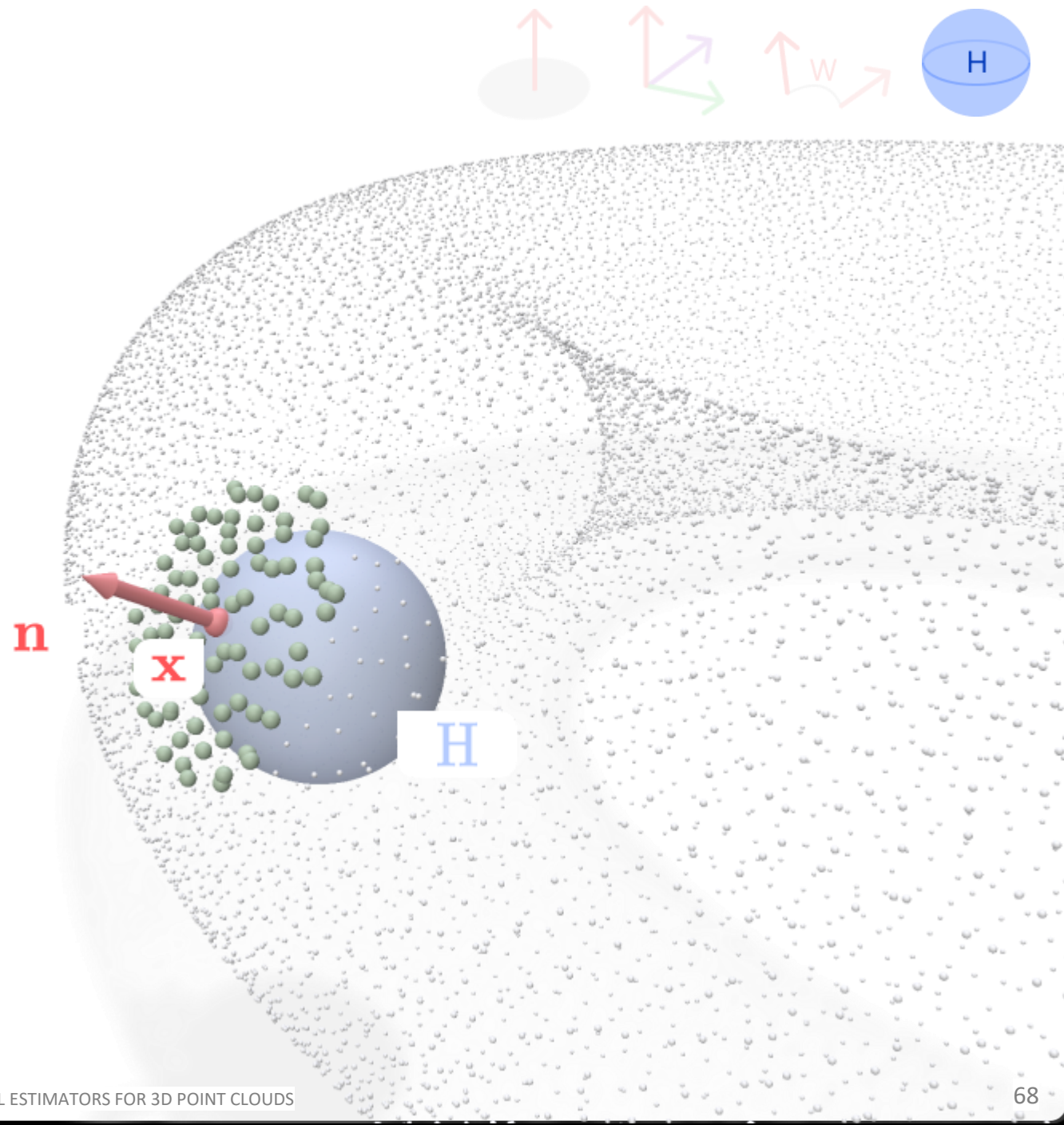
- The barycenter is linked to the mean curvature [PWY*07]

$$\bar{H}(\mathbf{x}) = \frac{4 \|\bar{\mathbf{p}}(\mathbf{x}) - \mathbf{x}\|}{r^2}$$



Methods – Mean

- Estimate
 - Mean curvature



Methods – ShapeOperator [KSNS07]



- Least-squares regression of the Weingarten Map
- Maps 2D directions in the tangent plane to 2D normal variations

$$W = F_I^{-1} F_{II}$$

Methods – ShapeOperator [KSNS07]



- Least-squares regression of the Weingarten Map
- Maps 2D directions in the tangent plane to 2D normal variations

$$W = F_I^{-1} F_{II}$$



$$\bar{W}(\mathbf{x}) = \left(\sum_{i \in \mathcal{N}_x} w_i \mathbf{d}_i \mathbf{d}_i^T \right)^{-1} \left(\sum_{i \in \mathcal{N}_x} w_i \mathbf{v}_i \mathbf{v}_i^T \right)$$

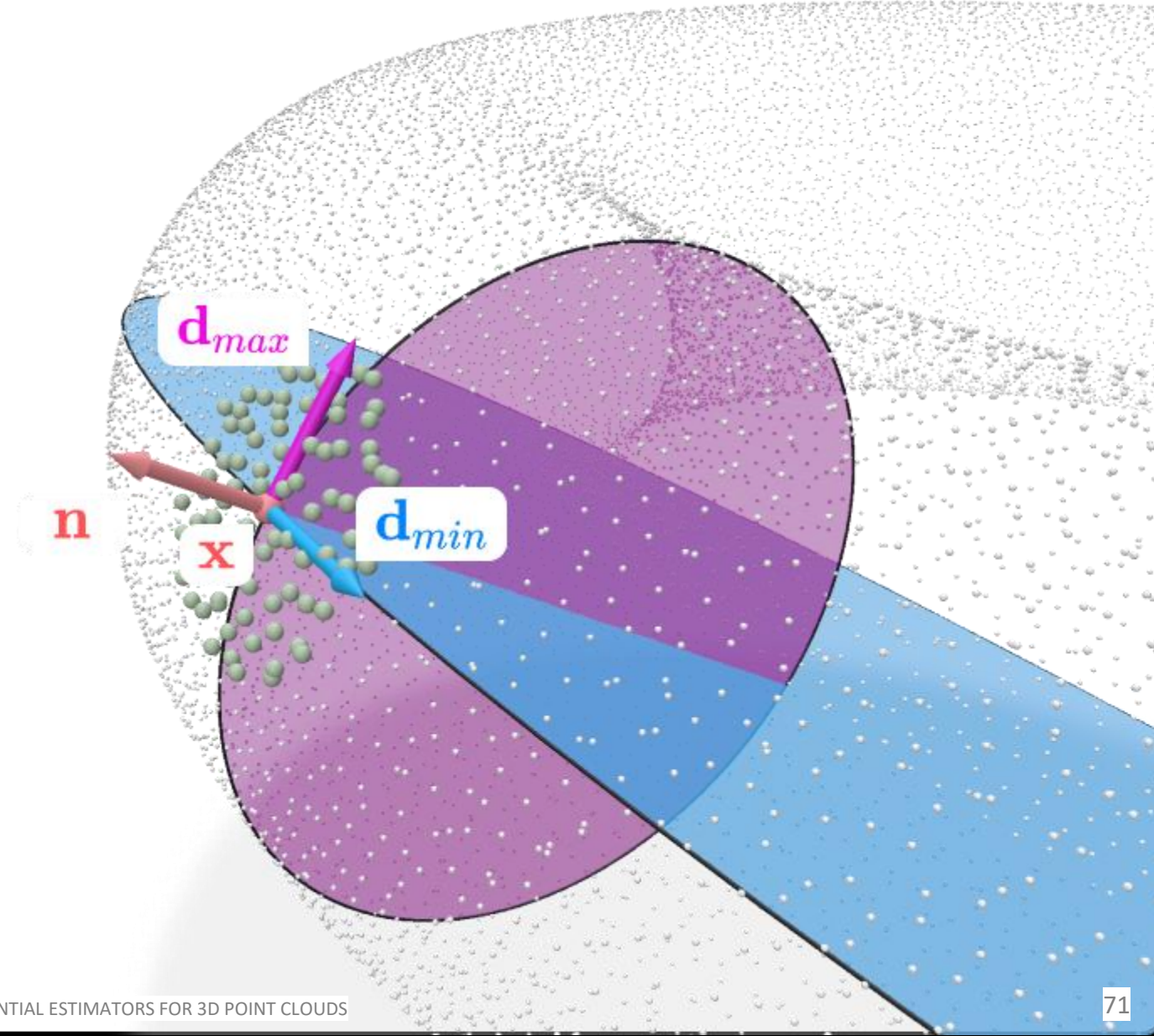
$$\mathbf{d}_i = \bar{P}(\mathbf{x})^T (\mathbf{p}_i - \mathbf{x}) \quad (\text{position variation})$$

$$\mathbf{v}_i = \bar{P}(\mathbf{x})^T (\mathbf{n}_i - \mathbf{n}) \quad (\text{normal variation})$$

Methods – ShapeOperator [KSNS07]

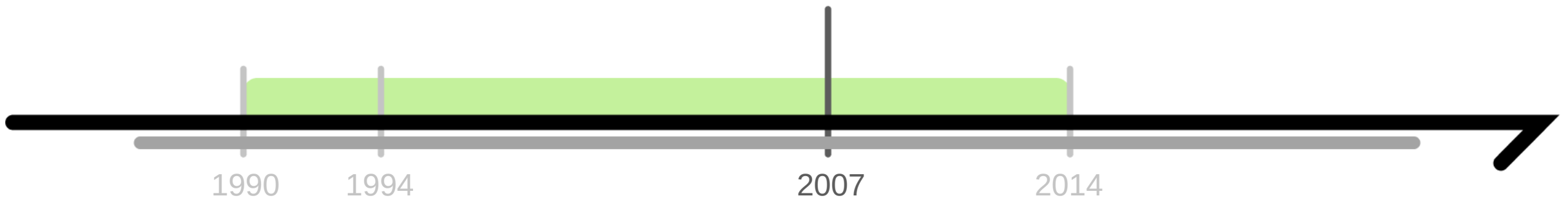


- Estimate
 - Weingarten Map
- Requirements
 - Oriented normal vectors



Methods – Direct Point-Based Approaches

- Capture the geometry directly without reconstructing the explicit surface



M

ethods – Direct Point-Based Approaches

Pros

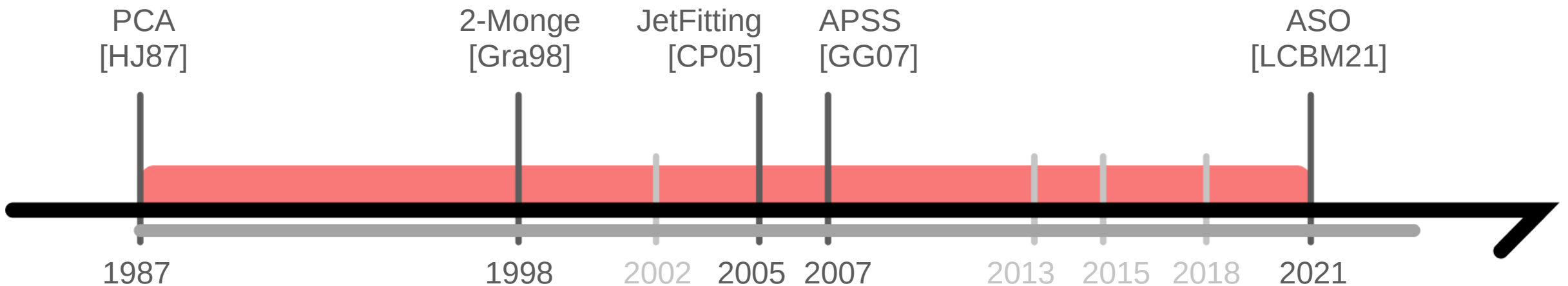
- Simple formulas, efficient
- Good to estimate principal directions

Cons

- Noise sensitive
- Lack of precision for curvature estimation (except for the ShapeOperator)

Methods – Local surface models

- 2.5D height field fitting
- 3D surface fitting

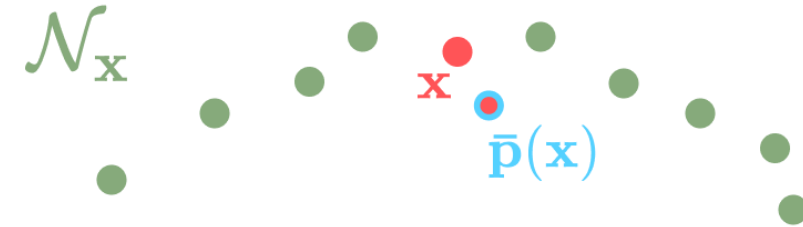


Methods – PCA [HJ87]



- Find orthogonal directions of position's variance

$$C(\mathbf{x}) = \frac{1}{\sum_{i \in \mathcal{N}_x} w_i} \sum_{i \in \mathcal{N}_x} w_i (\mathbf{p}_i - \bar{\mathbf{p}}(\mathbf{x})) (\mathbf{p}_i - \bar{\mathbf{p}}(\mathbf{x}))^T$$

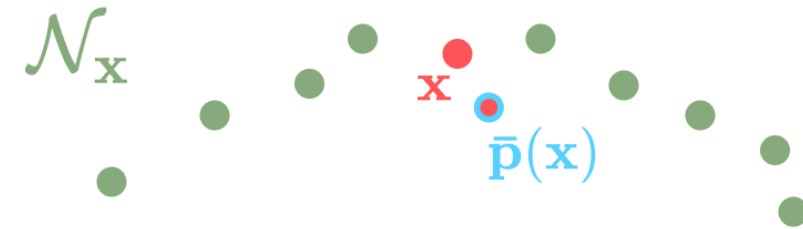


Methods – PCA [HJ87]



- Find orthogonal directions of position's variance
- Tangent frame / normal estimation

$$C(\mathbf{x}) = \frac{1}{\sum_{i \in \mathcal{N}_x} w_i} \sum_{i \in \mathcal{N}_x} w_i (\mathbf{p}_i - \bar{\mathbf{p}}(\mathbf{x})) (\mathbf{p}_i - \bar{\mathbf{p}}(\mathbf{x}))^T$$

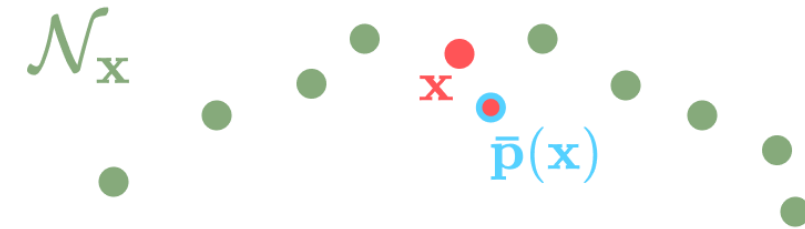


Methods – PCA [HJ87]



- Find orthogonal directions of position's variance
- Tangent frame / normal estimation

$$C(\mathbf{x}) = \frac{1}{\sum_{i \in \mathcal{N}_x} w_i} \sum_{i \in \mathcal{N}_x} w_i (\mathbf{p}_i - \bar{\mathbf{p}}(\mathbf{x})) (\mathbf{p}_i - \bar{\mathbf{p}}(\mathbf{x}))^T$$



$$\lambda_0^C \leq \lambda_1^C \leq \lambda_2^C$$
$$\mathbf{u}_0^C \quad \mathbf{u}_1^C \quad \mathbf{u}_2^C$$

eigenvalues (related to position's variance)

eigenvectors (normal and frame directions)

Methods – PCA [HJ87]



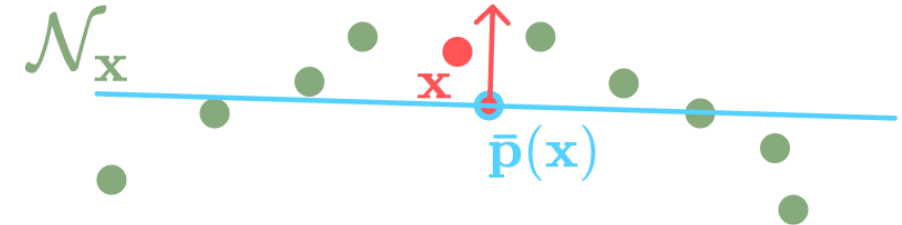
- Find orthogonal directions of position's variance
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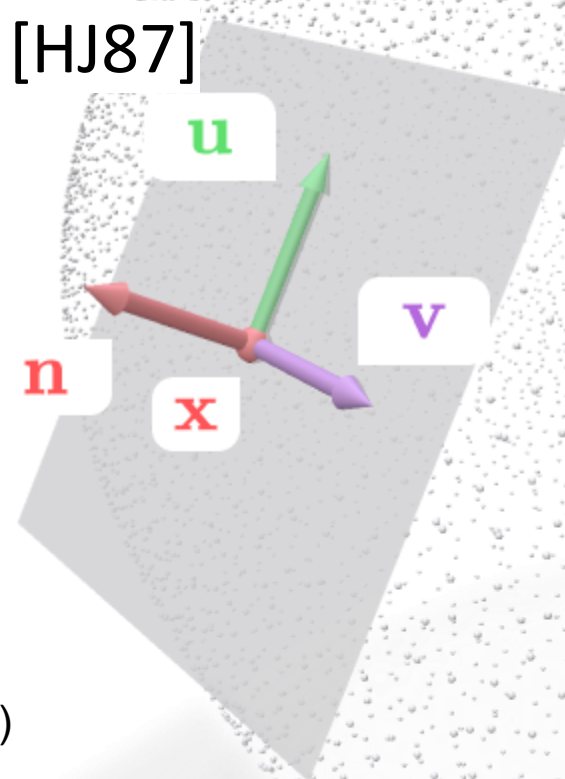
eigenvectors (normal and frame directions)



Methods – PCA [HJ87]



- Find orthogonal directions of position's variance
- Tangent frame / normal estimation [HJ87]



$$\lambda_0^C \leq \lambda_1^C \leq \lambda_2^C$$
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eigenvalues (related to position's variance)

eigenvectors (normal and frame directions)

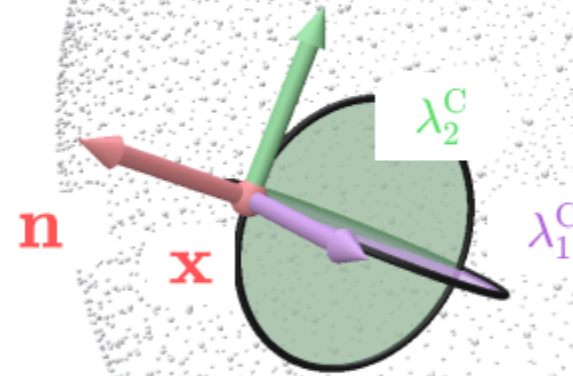
Methods – PCA [HJ87]

- Analysis of the eigenvalues
 - Not related to curvatures
 - Over-estimate the curvatures

$$\lambda_0^C \leq \lambda_1^C \leq \lambda_2^C$$
$$\mathbf{u}_0^C \quad \mathbf{u}_1^C \quad \mathbf{u}_2^C$$

eigenvalues (related to position's variance)

eigenvectors (normal and frame directions)



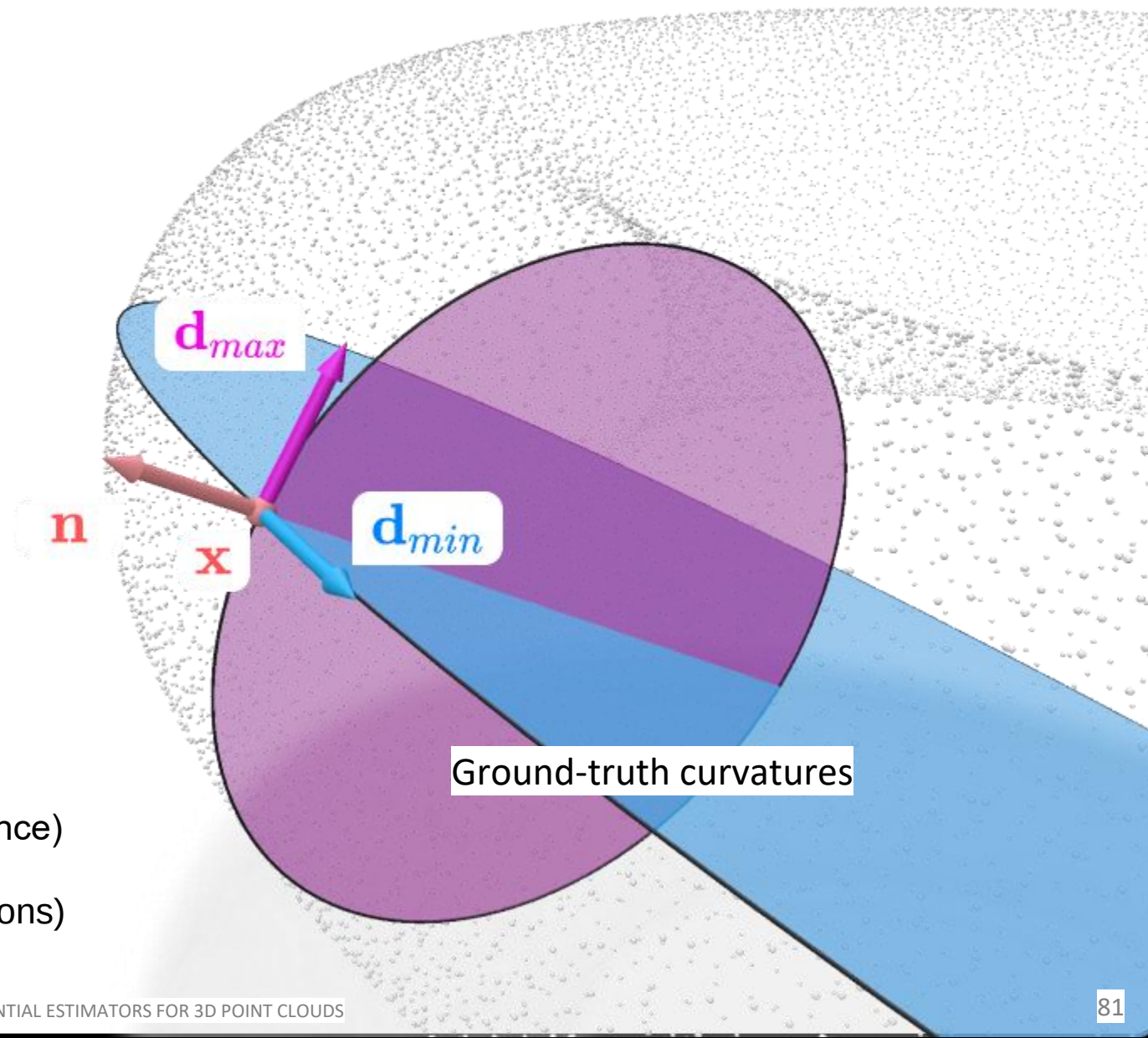
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eigenvalues (related to position's variance)

eigenvectors (normal and frame directions)

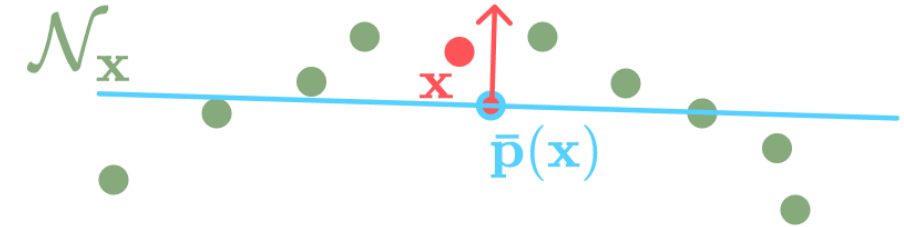


Methods – 2-Monge [Gra98]



- Monge patch fitting
 - 2nd order bivariate polynomial

$$h(\mathbf{q}) = u_c + \mathbf{u}_\ell^T \mathbf{q} + \mathbf{q}^T \mathbf{U}_q \mathbf{q}$$



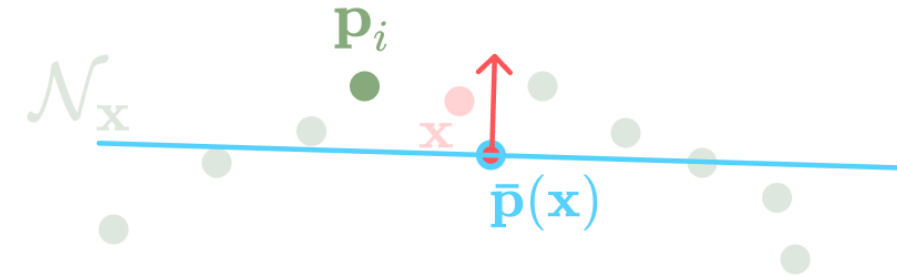
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$$h(\mathbf{q}) = u_c + \mathbf{u}_\ell^T \mathbf{q} + \mathbf{q}^T \mathbf{U}_q \mathbf{q}$$

height to the plane $\begin{pmatrix} h_i \\ \mathbf{q}_i \end{pmatrix} = \bar{\mathbf{B}}(\mathbf{x})^T (\mathbf{p}_i - \bar{\mathbf{p}}(\mathbf{x}))$
2D point coordinates



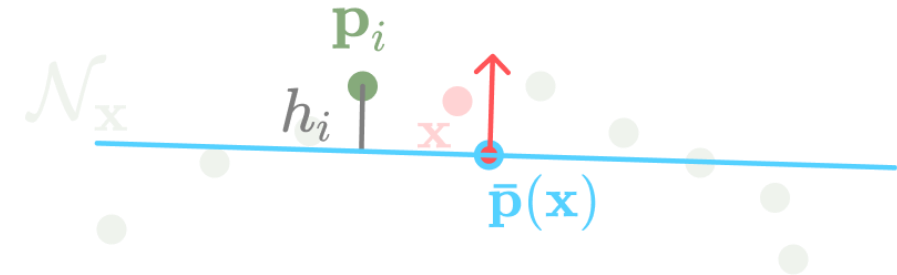
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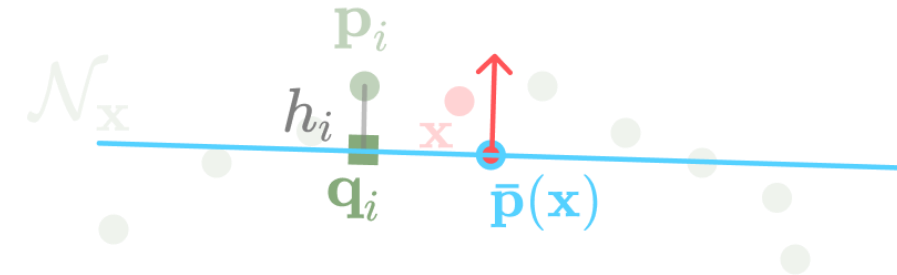
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2D point coordinates



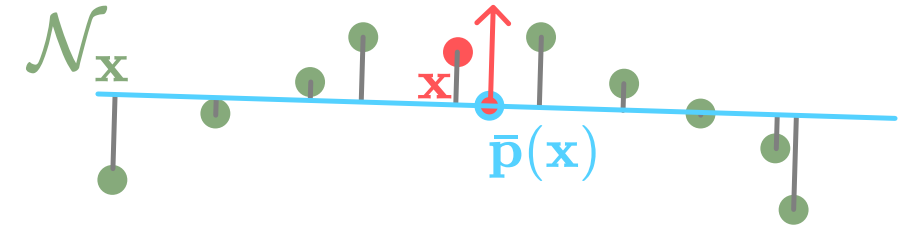
Methods – 2-Monge [Gra98]



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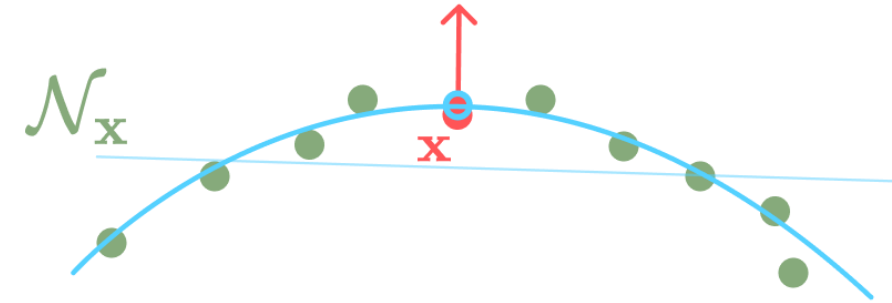


- Monge patch fitting
 - 2nd order bivariate polynomial

$$h(\mathbf{q}) = u_c + \mathbf{u}_\ell^T \mathbf{q} + \mathbf{q}^T \mathbf{U}_q \mathbf{q}$$

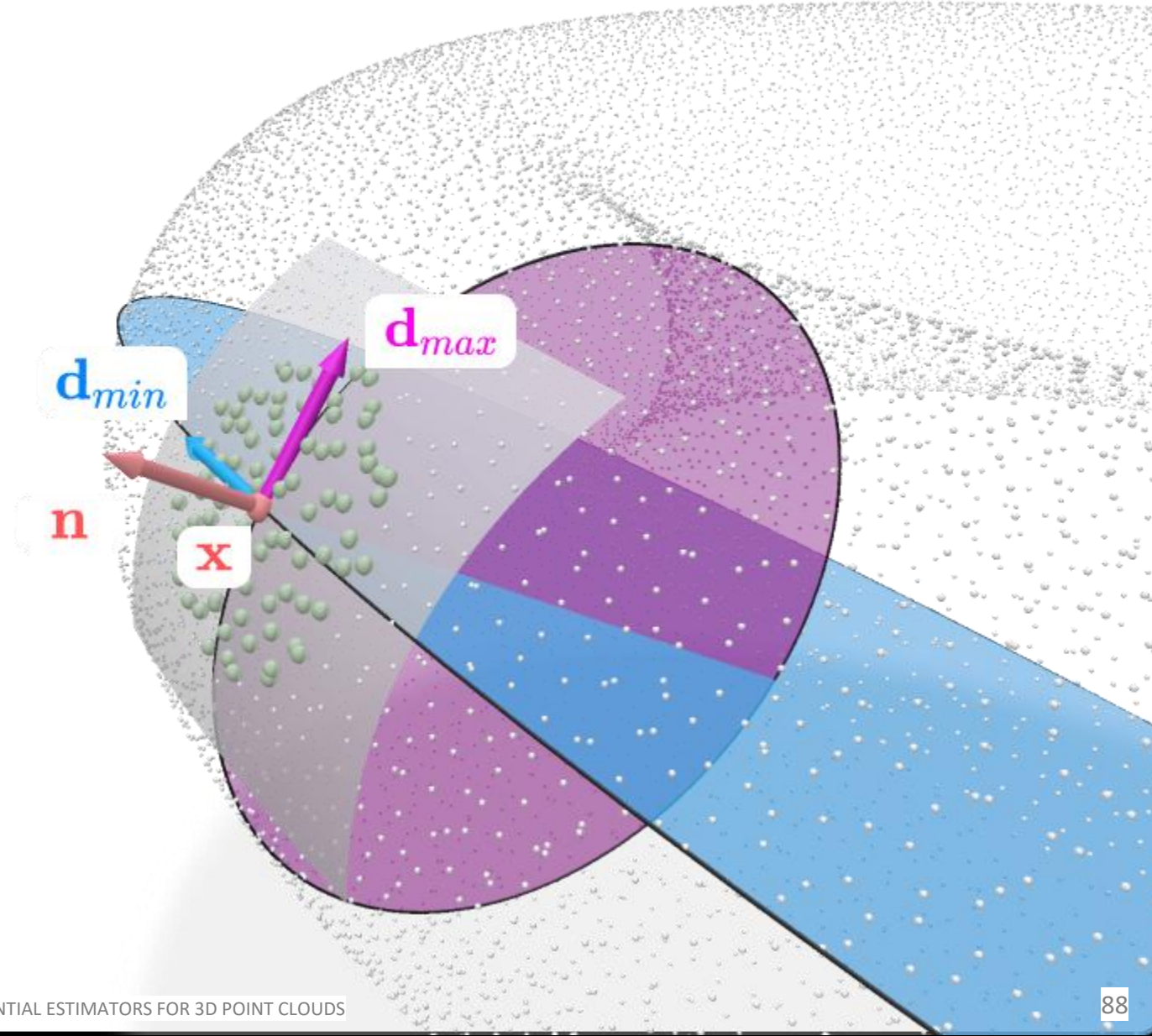
height to the plane $\begin{pmatrix} h_i \\ \mathbf{q}_i \end{pmatrix} = \bar{\mathbf{B}}(\mathbf{x})^T (\mathbf{p}_i - \bar{\mathbf{p}}(\mathbf{x}))$

2D point coordinates



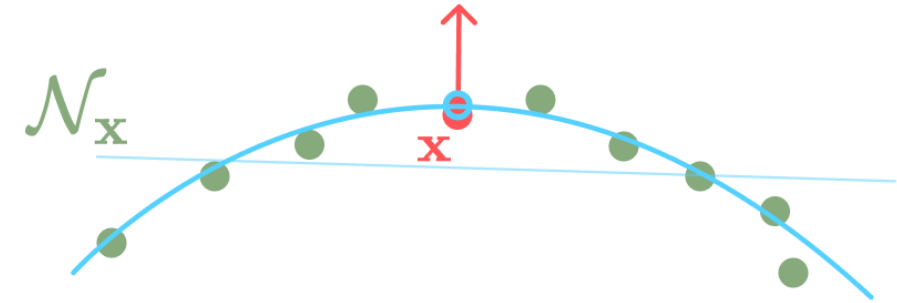
Methods – 2-Monge [Gra98]

- Estimate
 - Weingarten Map
- Requirements
 - Initial tangent frame



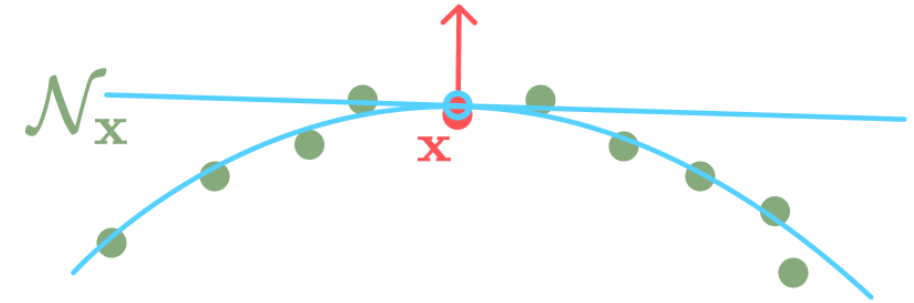
Methods – JetFitting [CP05]

- Monge patch fitting
 - Higher order bivariate polynomial
- Normal refinement



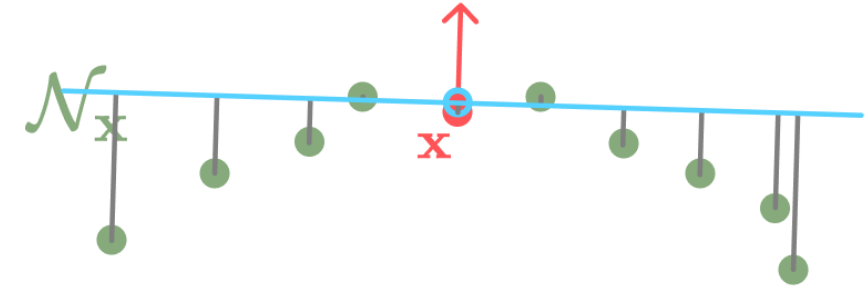
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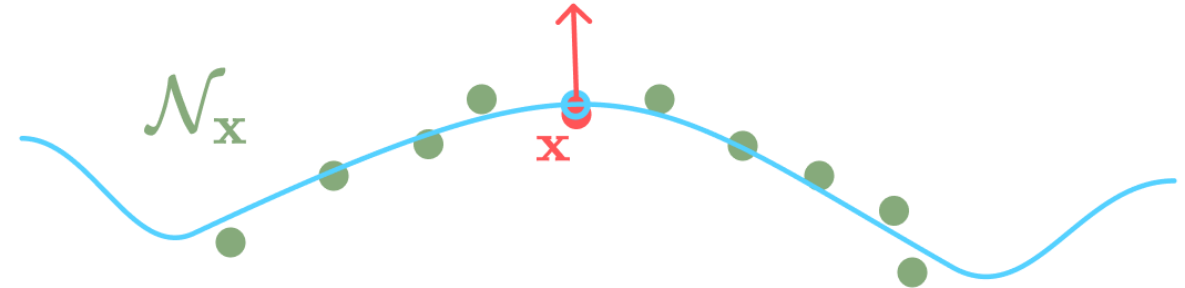
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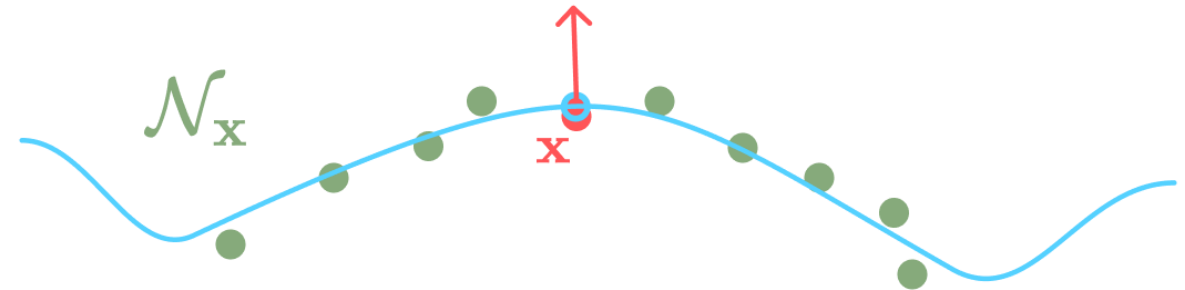
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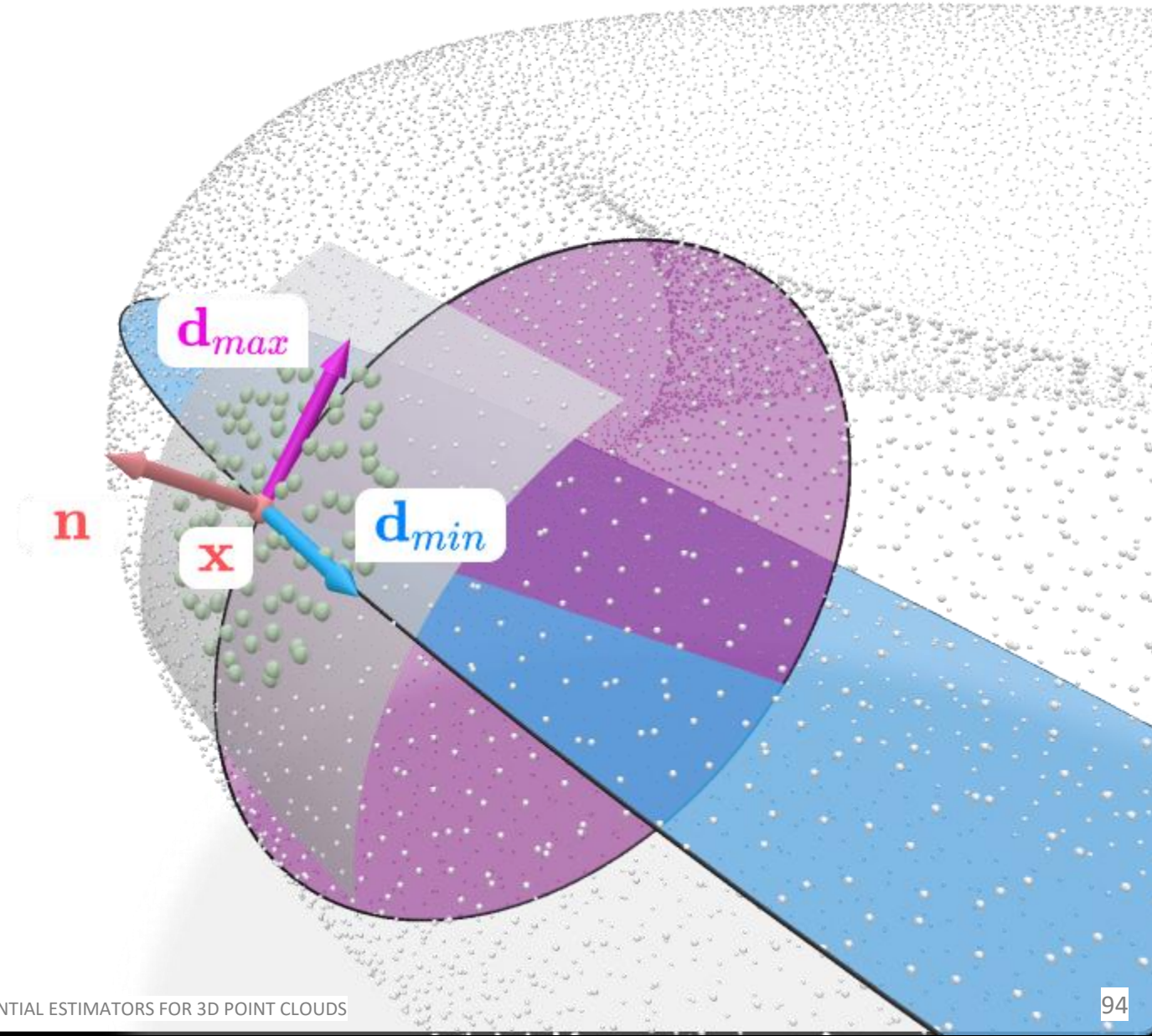
Methods – JetFitting [CP05]

- Estimate
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- Requirements
 - Initial tangent frame



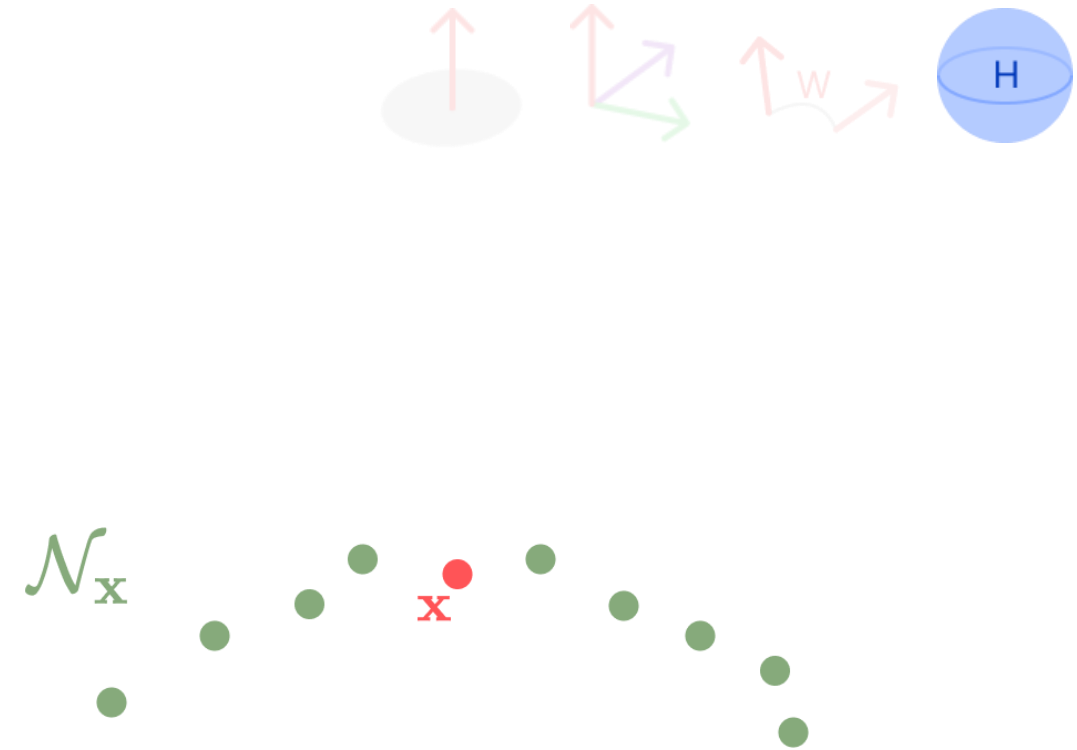
Methods – JetFitting [CP05]

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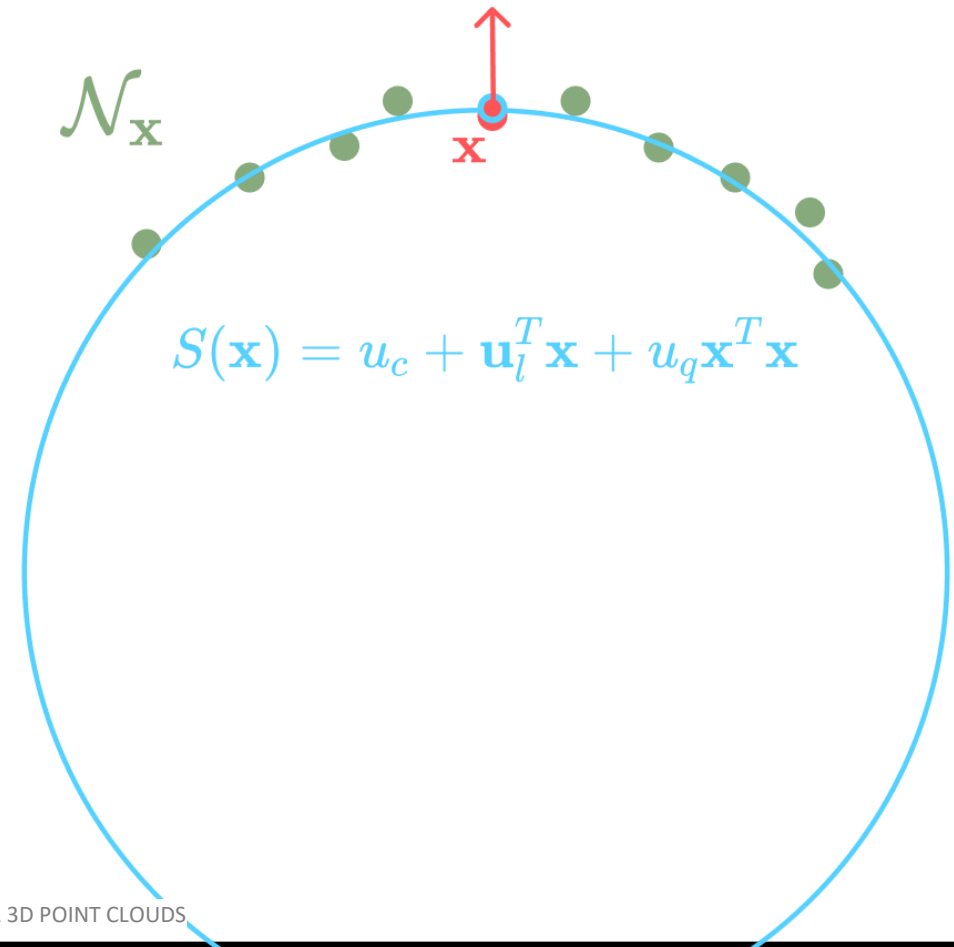
Methods – APSS [GG07]

- Algebraic sphere fitting
- Based on Pratt's work [Pra87]
- Closed-form formulation



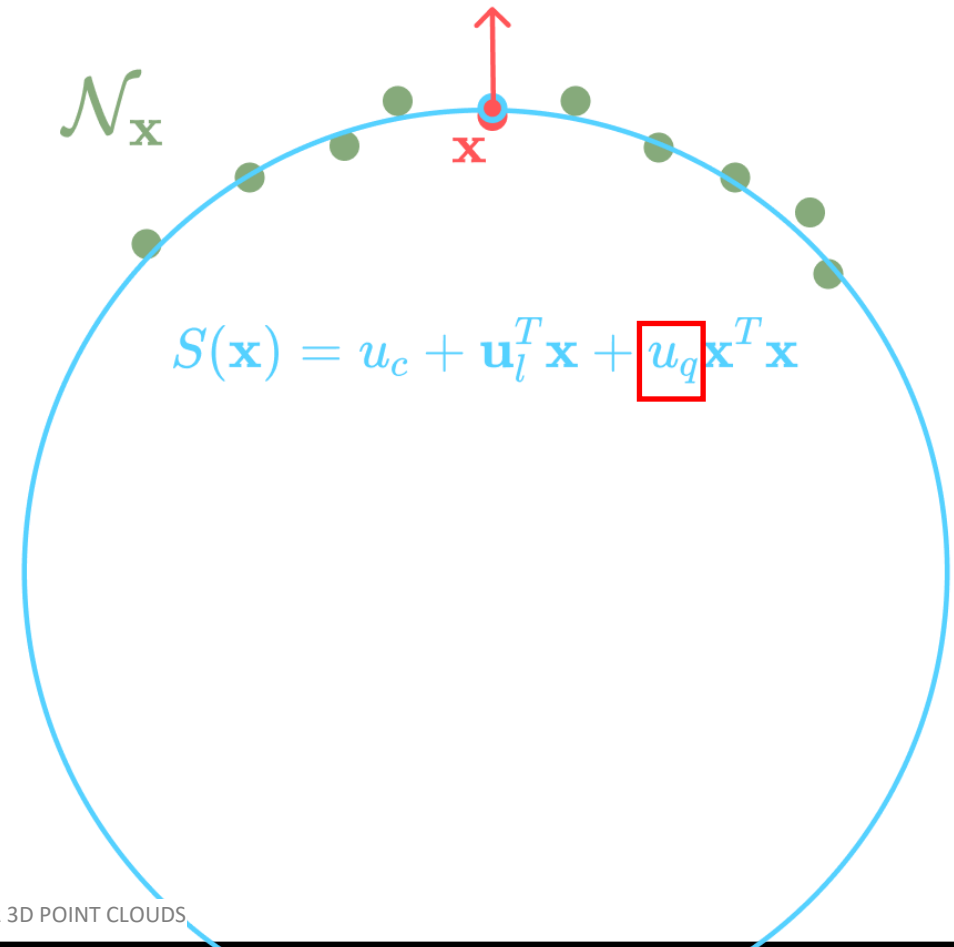
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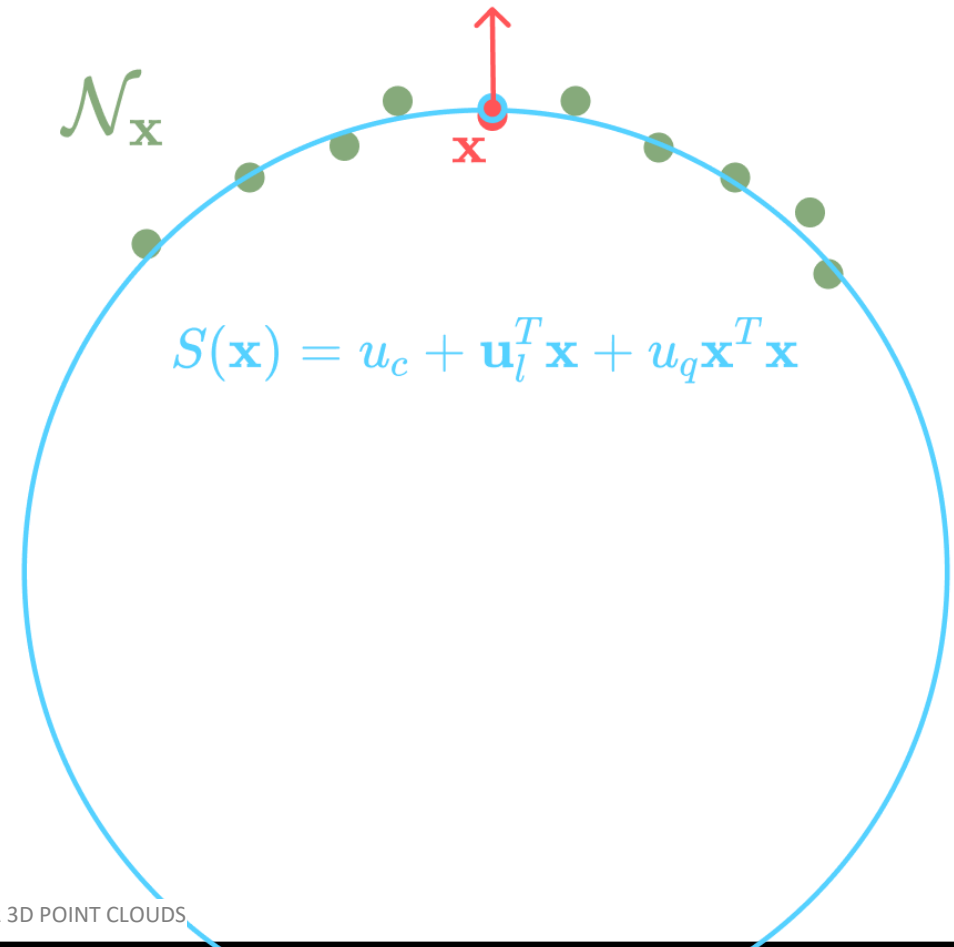
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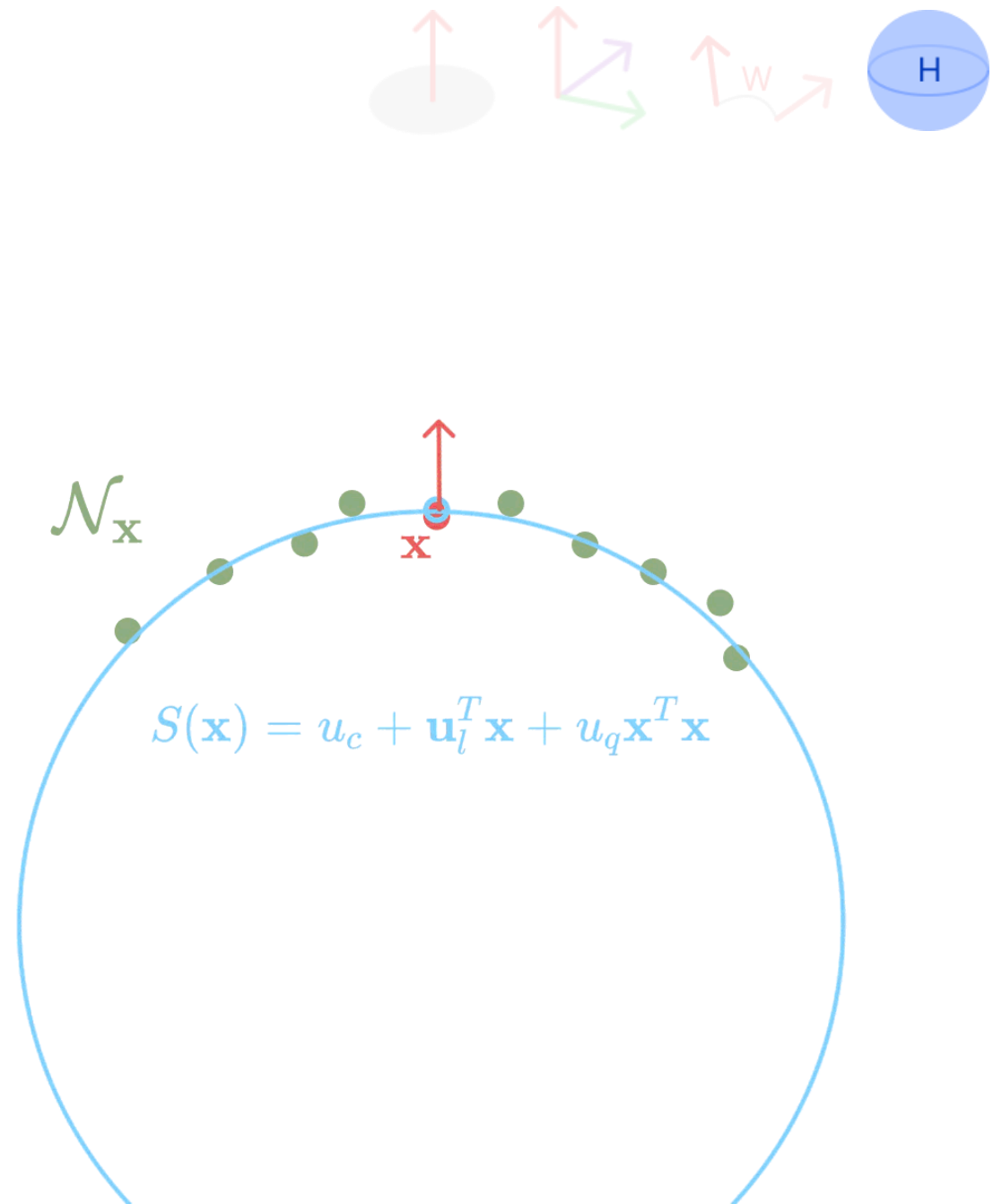
Methods – APSS [GG07]

- Algebraic sphere fitting
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- Closed-form formulation
- Second-order derivative
 - $\nabla^2 S(\mathbf{x}) = 2 u_q$



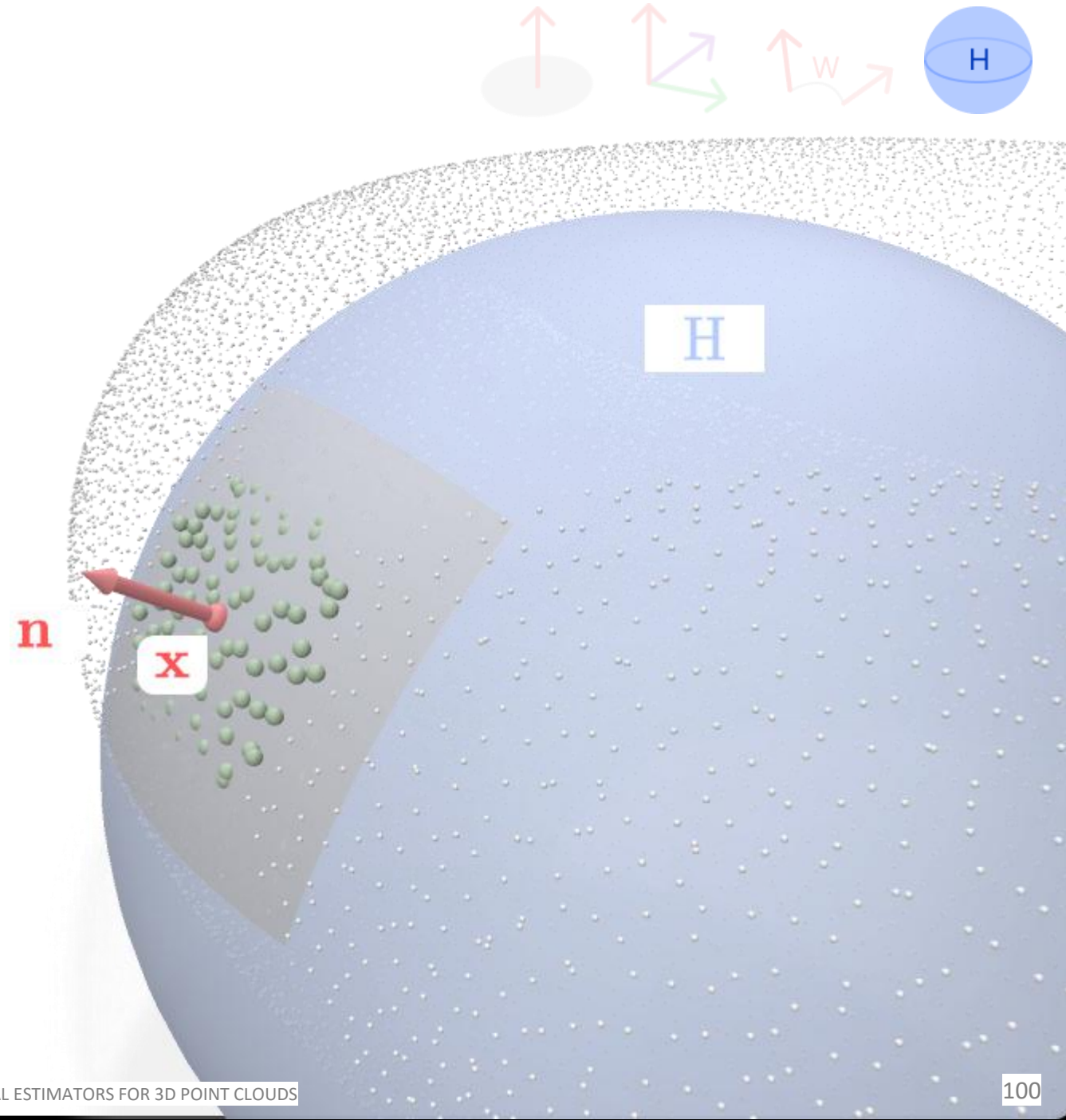
Methods – APSS [GG07]

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- Second-order derivative
 - $\nabla^2 S(\mathbf{x}) = 2 u_q$
 - Mean curvature



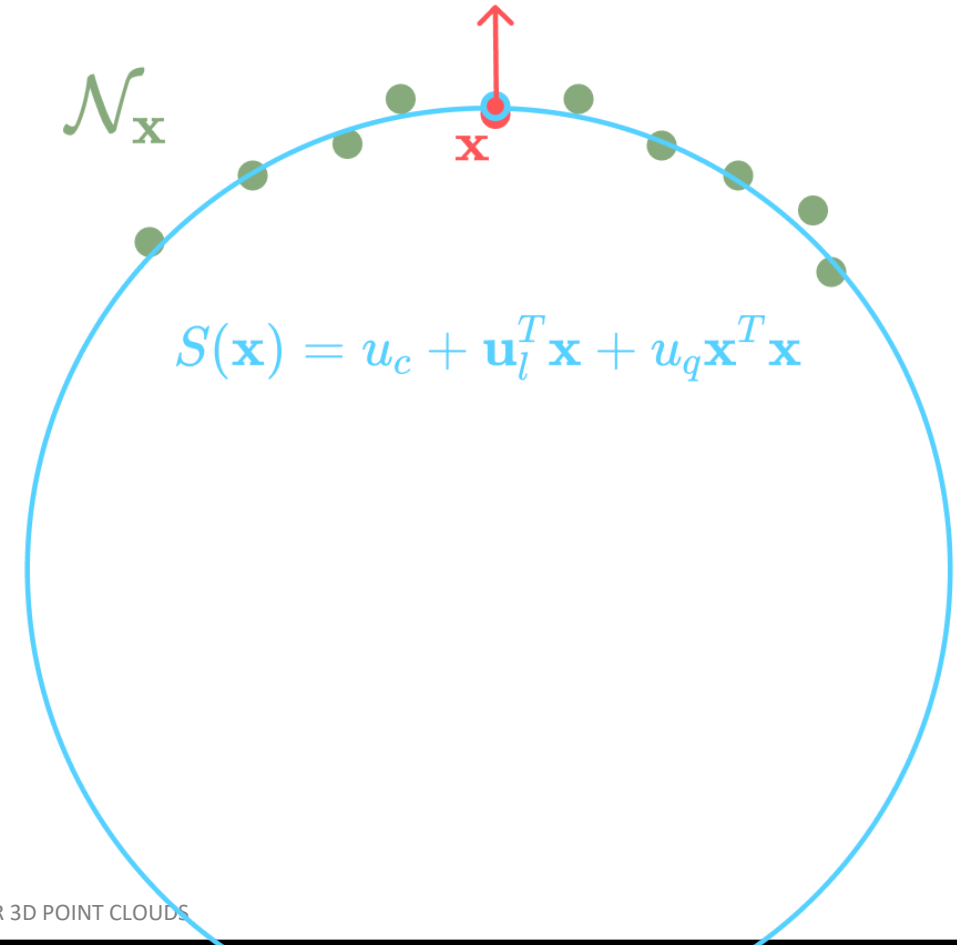
Methods – APSS [GG07]

- Estimate
 - Mean curvature H
- Requirements
 - Input normals
- Other sphere fitting variants
 - 3D positions only [Pra87]
 - non-oriented normals [CGBG13]





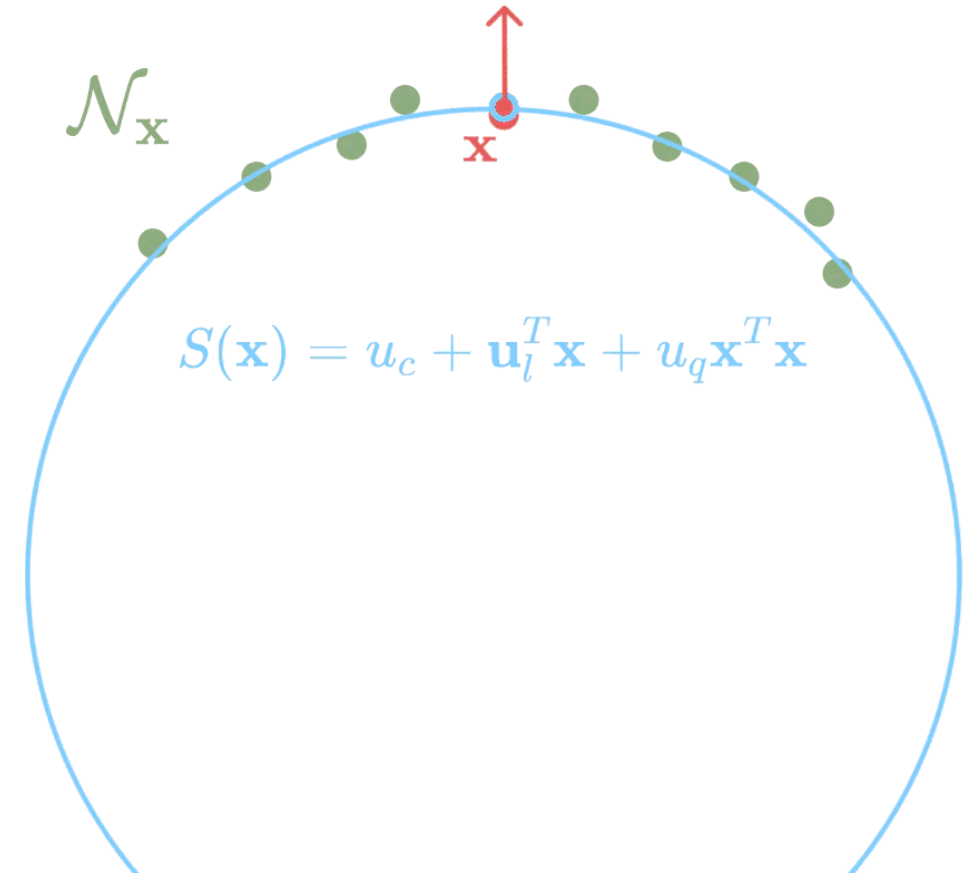
- Extension to the APSS for Weingarten Map estimation



Methods – ASO [LCBM21]

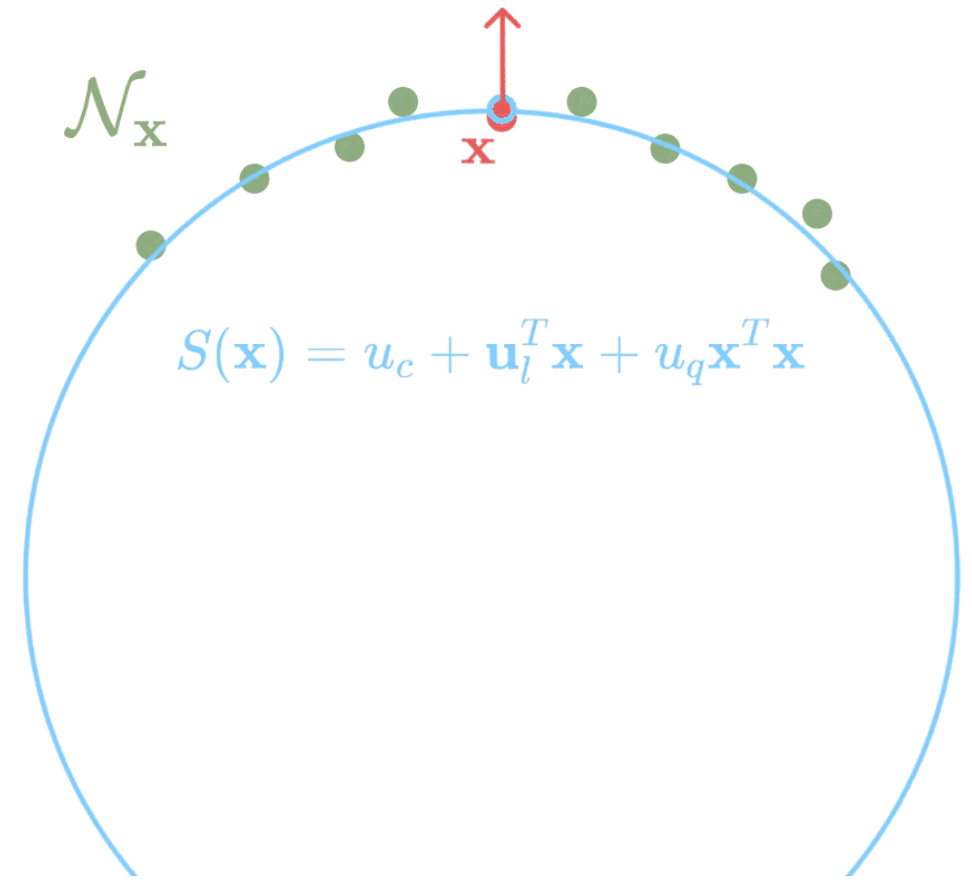
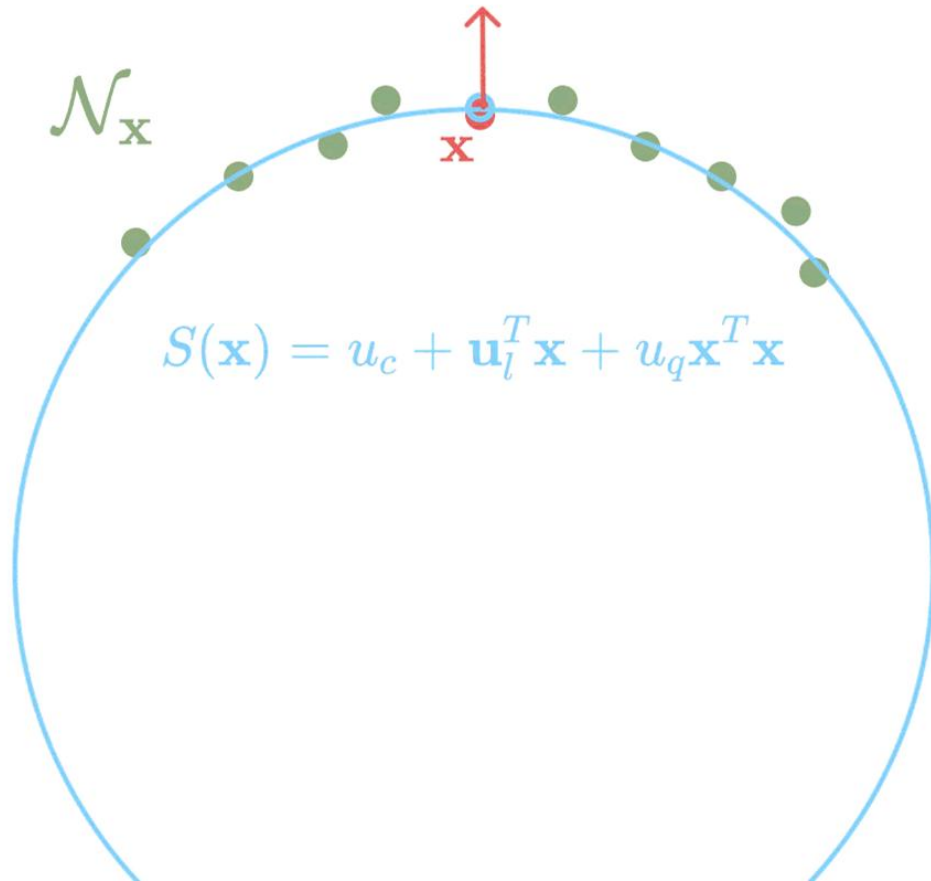


- Extension to the APSS for Weingarten Map estimation
- Weight kernel differentiation
 - Variations of the normals on the estimated surface



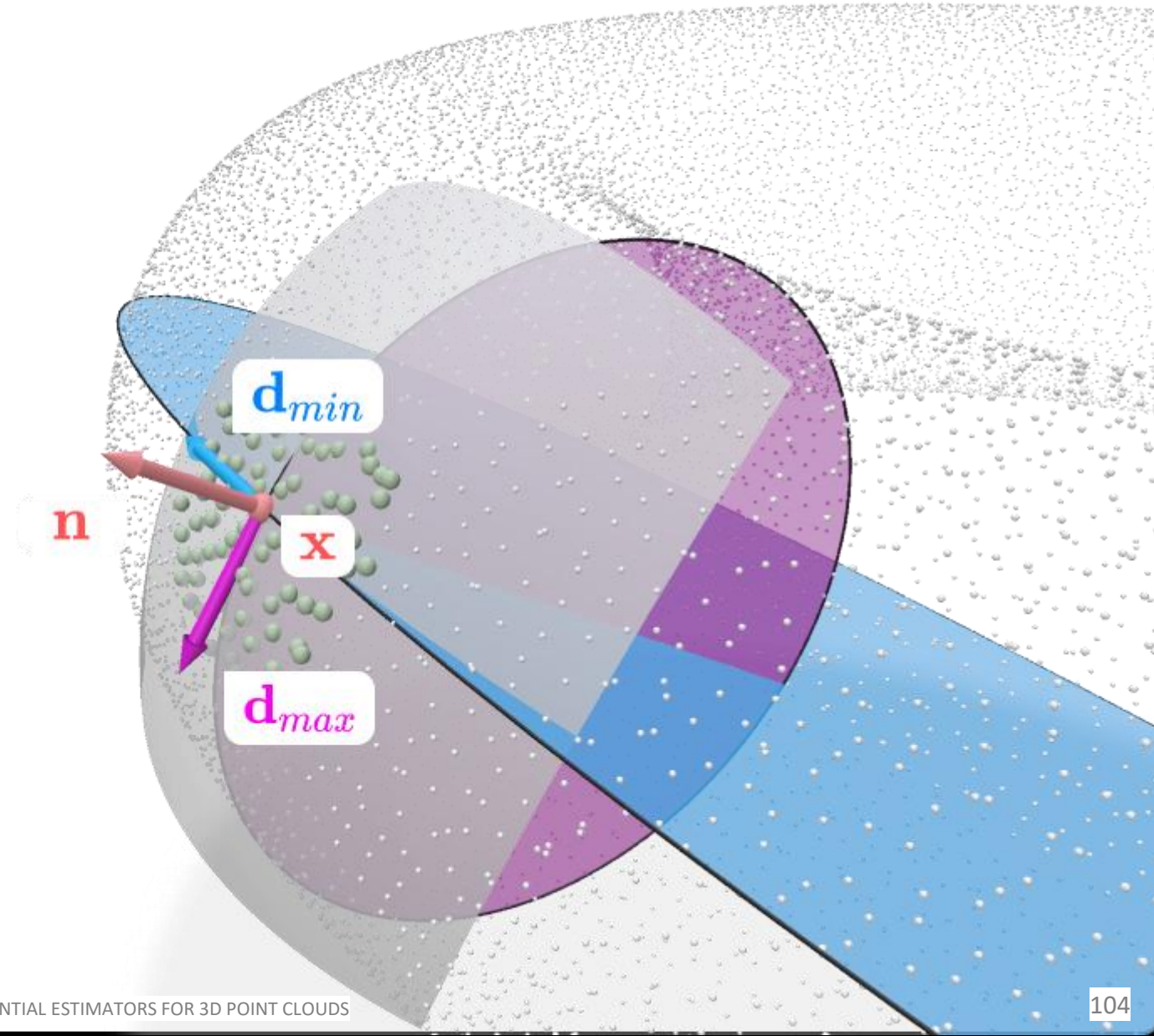
Methods – ASO [LCBM21]

- Difference between APSS and ASO



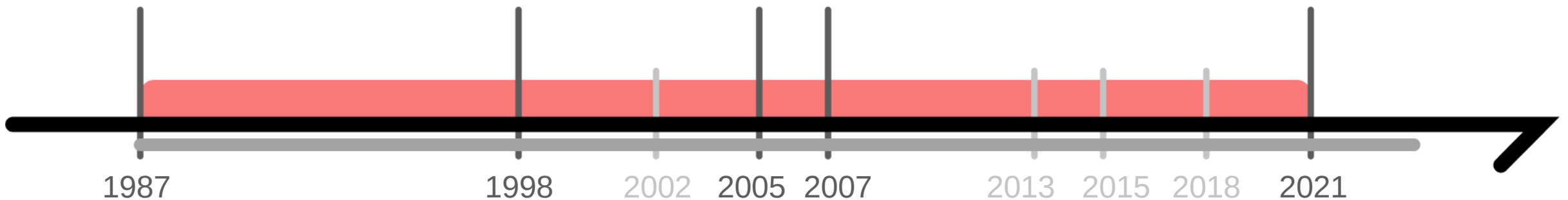
Methods – ASO [LCBM21]

- Estimate
 - Weingarten Map
- Requirements
 - Oriented normals



Methods – Local surface models

- Analytic curvature estimation through geometric primitive fitting



Methods – Local surface models

Pros

- Stable
- « Gold standard » on perfect data (JetFitting)
- 3D methods avoid the two pass fitting (using tangent plane)

Cons

- 2.5D methods : Tangent plane - dependent
- The more the order of the primitive is, the more it tends to overfit the noise
- May be slower than direct point-based approaches

Methods – Measure theory approaches

- Geometric Measure Theory



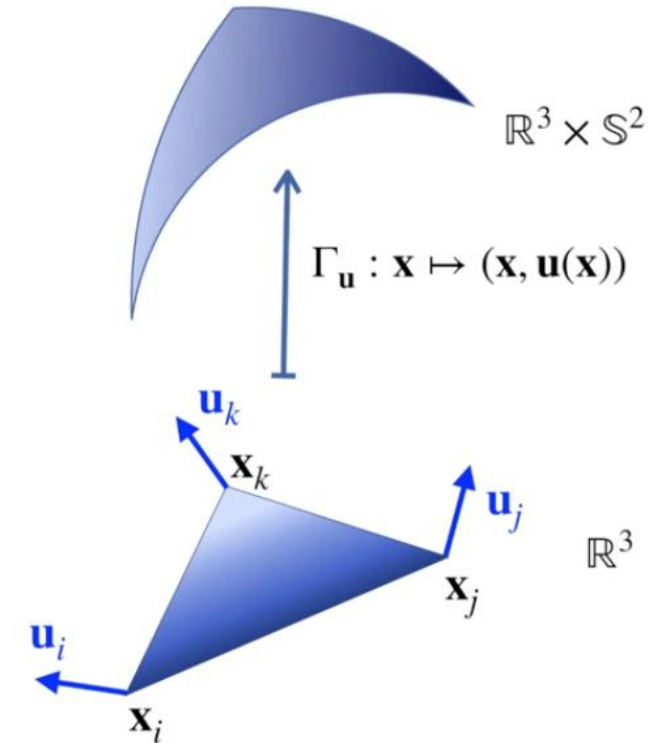
Methods – AvgHexagram [LCL*23]



- Based on normal cycles [CSM03]
- Summations over locally generated triangles $\{i,j,k\}$

$$\begin{aligned}\mu_{\tau}^{(0)} &= \frac{1}{2} \bar{\mathbf{n}} \cdot ((\mathbf{p}_j - \mathbf{p}_i) \times (\mathbf{p}_k - \mathbf{p}_i)), \\ \mu_{\tau}^{(1)} &= \frac{1}{2} \bar{\mathbf{n}} \cdot ((\mathbf{n}_k - \mathbf{n}_j) \times \mathbf{p}_i \\ &\quad + (\mathbf{n}_i - \mathbf{n}_k) \times \mathbf{p}_j + (\mathbf{n}_j - \mathbf{n}_i) \times \mathbf{p}_k), \\ \mu_{\tau}^{(2)} &= \frac{1}{2} \mathbf{n}_i \cdot (\mathbf{n}_j \times \mathbf{n}_k), \\ \mu_{\tau}^{(\text{XY})} &= \frac{1}{2} \bar{\mathbf{n}} \cdot ((\mathbf{Y} \cdot (\mathbf{n}_k - \mathbf{n}_i)) \mathbf{X} \times (\mathbf{p}_j - \mathbf{p}_i) \\ &\quad - (\mathbf{Y} \cdot (\mathbf{n}_j - \mathbf{n}_i)) \mathbf{X} \times (\mathbf{p}_k - \mathbf{p}_i)),\end{aligned}$$

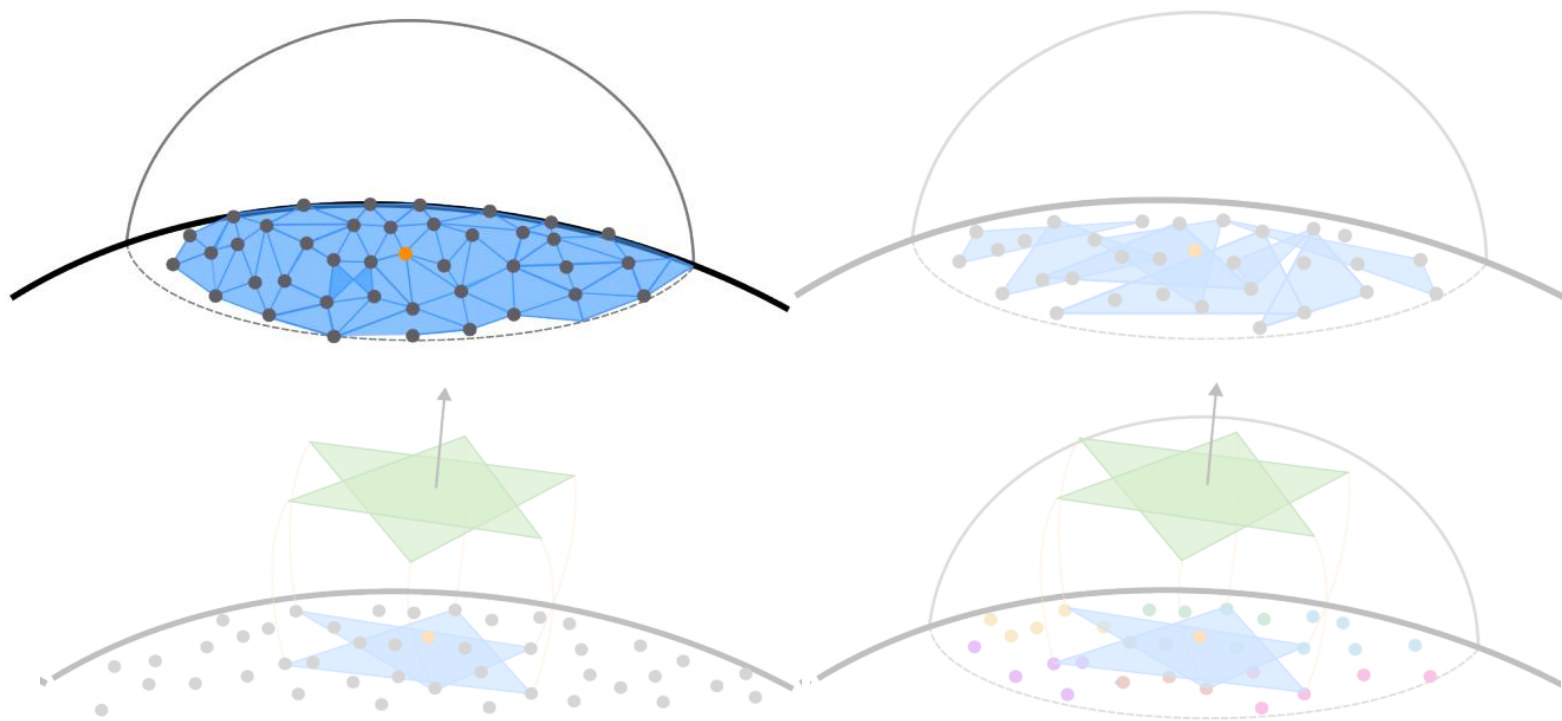
$$\begin{aligned}\bar{H} &= \sum_{\tau \in T} \mu_{\tau}^{(1)} / \sum_{\tau \in T} \mu_{\tau}^{(0)}, \\ \bar{K} &= \sum_{\tau \in T} \mu_{\tau}^{(2)} / \sum_{\tau \in T} \mu_{\tau}^{(0)},\end{aligned}$$



Methods – AvgHexagram [LCL*23]



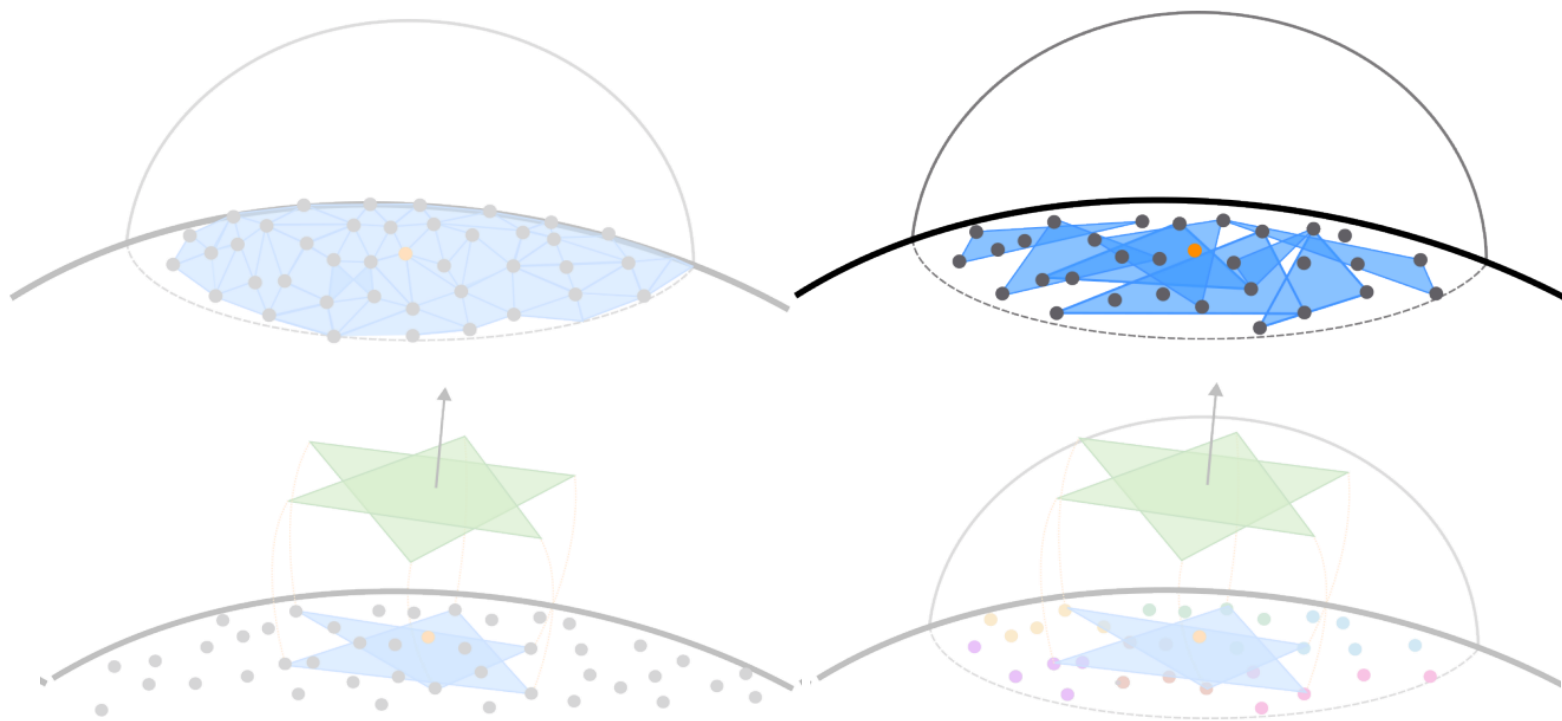
- Generation of the triangles in different ways



Methods – AvgHexagram [LCL*23]



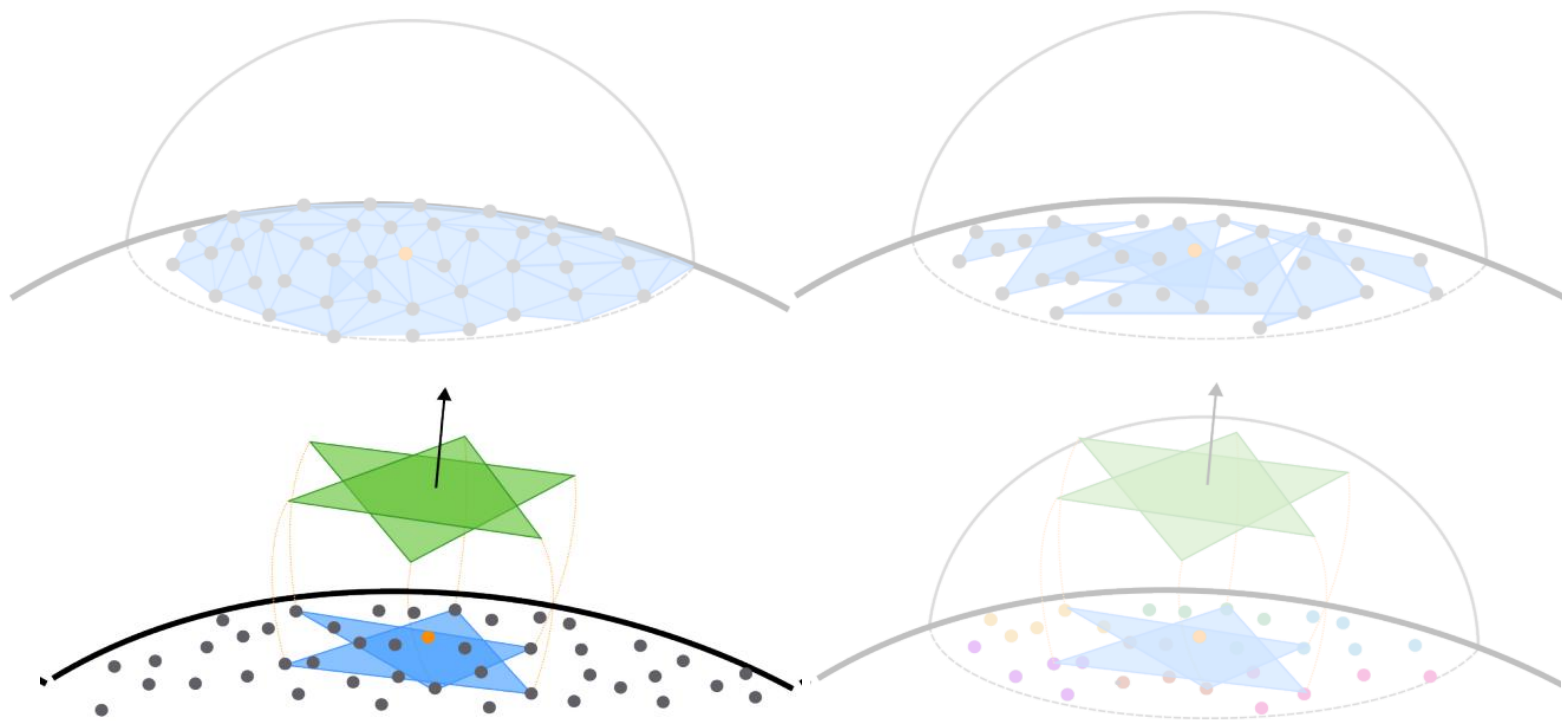
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Methods – AvgHexagram [LCL*23]



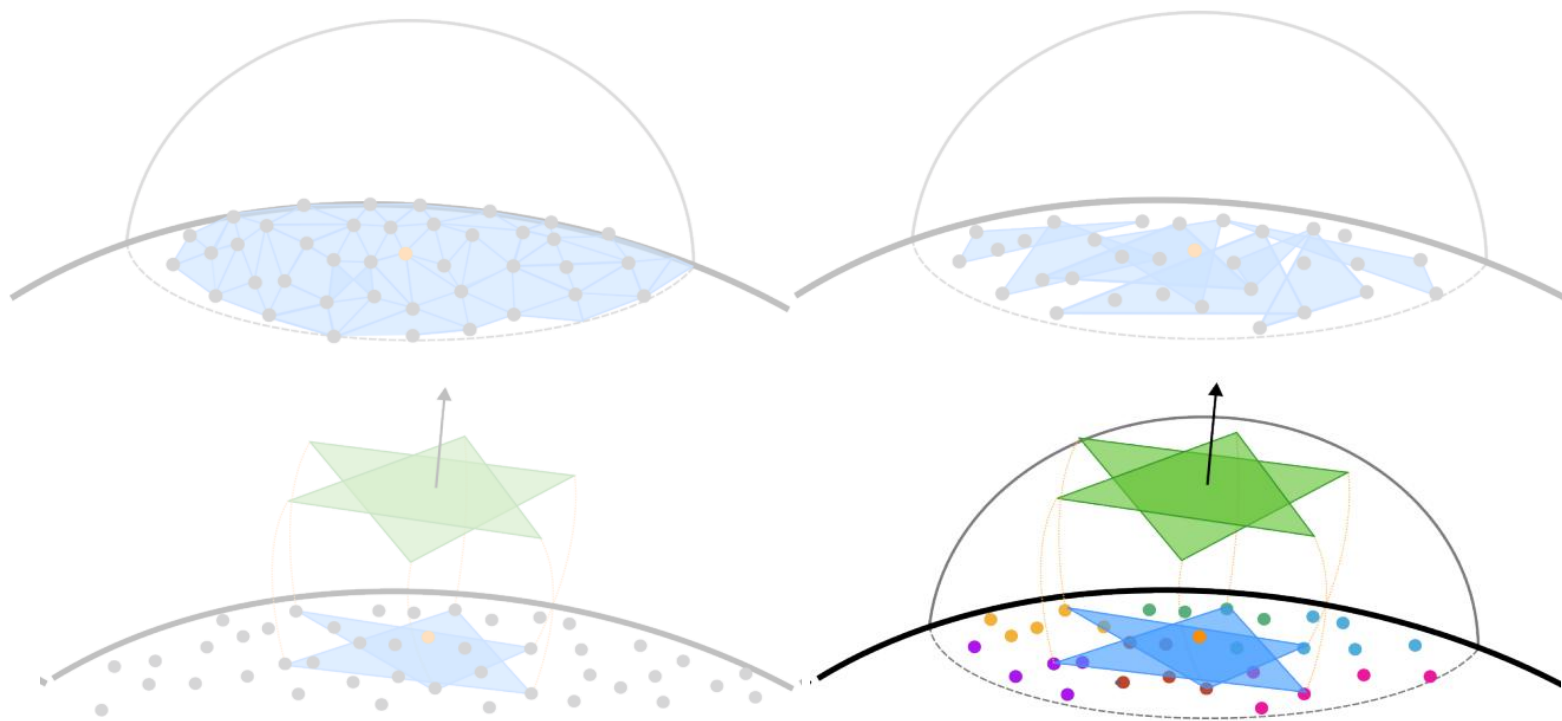
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Methods – AvgHexagram [LCL*23]



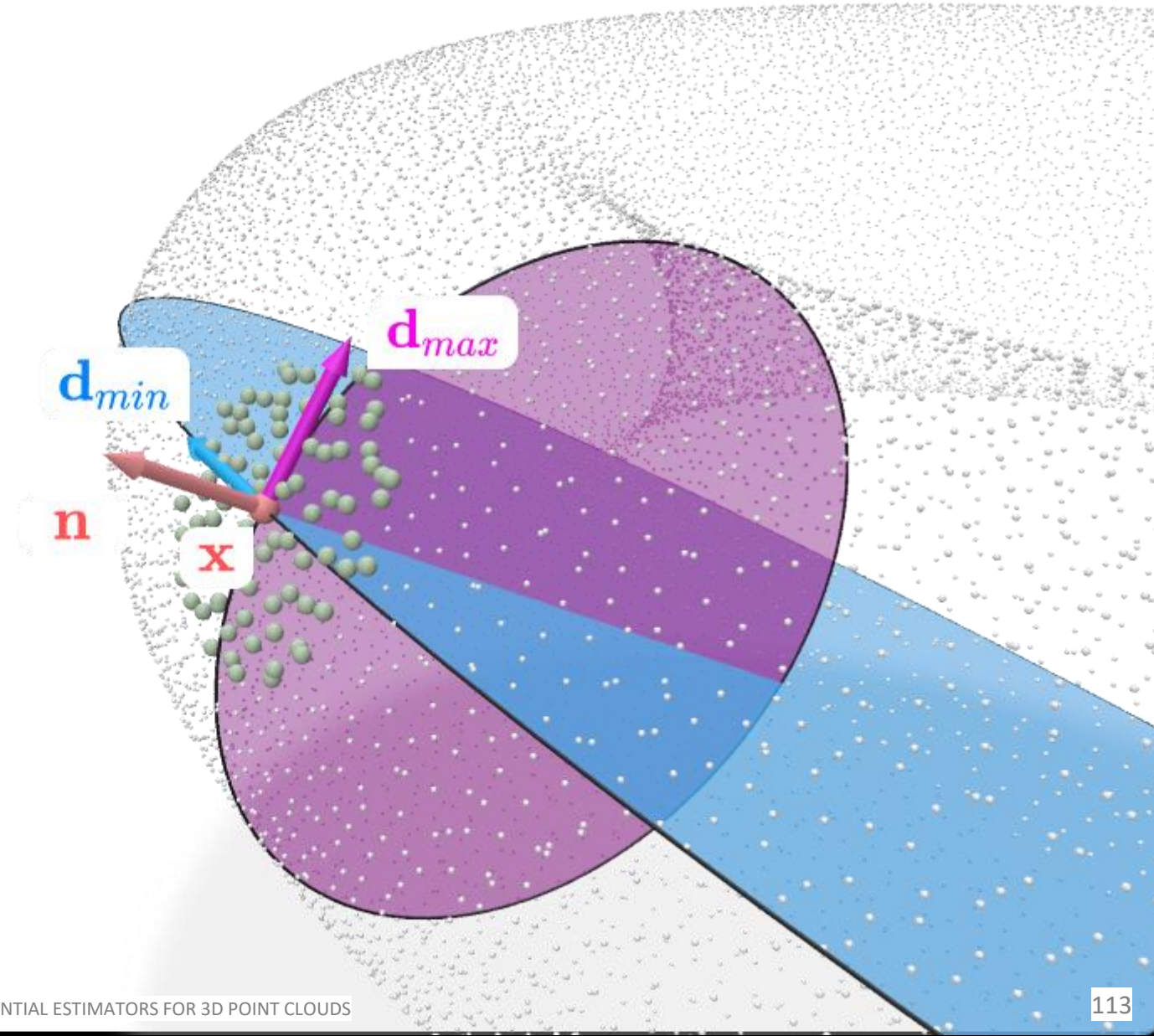
- Generation of the triangles in different ways



Methods – AvgHexagram [LCL*23]



- Estimate
 - Weingarten Map
- Requirements
 - Oriented normals



Methods – Measure theory approaches

- Define curvatures as measures on the bundle $\{x, n\}$



Methods – Measure theory approaches

Pros

- Theoretical guarantees (convergence and stability)
- Robustness to noise position
- AvgHexagram time-efficient

Cons

- Normal dependent

Methods – Overview

What about Machine Learning approaches?

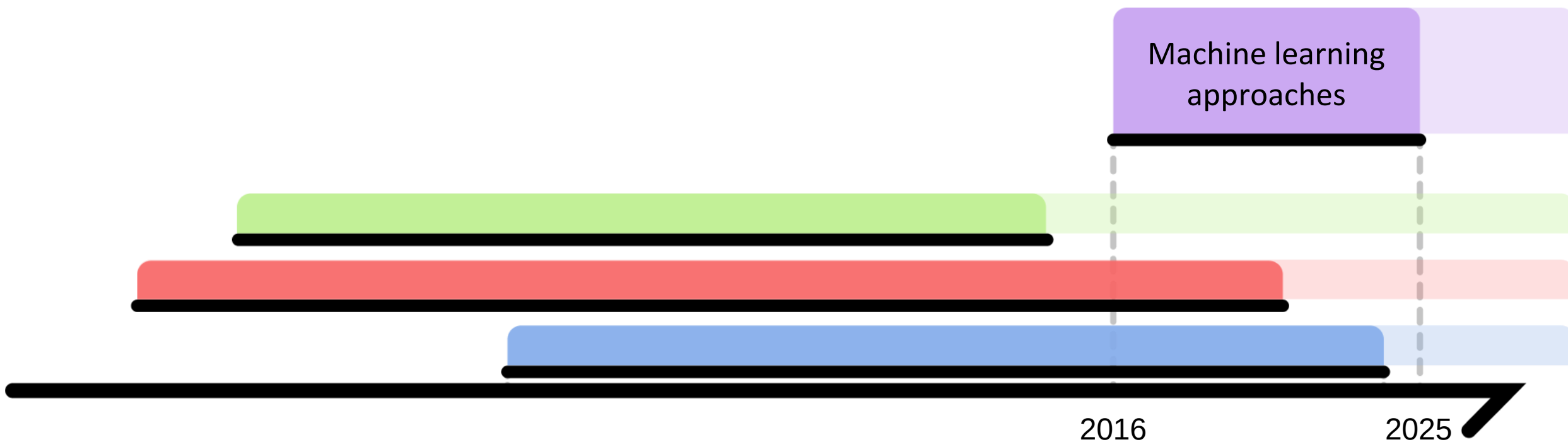
1987 1990

1998

2014

2021 2024

Methods – Overview



Methods – Machine learning approaches

- Regression-based
 - Learn features on the point cloud directly
- [BM16, QSMG17, GKOM18, ZNZ*25]

Methods – Machine learning approaches

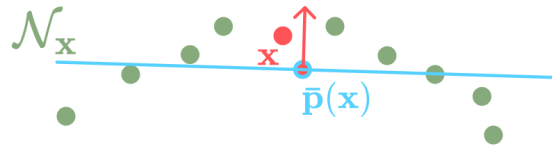
- Weight estimation
 - Offset points [ZLD*21, LZW*22, ZJW*23]
 - Weights [LOM20, BSG20, ZLD*21, ZJW*23]
 - Primitive fitting (plane or jet)

Methods – Machine learning approaches

- Local surface estimation
 - Differential RANSAC model [CZB*22]
 - Hypersurface fitting on the feature space [LLC*22, LFS*23]
 - Derivatives of implicit surface [LSY24]

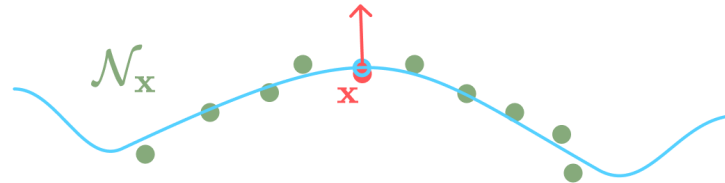
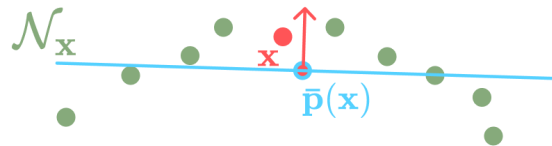
Methods – Machine learning approaches

- Most of these are dedicated (but not limited) to normal estimation
- The better ones often rely on classical primitive fitting
 - Plane [LOM20]



Methods – Machine learning approaches

- Most of these are dedicated (but not limited) to normal estimation
- The better ones often rely on classical primitive fitting
 - Plane [LOM20]
 - JetFitting [BSG20, ZLD*21, ZJW*23]



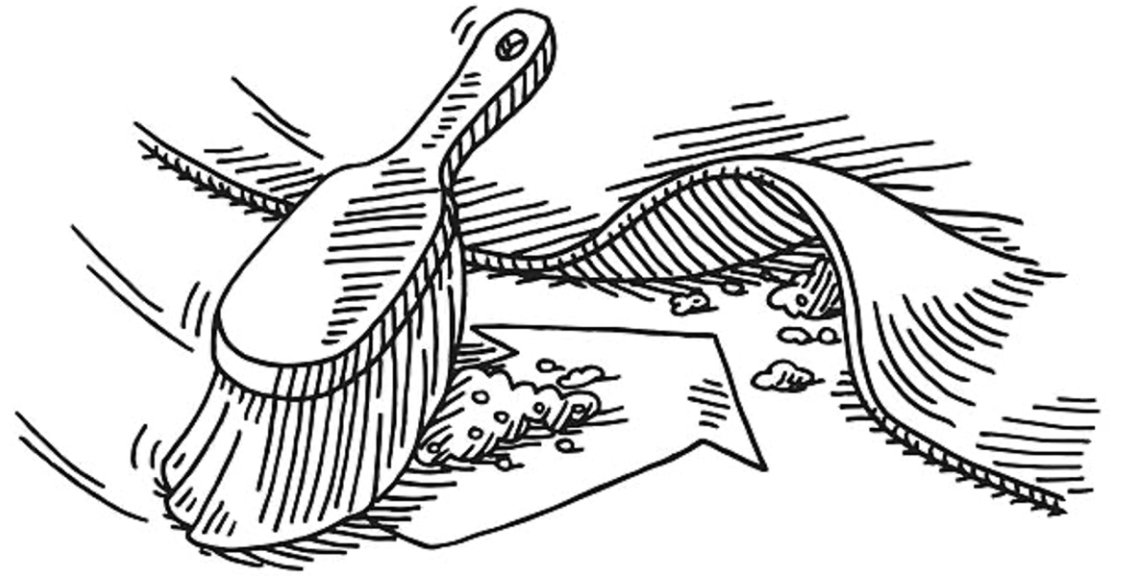
Methods – Machine learning approaches

- Most of these are dedicated (but not limited) to normal estimation
- The better ones often rely on classical primitive fitting
 - Plane [LOM20]
 - JetFitting [BSG20, ZLD*21, ZJW*23]
- Why choose these primitive instead of others?



QUESTION TIME

What's under the rug?



What's under the rug? – Weighting schemes

- Neighborhood weighting kernels

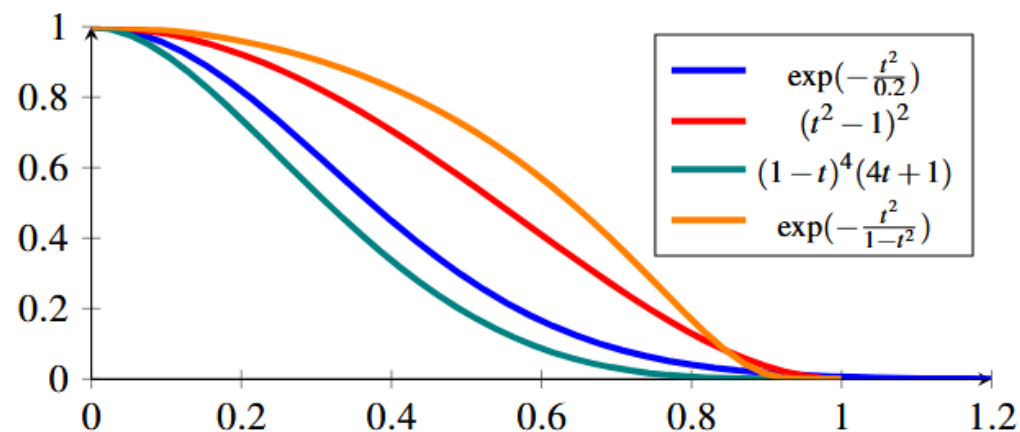
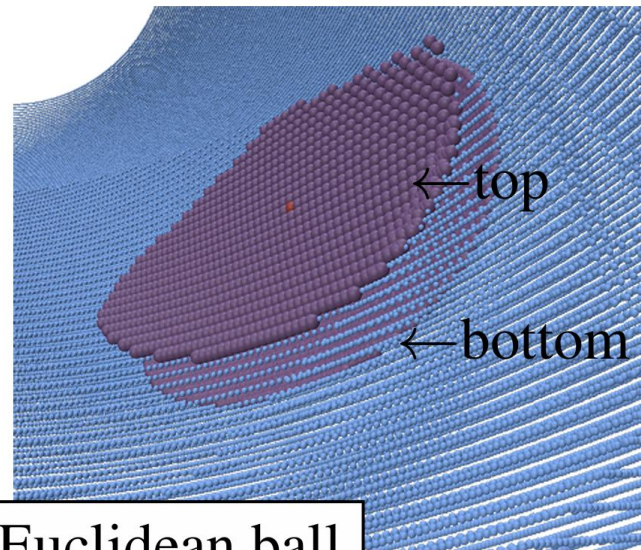


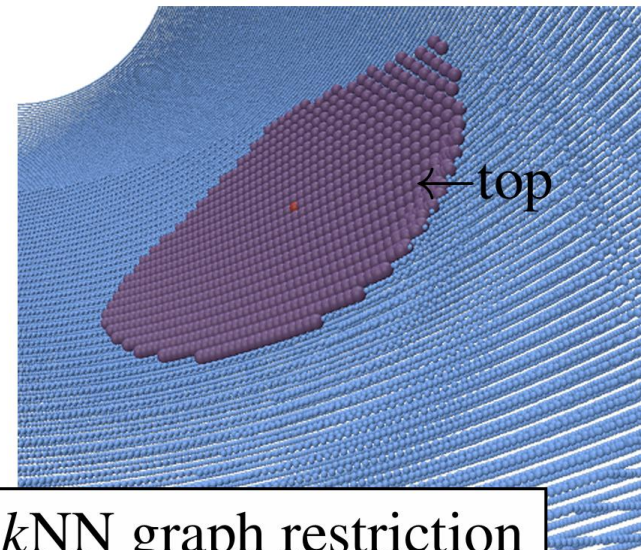
Figure 6: Standard weighting kernels Ω : *Gaussian*, *quartic*, *Wendland* [Wen95], and *bump* [BLM22].

What's under the rug? – Neighborhood collection

- Neighborhood weighting kernels
- Neighborhood collection



Euclidean ball



k NN graph restriction

Benchmark

04

Benchmark

- Benchmark ecosystem
 - Datasets
 - Quantitative analysis tools
 - Qualitative analysis tools
 - Visualization
 - Website



[Link to the website](#)

Benchmark

- Benchmark ecosystem

- Datasets
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→ Open

→ Open

→ Open

→ Open

→ Open



[Link to the website](#)

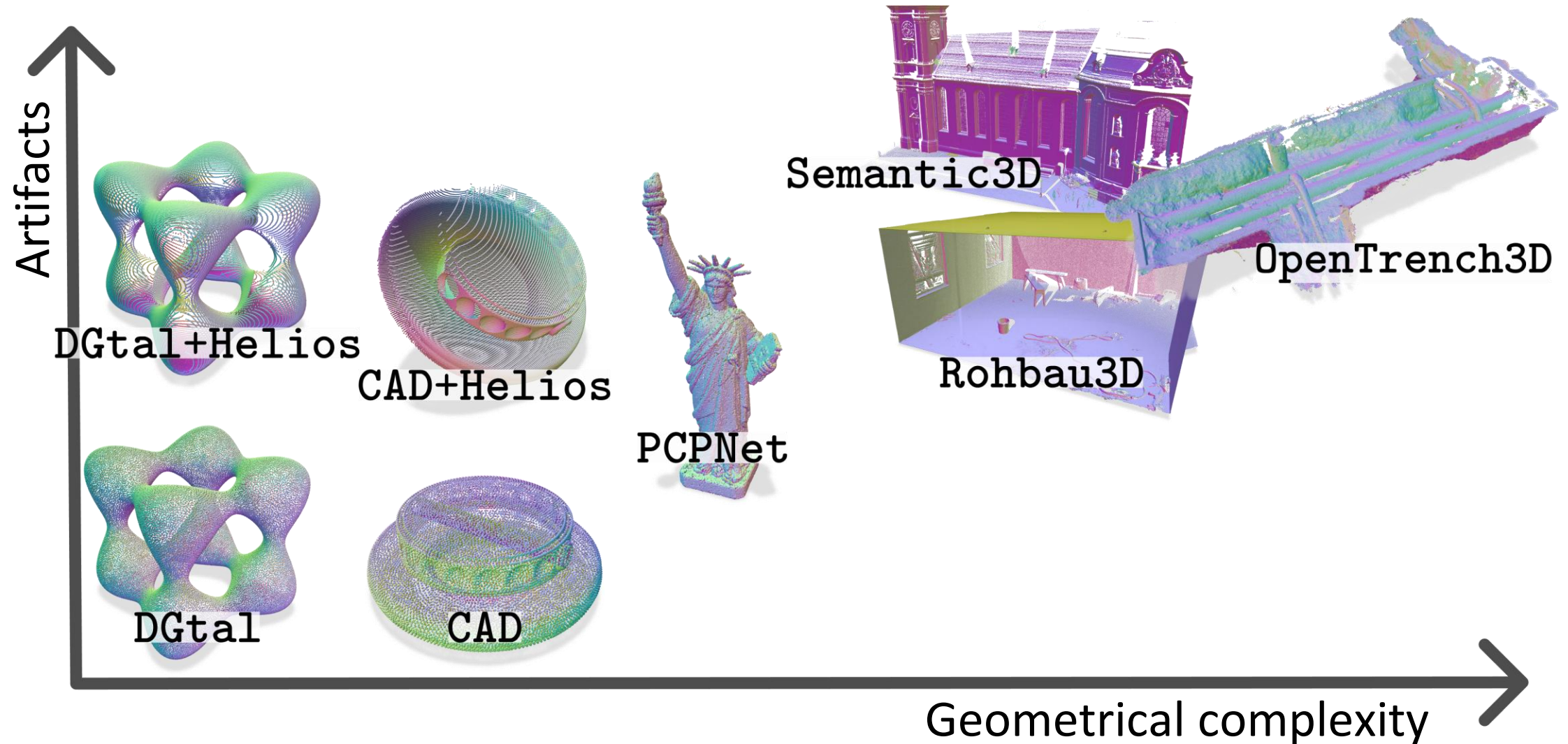
Benchmark

- Estimators
 - Ponca [MLG*20] (also for kD-Tree)
 - CGAL [PC24] for JetFitting
 - Eigen for the maths
- Visualization tools
 - Polyscope
- Dataset generation
 - DGtal [DGt]
 - Helios++ [WEW*22] for LiDAR-like noise

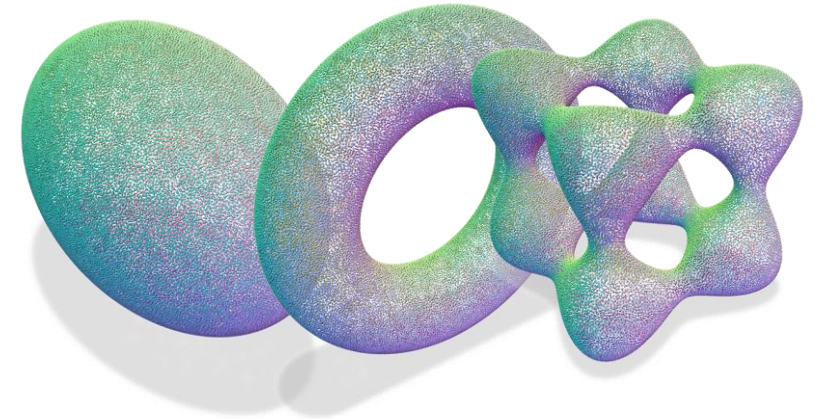


[Link to the website](#)

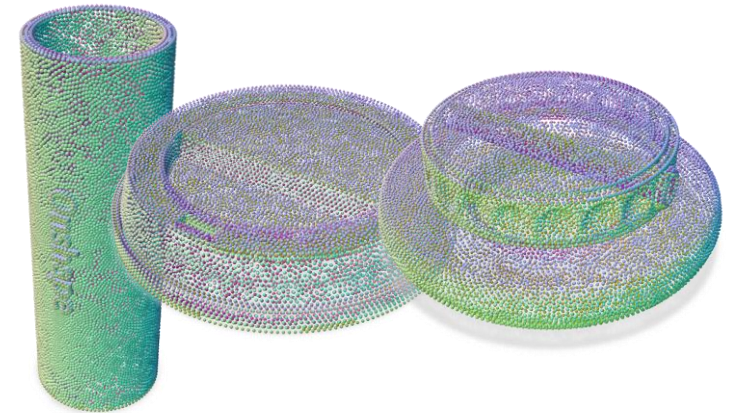
Benchmark - Datasets



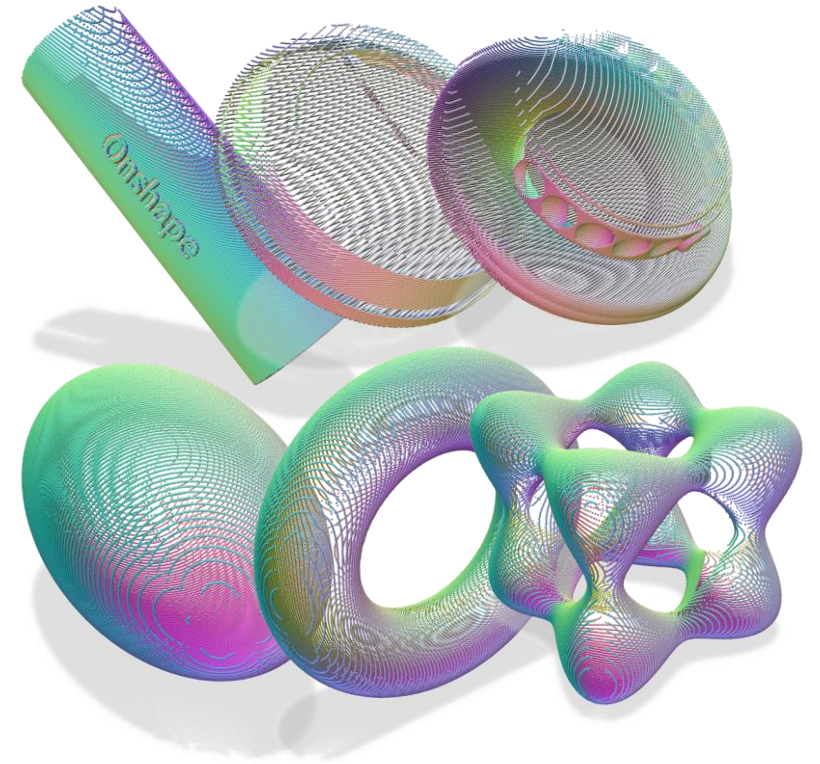
- Base of our benchmarking tool
- Samples of implicit polynomial surfaces
 - Generated with DGtal [DGt]
- Ground-truth
 - Weingarten maps

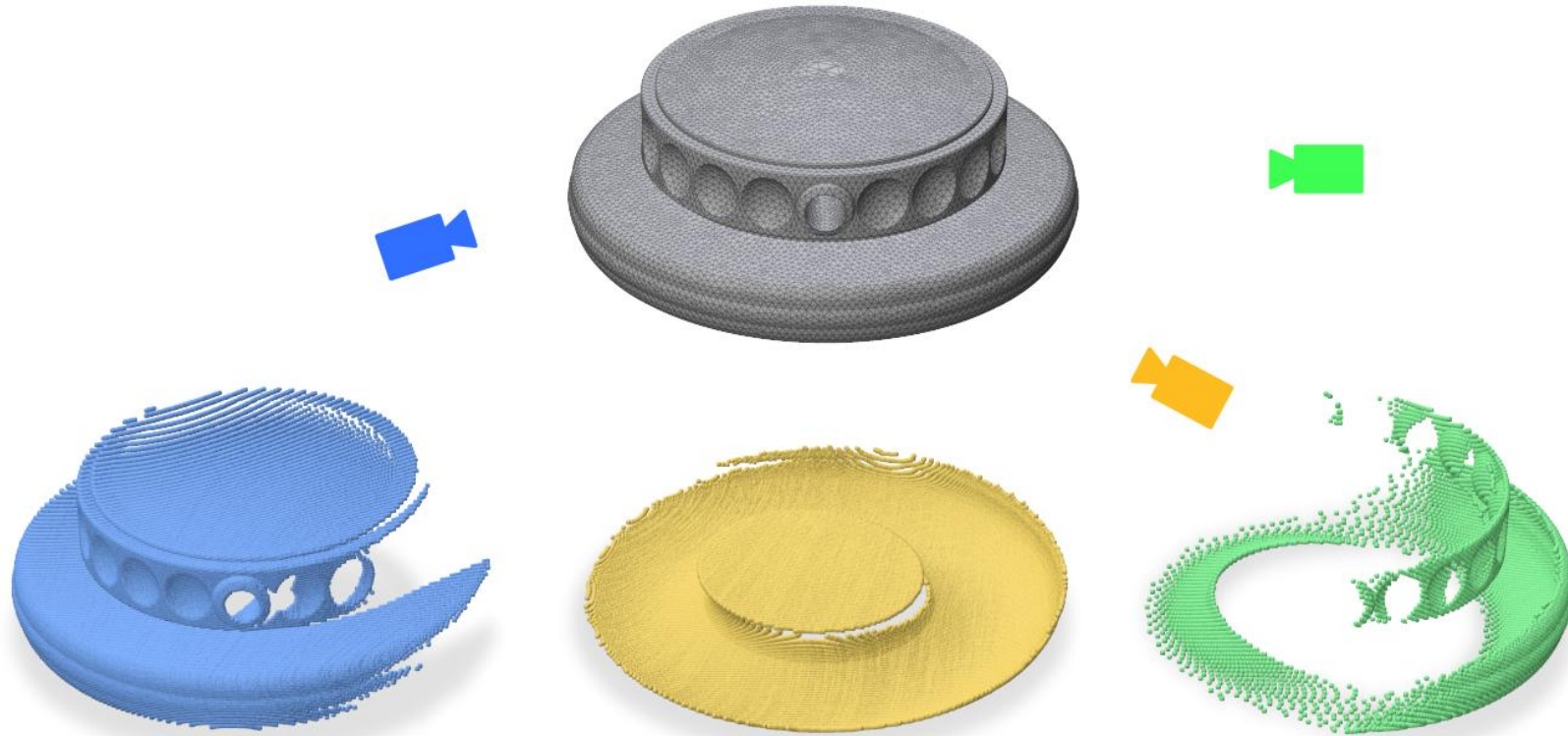


- Coming from the ABC dataset [KMJ*19]
- Sampled from the meshes
 - Poisson disk sampling [Yuk15]
- Ground-truth
 - Normals

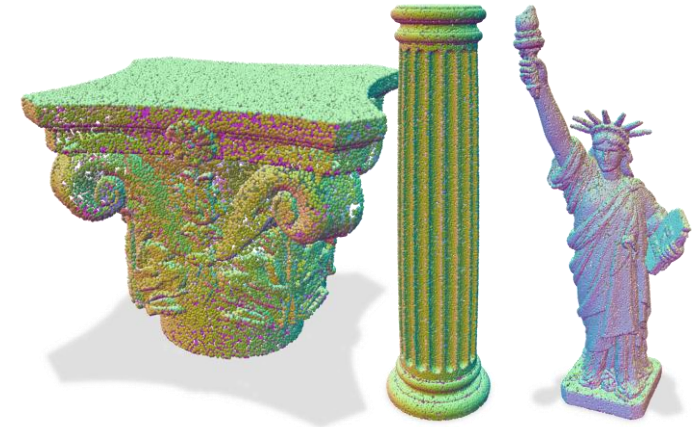


- LiDAR-like density variation
 - Helios++ [WEW*22]
- Over the DGtal dataset
- Over the ABC dataset

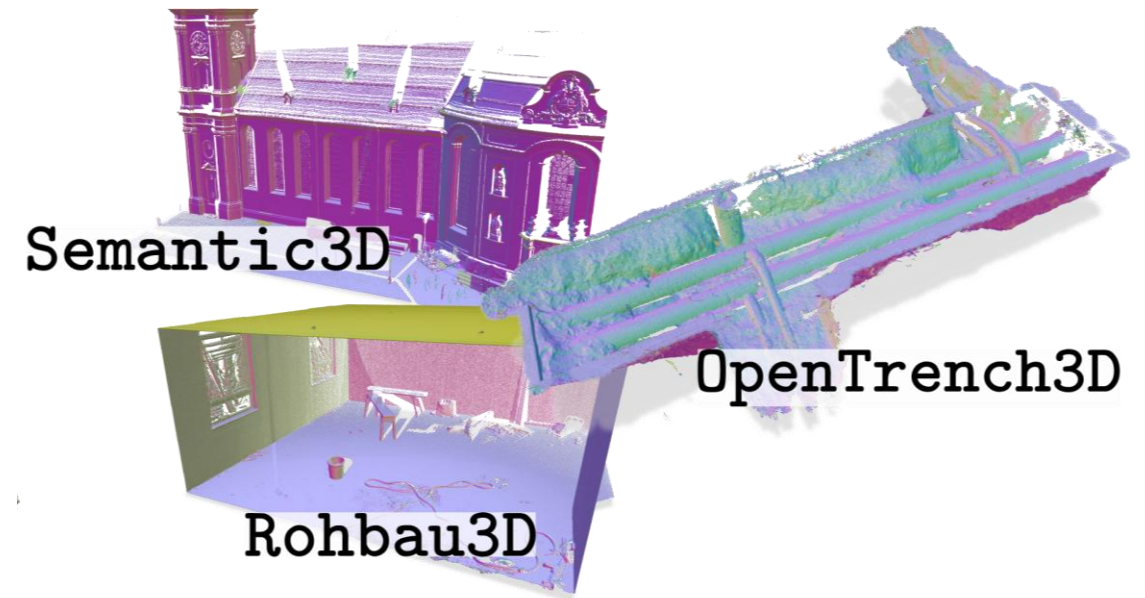


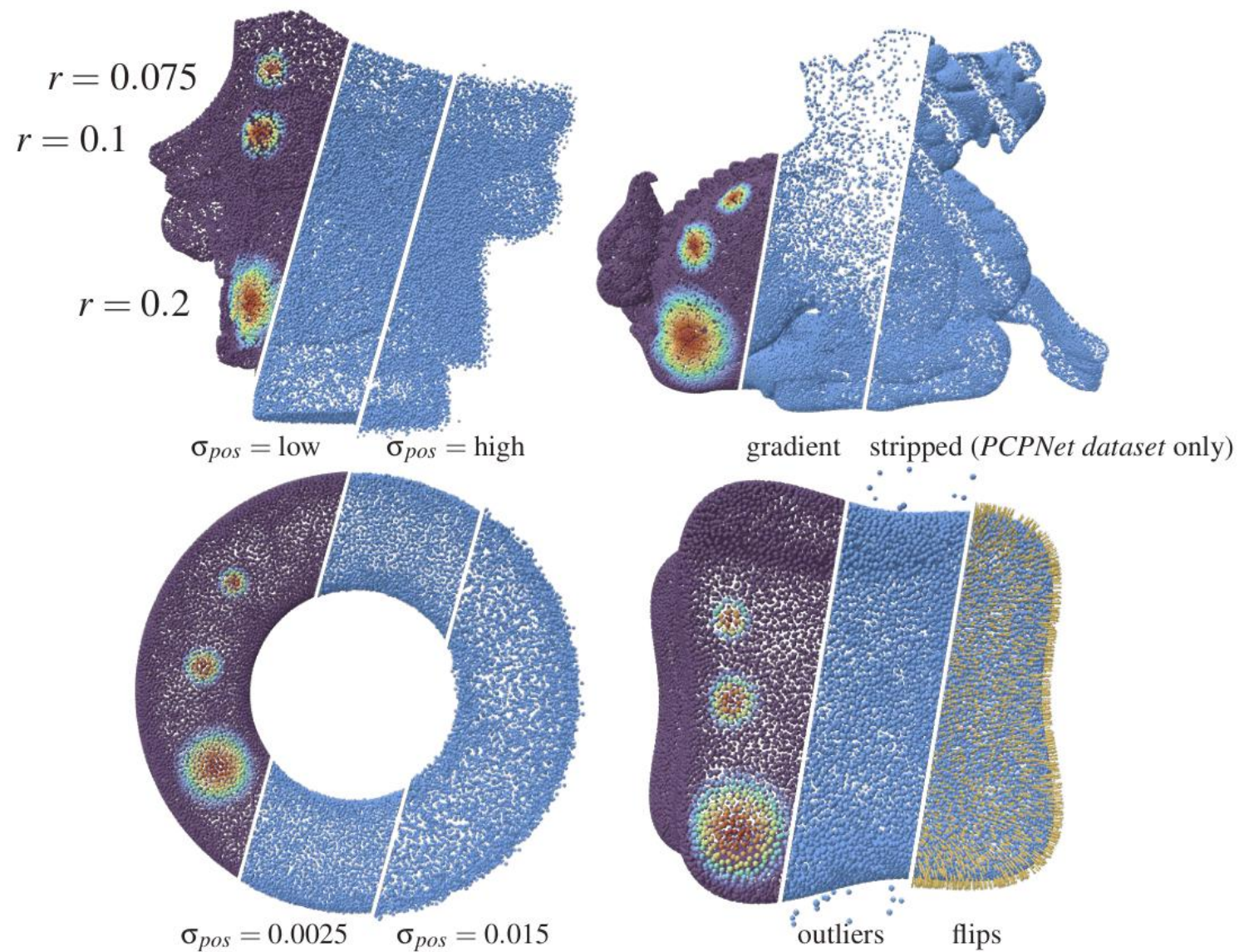


- Widely used in deep learning
- Test set of PCPNet Dataset
- Ground-truth
 - (mesh-estimated) curvatures
 - Normals



- Used for qualitative evaluation
- Rohbau3D [RB25]
- OpenTrench3D [HJP*24]
- Semantic3D [HSL*17]
- PCA-estimated normals
 - Z-oriented

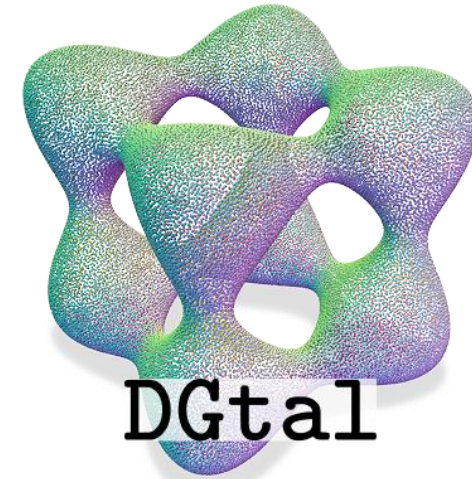




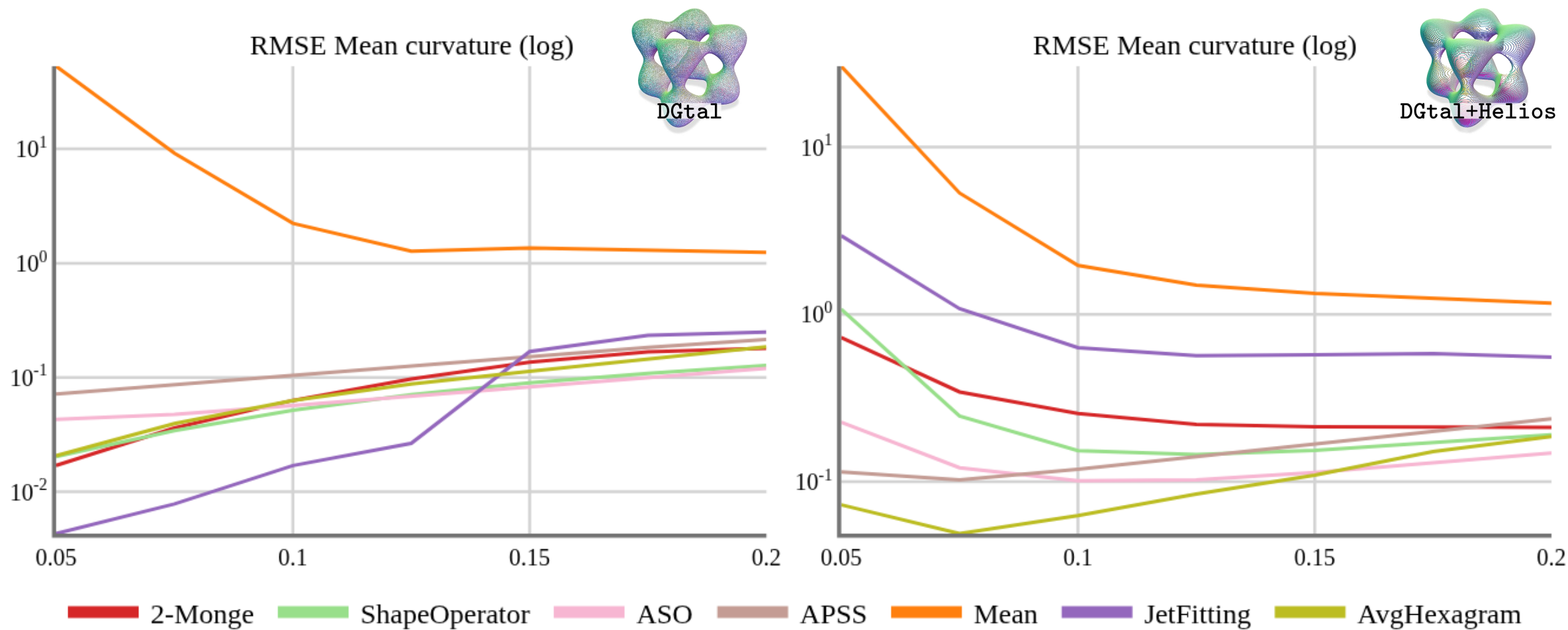
Benchmark – Evaluation

- Types de métriques
 - RMSE
 - RMS Angular E

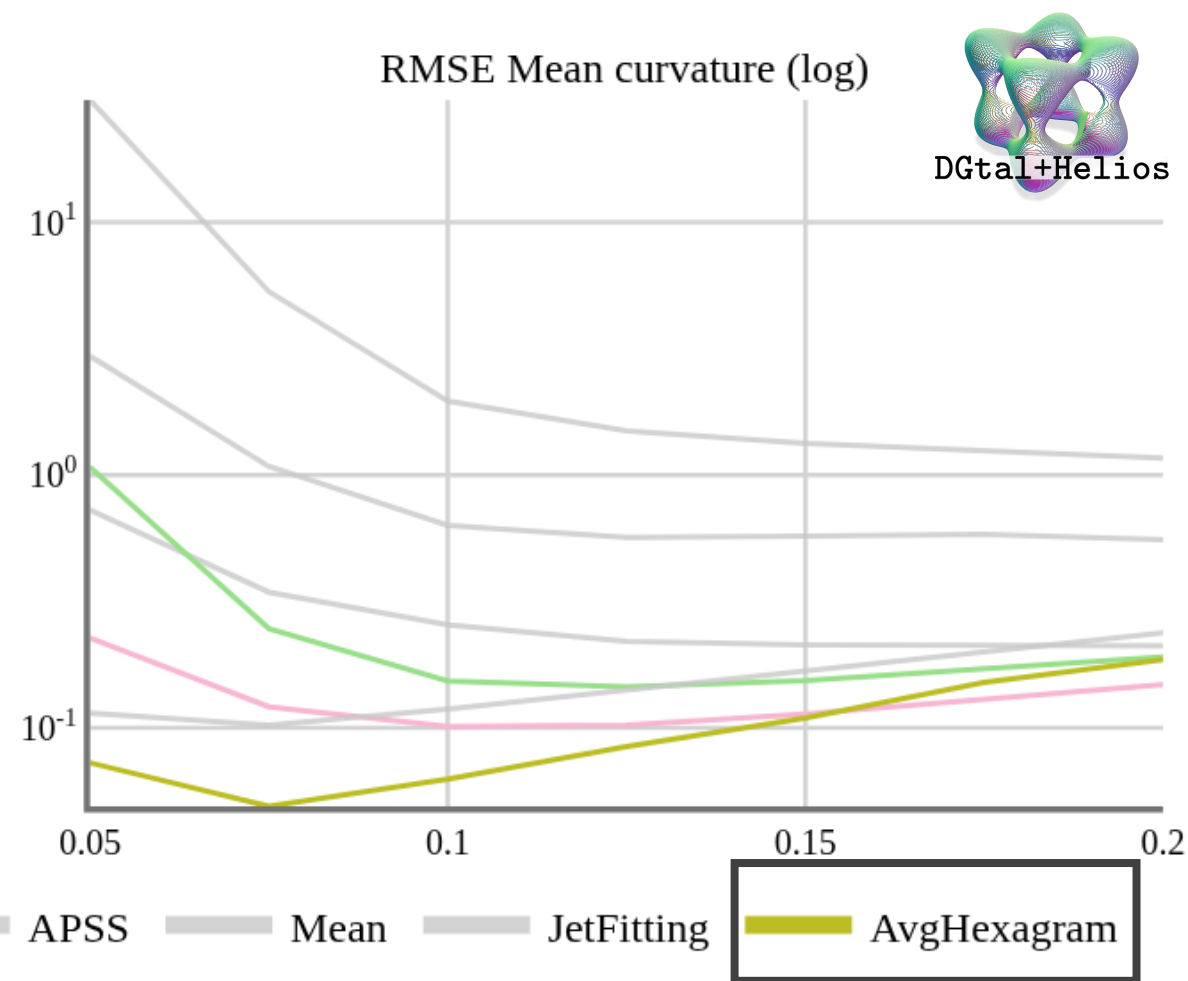
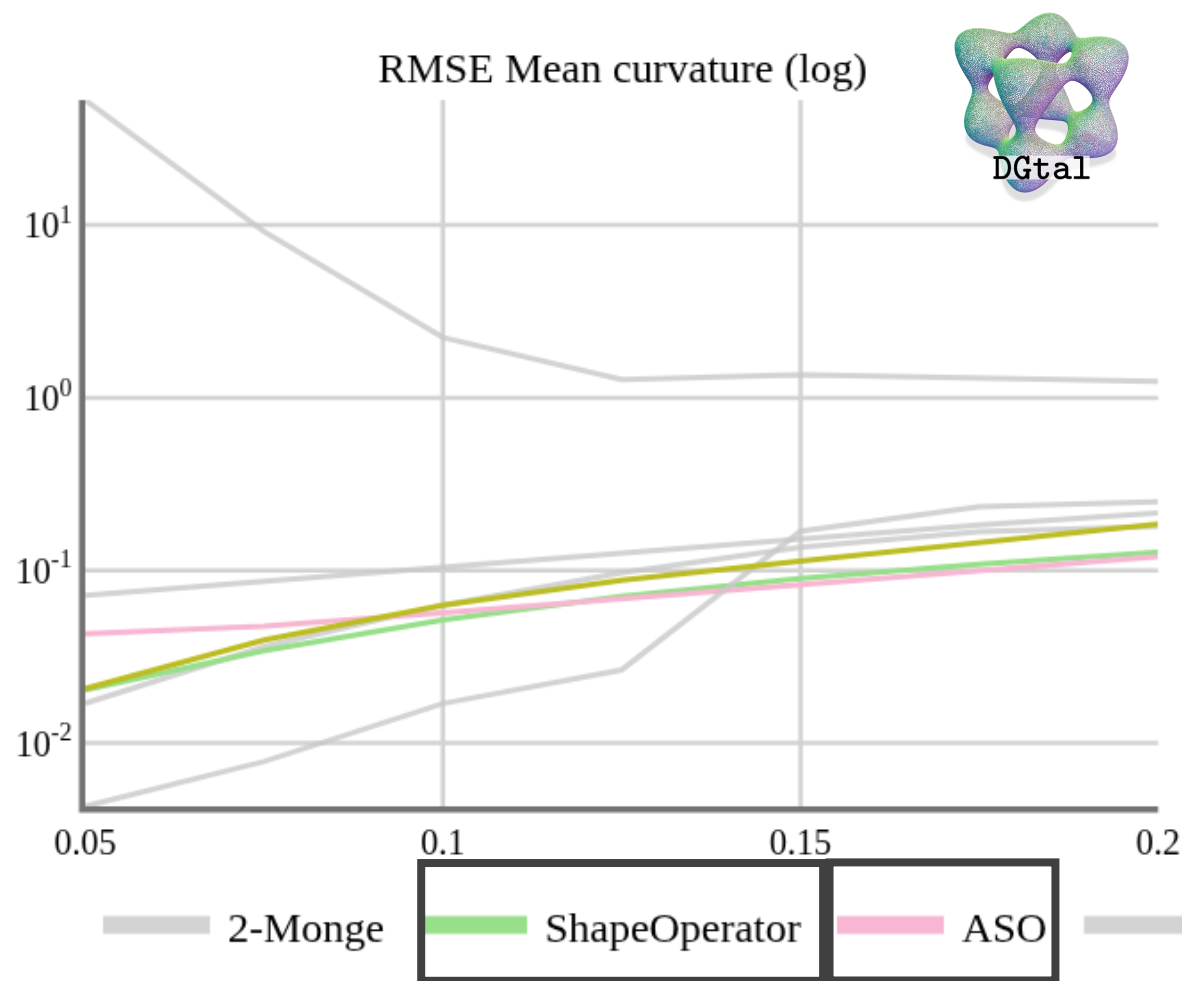
- DGtal dataset (left)
 - Clean data (no noise)
 - 50 000pts per shape
- DGtal + Helios dataset (right)
 - No added noise
 - Density variation with Helios



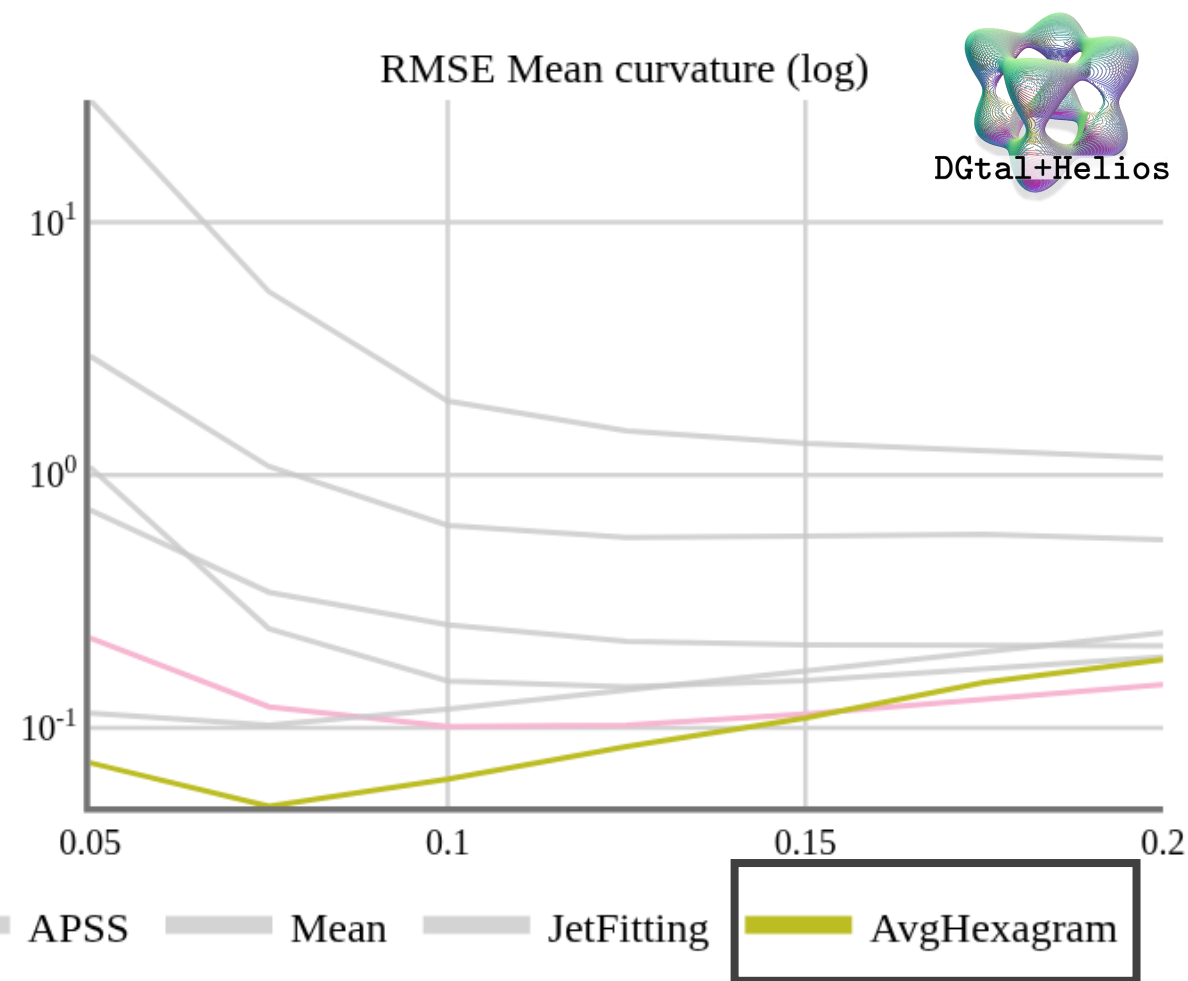
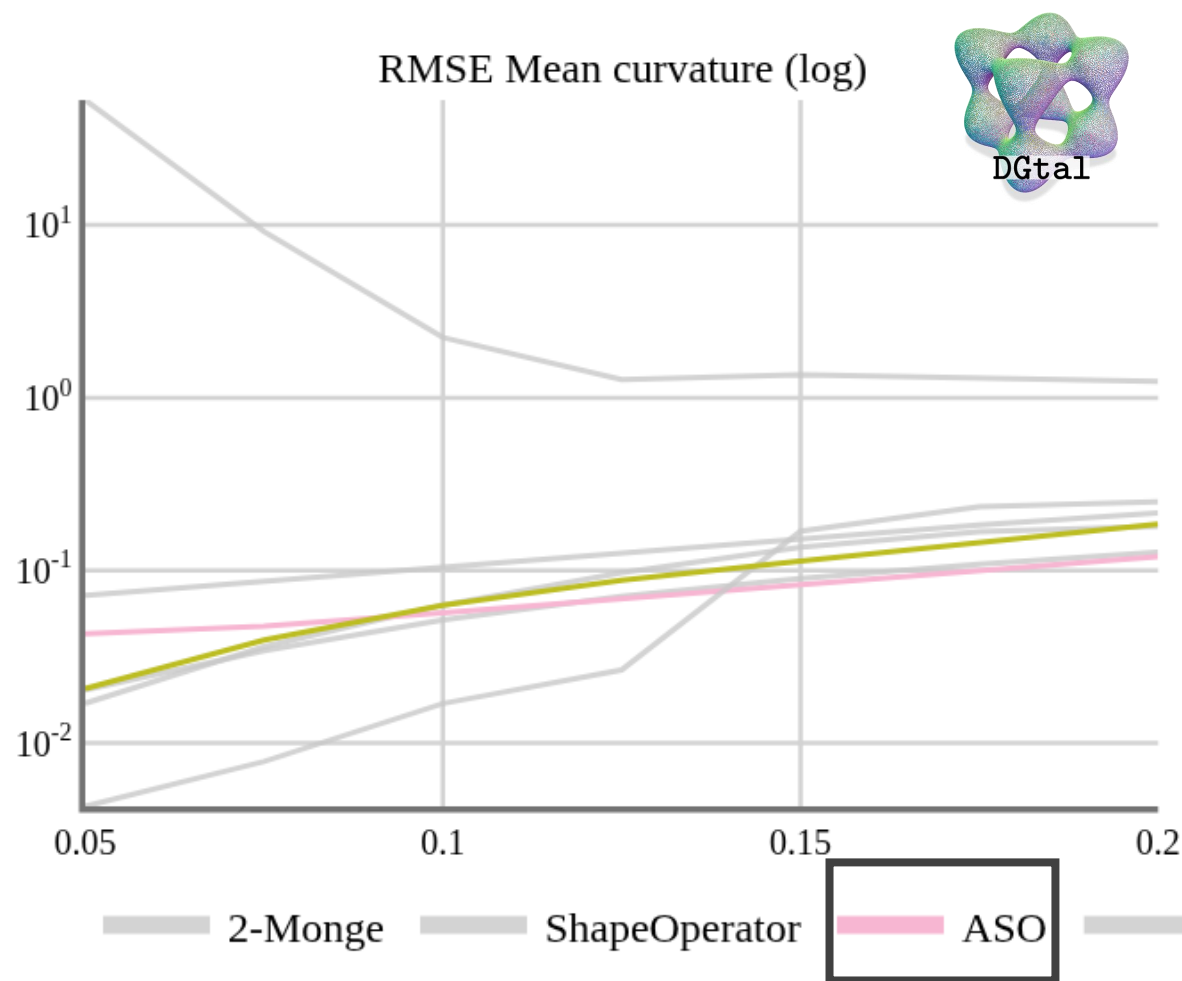
Results and take away messages



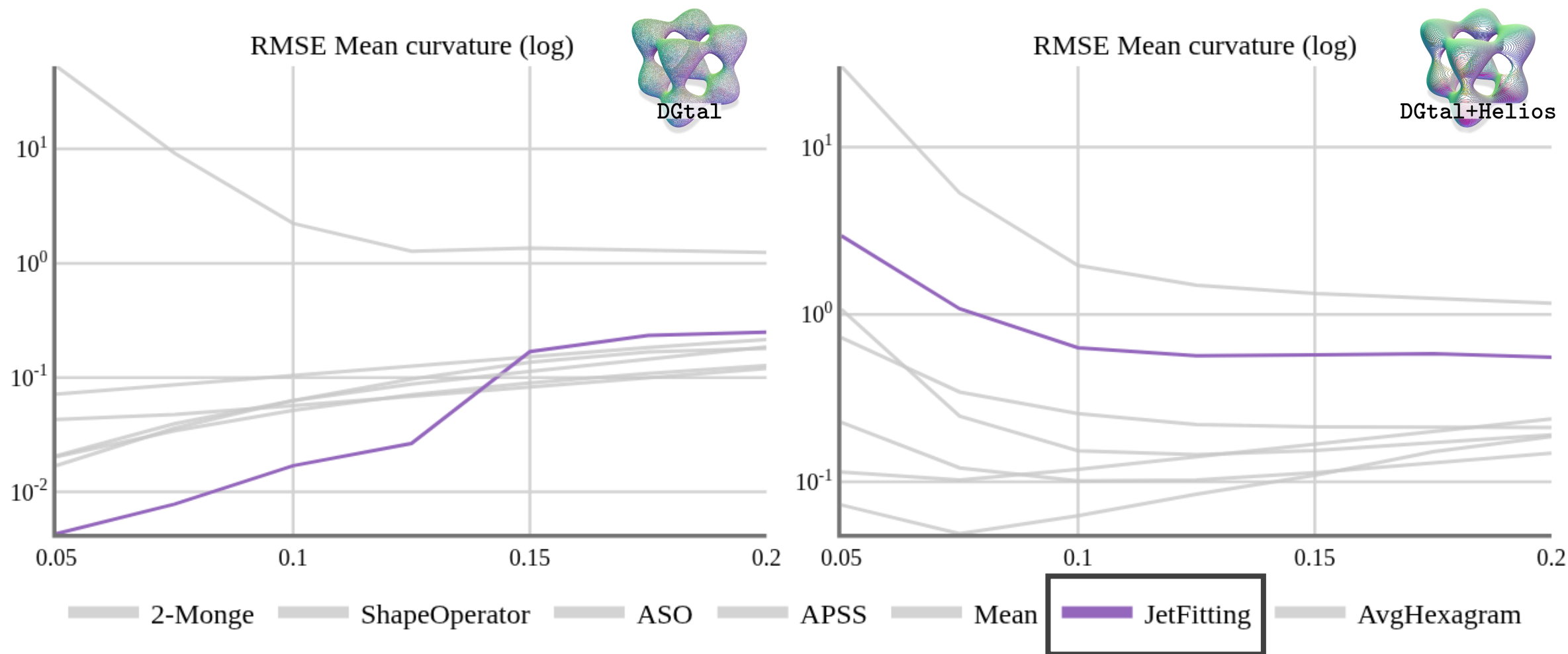
Results and take away messages



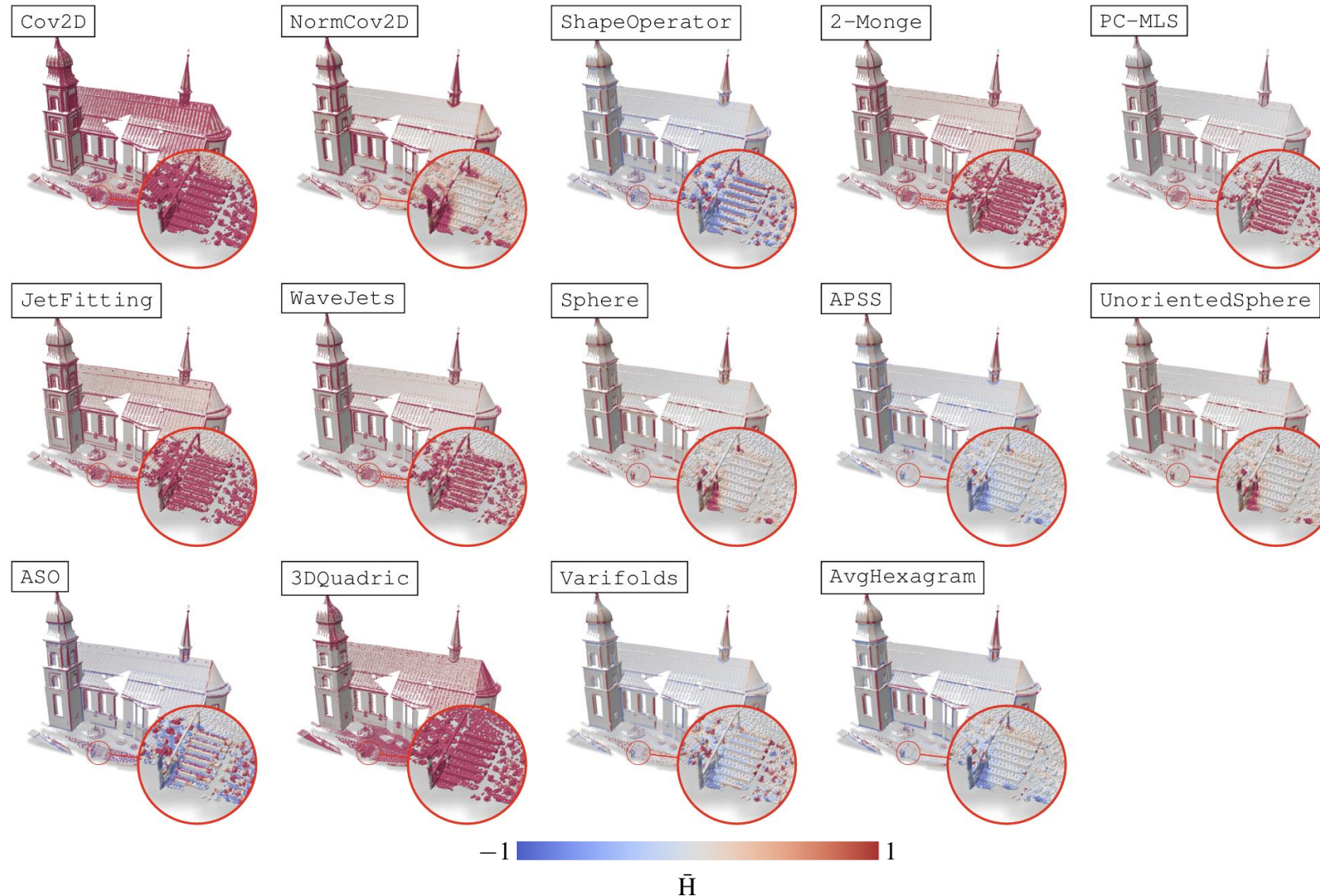
Results and take away messages



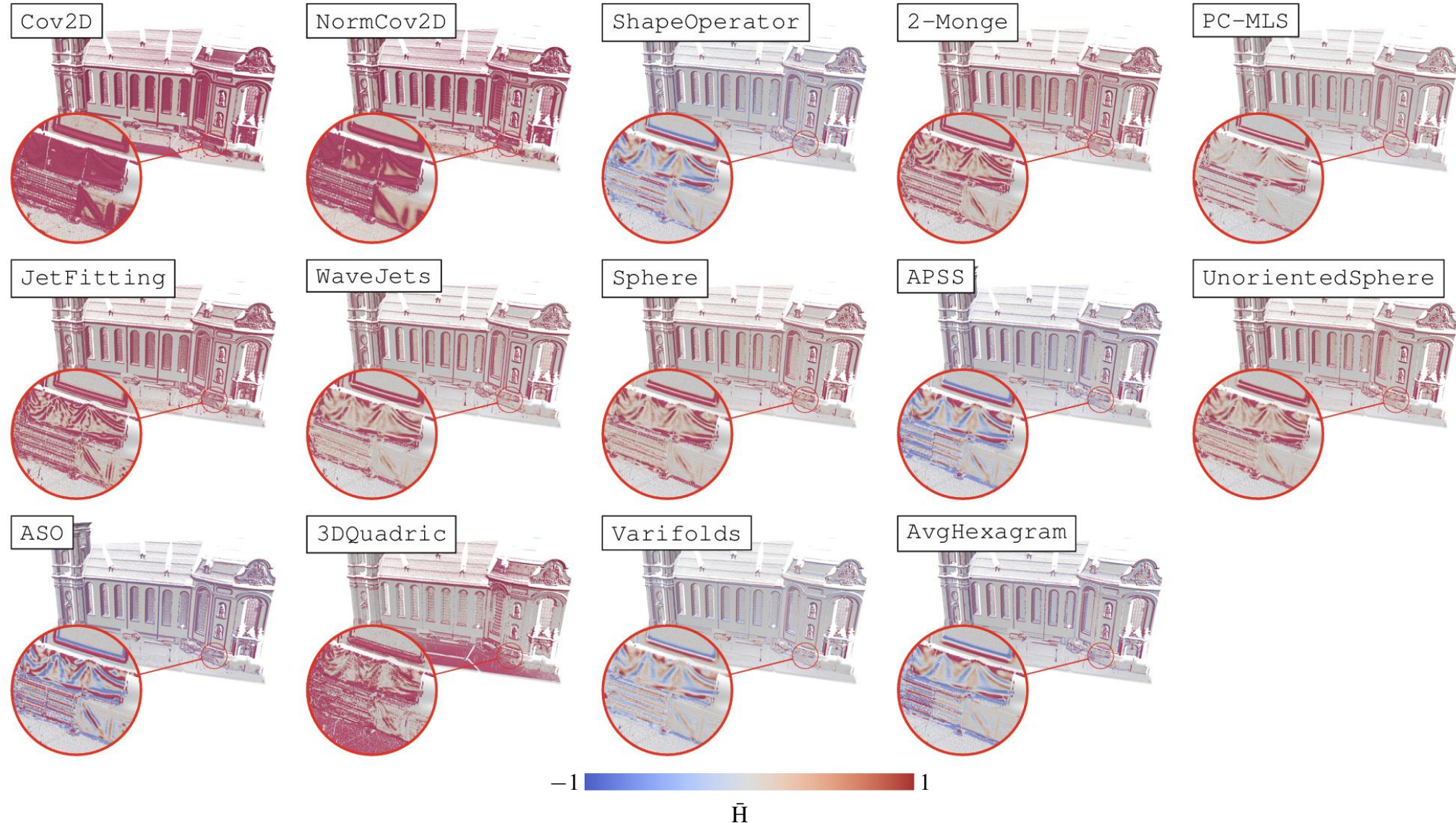
Results and take away messages



Results and take away messages



Results and take away messages



Future works

05

Future works

- Deeper analysis of the dependence on normals and orientation
- Unified benchmark including deep learning and 'classical' approaches
- Include prior knowledge of sensors
- Higher order differential properties

Conclusion

06

Summary

- Survey of differential estimators on 3d point cloud

Direct point-based approaches

Local surface models

Measure theory approaches

Machine learning approaches

- Open source benchmark

- Several datasets
- Ecosystem (data analysis, evaluation, website)

- Results

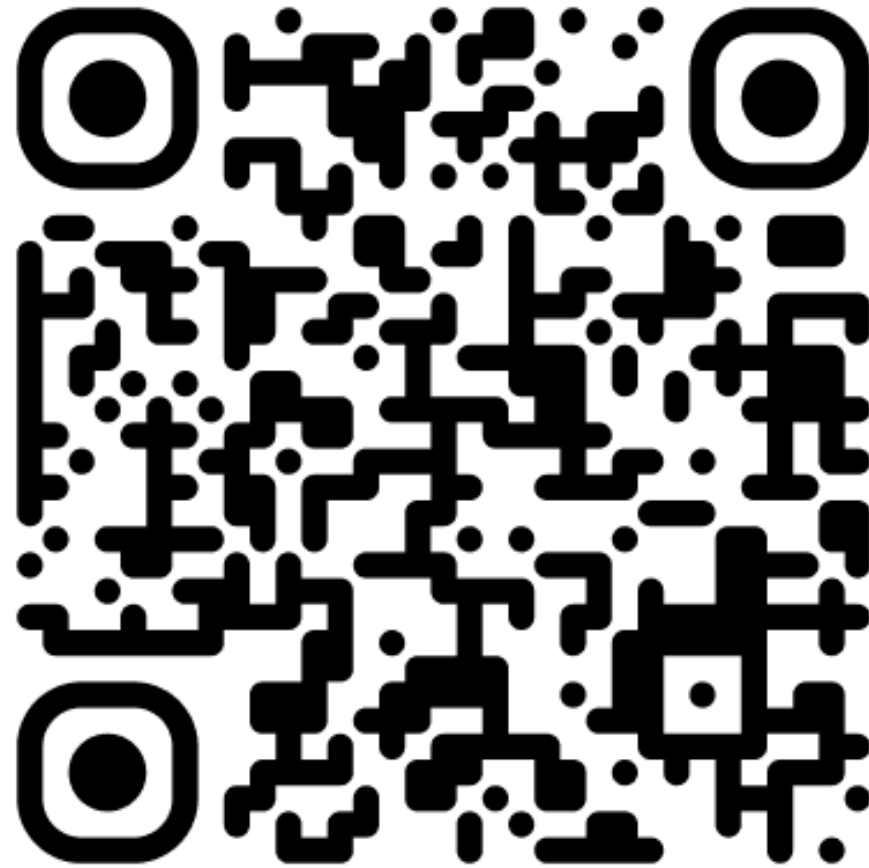
- No « best » method
- Depends on radius, type of surface, differential property, input requirements ...

Thanks!



storm-irit.github.io/pcloud-differential-estimation-benchmark-website

Thanks!



Léo is looking for a post doc!

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